

Universal Coset Field Theory:

From Axioms to the Standard Model and Gravitation

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June 6, 2026

Abstract

We formulate Universal Coset Field Theory (UCFT) on a foundational graph–Stone substrate and the exceptional Jordan algebra $J_3(\mathbb{O})$ with compact E_6 structure. A pre-axiomatic layer (BAMLP: Barrier-Aware Metric–Laguerre Partitioning) posits a weighted cosmic graph, a division-algebra ladder $0 \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$, and an exceptional cascade $0 \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \cdots \rightarrow 0$. Two axioms specify the order parameter and quantum kinematics; six postulates **P1–P6** supply the E_6 -invariant potential, vacuum manifold, Lorentzian soldering, chiral matter content, three generations, and GUT–Yukawa sector. Under these hypotheses we prove that the global minimum of the potential is the rank-one orbit $E_6/(\text{Spin}(10) \times U(1))$ (EIII), with 32 Goldstone modes and positive-definite coset metric. Postulate **P3** yields four-dimensional Lorentzian spacetime; one-loop fluctuations induce a positive Planck mass $M_{\text{Pl}}^2 = (28/6) f^2 \mu^2 \log(\Lambda^2/\mu^2)/(16\pi^2)$ and an asymptotically free $\text{Spin}(10) \times U(1)$ sector with $b_0 = 12$ for three generations. With **P4–P6** the theory embeds the Standard Model, including neutrino masses via the type-I seesaw. Conjectural results include a gravitational ultraviolet fixed point $\kappa_* = 96\pi^2/5$ and heterotic string correspondences. Octonionic algebra, soldering, representation theory, and heat-kernel computations are collected in appendices.

Keywords: BAMLP; Stone spaces; graph universe; exceptional ladder; exceptional Jordan algebra; E_6 coset; induced gravity; $\text{Spin}(10)$ grand unification; Standard Model embedding.

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Part I

Foundations and the Master Chain

1 Foundations: Graphs, Stone Spaces, and the Exceptional Ladder

Universal Coset Field Theory (UCFT) is not formulated in a vacuum. The present section records the *pre-axiomatic* substrate from which the E_6 /Albert arena is meant to descend: a combinatorial graph description of configuration space, a Stone-spectral ladder through the normed division algebras, and an exceptional-group cascade terminating in the Standard Model. The assignment mechanism that binds these layers is **BAMLP** (Barrier-Aware Metric–Laguerre Partitioning): a capacity-balanced dual on a metric–measure space that partitions workload into additively weighted cells.

Throughout we classify claims as **theorem** (standard mathematics), **postulate** (added structural identification), or **conjecture** (motivated physics correspondence).

1.1 The cosmic graph as primitive data

Postulate 1.1 (Graph arena). Physical configuration space is presented first as a locally finite weighted graph $\mathcal{G} = (V, E, w)$ together with a finite or σ -finite workload measure μ on V (or on edges/nodes after projection). Operational cost is a graph-geodesic (or barrier-aware) distance $c_i(x)$ from site x to territory generator $z_i \in V$. The *graph description of the universe* is this triple $(\mathcal{G}, \mu, \{c_i\})$; continuous geometry and Lie-group structure are derived, not assumed at this level.

In the BAMLP framework [1], territory assignment is the deterministic rule

$$x \mapsto \arg \min_i (c_i(x) - \omega_i), \quad (1)$$

with additive capacity weights ω_i adjusted until cell masses match prescribed targets τ_i . On a graph, the cells are additively weighted graph-Voronoi regions; capacities ω_i are Lagrange multipliers for the mass constraints. UCFT interprets this dual not as a political districting algorithm but as the *selection principle* that balances degrees of freedom across symmetry channels.

Theorem 1.2 (BAMLP capacity dual (finite, bounded cost)). *Let $c_i : V \rightarrow \mathbb{R}$ be bounded below and μ a finite measure on V . For target masses $\tau_i > 0$ with $\sum_i \tau_i = \mu(V)$, there exist weights $\omega \in \mathbb{R}^K$ such that the partition induced by (1) satisfies $|\mu(\text{cell}_i) - \tau_i| \leq \varepsilon$ for any $\varepsilon > 0$ under capacity balancing. The map $\omega \mapsto$ cell masses is concave; in the squared-Euclidean specialization the cells are Laguerre (power) diagrams.*

Remark 1.3 (UCFT reading). The 32 clock fields of UCFT are hypothesized to arise as the *boundary data* of a BAMLP partition on a configuration graph whose generators label symmetry-breaking channels. The rank-one EIII vacuum is then the unique capacity-balanced cell at the E_6 level, not an arbitrary minimum of an unconstrained potential. This identification is **conjectural**; the E_6 -invariant potential of postulate P1 supplies the proved dynamics once the arena is fixed.

1.2 Stone spaces and the division-algebra ladder

Stone duality links zero-dimensional compact Hausdorff spaces (Stone spaces) with Boolean algebras. The profinite spectrum of a graph’s clique complex supplies the combinatorial skeleton on which the normed division-algebra tower is mounted.

Definition 1.4 (Stone spectrum of a graph). Let $\mathcal{G}$ be a graph and $\mathcal{B}(\mathcal{G})$ the Boolean algebra generated by its vertex cuts (sets of vertices separated by removing edge subsets). The *Stone space* is $S(\mathcal{G}) = \text{Spec}(\mathcal{B}(\mathcal{G}))$, a compact, Hausdorff, totally disconnected space. A morphism of graphs inducing a Boolean homomorphism gives a continuous map of Stone spaces.

Theorem 1.5 (Stone duality). *The category of Stone spaces is contravariantly equivalent to the category of Boolean algebras. In particular, finite Boolean algebras correspond to finite discrete Stone spaces; profinite limits correspond to inverse systems of finite graphs.*

Theorem 1.6 (Hurwitz–Radon division algebra ladder). *The only finite-dimensional normed division algebras over $\mathbb{R}$ are, up to isomorphism, the tower*

$$0 \longrightarrow \mathbb{R} \longrightarrow \mathbb{C} \longrightarrow \mathbb{H} \longrightarrow \mathbb{O}, \quad (2)$$

of dimensions 1, 2, 4, 8. The passage $\mathbb{O} \rightarrow \mathbb{S}$ (sedenions) fails: $\mathbb{S}$ is not a division algebra. The Albert algebra $J_3(\mathbb{O})$ and the cubic norm $N(X) = \det X$ are the canonical exceptional-Jordan packaging of the terminal algebra $\mathbb{O}$ in this tower.

Postulate 1.7 (Stone–division lift). The profinite Stone limit of the cosmic graph $\mathcal{G}$ carries a sequential lift through the division-algebra rungs (2). The zero object 0 at the base denotes the trivial Boolean/Stone sector (no observable localisation); each rung adds a multiplicative structure compatible with the graph’s capacity measure:

Rung	Algebra	dim	Automorphism / structure group (typical)
0	$\{0\}$	0	trivial
$\mathbb{R}$	reals	1	order, $\text{Aut} = \{1\}$
$\mathbb{C}$	complexes	2	$U(1)$ phase
$\mathbb{H}$	quaternions	4	$SU(2)$
$\mathbb{O}$	octonions	8	G_2
$J_3(\mathbb{O})$	Albert	27	$F_4 \subset E_6$

The lift is functorial with respect to graph refinements: each BAMLP capacity rebalancing that refines $\mathcal{G}$ induces a compatible map of Stone spectra and preserves the division-algebra rung labels.

1.3 The exceptional ladder

Above the octonionic/Jordan rung, the exceptional Lie algebras form a second ladder linked to heterotic string theory and grand unification. UCFT occupies the E_6 rung; the full conjectured cascade is:

$$0 \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \text{Spin}(10) \times U(1) \rightarrow SU(5) \rightarrow SU(3)_c \times SU(2)_L \times U(1)_Y \rightarrow SU(3)_c \times U(1)_{\text{em}} \rightarrow \quad (3)$$

Theorem 1.8 (Steps proved or embedded in UCFT). *The following steps of (3) are established in the body of this manuscript (under postulates P1–P6 where indicated):*

- (i) $E_6 \rightarrow \text{Spin}(10) \times U(1)$: *vacuum selection* ([Theorem 6.7](#)).
- (ii) $\text{Spin}(10) \times U(1) \rightarrow SU(5) \times U(1)_X \rightarrow \cdots \rightarrow SU(3)_c \times U(1)_{\text{em}}$: *GUT descent* ([Theorem 13.2](#)).

- (iii) $E_8 \times E_8$ central charges $(c_L, c_R) = (16, 10)$ and $h^1(V_{16}) = 3$: heterotic correspondence ([section 14](#)).

Conjecture 1.9 (BAMLP–exceptional closure). *The entire ladder (3), including the initial $0 \rightarrow E_8$ lift from the Stone–division tower and the final $SU(3)_c \times U(1)_{\text{em}} \rightarrow 0$ collapse, is the capacity-balanced partition of a single cosmic graph $(\mathcal{G}, \mu)$ under BAMLP with generators labelled by the groups in (3). In particular:*

- E_8 is the maximal symmetry of the $\mathbb{O}$ -valued graph measure before Jordan projection;
- $E_8 \times E_8$ is the product graph for left/right heterotic sectors;
- E_6 is the rank-one balanced cell selected by (1) with E_6 -invariant capacity weights;
- the terminal 0 is the trivial gauge group after electroweak breaking.

1.4 Logical position in UCFT

The dependency order becomes:

Graph $(\mathcal{G}, \mu)$ + Stone lift $\Rightarrow$ Division tower (2)
 $\Downarrow$ **BAMLP capacity dual (1)**
 $\Downarrow$ **Exceptional ladder (3)** (hits E_6)
 $\Downarrow$ **UCFT axioms I–II + P1–P6 $\Rightarrow$ Standard Model + gravitation**

Remark 1.10 (Epistemic status). The graph postulate ([Theorem 1.1](#)), Stone–division lift ([Theorem 1.7](#)), and exceptional closure ([Theorem 1.9](#)) are *foundational conjectures*. They are not used as hypotheses in any theorem of [section 6–section 13](#). Their role is to supply a *generative narrative*: if the universe is a capacity-balanced graph partition, the E_6 /Albert coset and its proved SM embedding are the natural intermediate rungs, not ad hoc choices. Deriving P1–P6 from BAMLP remains open work.

1.5 Open problems

1. Construct an explicit cosmic graph $\mathcal{G}$ whose BAMLP partition at the E_6 level reproduces the EIII vacuum manifold $\dim_{\mathbb{R}} = 32$.
2. Prove that the Stone profinite limit of $\mathcal{G}$ inherits an F_4 action descending to the compact E_6 structure group on $J_3(\mathbb{O})$.
3. Derive the E_6 -invariant potential (6) as the capacity-dual functional of a graph measure, not as a postulate.
4. Extend the ladder (3) below E_8 with a controlled $E_8 \rightarrow E_6$ breaking mechanism compatible with the heterotic $E_8 \times E_8$ product.

2 Overview of the Construction

This section summarizes the logical structure of the paper. The foundational layer ([section 1](#)) posits a graph description of the universe, a Stone-spectral ladder through the division algebras, and BAMLP capacity balancing on the exceptional cascade (3). The axioms and postulates P1–P6 then yield a four-dimensional quantum field theory whose low-energy limit contains the Standard Model coupled to Einstein–Hilbert gravitation.

Step	Input	Output	Reference
−3	Graph $\mathcal{G}$, measure μ	Stone spectrum, Boolean cuts	subsection 1.1
−2	Stone lift	$0 \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$	Theorem 1.6
−1	BAMLP dual	Exceptional ladder to E_6	Theorem 1.9
0	Axioms I–II	$J_3(\mathbb{O})$, compact E_6	§5
1	P1	Rank-one minimum	Theorem 6.7
2	P2	$\mathcal{M} = \text{EIII}$, 32 modes	section 7
3	Geometry	$\mathbf{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$	Theorem 7.2
4	Maurer–Cartan	Composite $\text{Spin}(10) \times U(1)$	Theorem 9.1
5	P3	Lorentzian M^4 ; spectrum $28 + 4$	section 8
6	Axiom II, P3	Emergent time	subsection 8.4
7	Massive modes	$M_{\text{Pl}}^2 > 0$	section 11
8	Matter content	Asymptotic freedom; $b_0(3) = 12$	Theorem 13.7
9	FRG	$\kappa_\star = 96\pi^2/5$	section 12
10	P4–P6	Standard Model	section 13
11	–	Heterotic correspondence	section 14

Table 1: Main logical dependencies.

2.1 Axioms

Axiom 2.1 (I: Order parameter). The field $\Phi \in \mathbf{27} = J_3(\mathbb{O})$ transforms under compact E_6 . The origin $\Phi = 0$ is a strict local maximum of the potential of P1 ([Theorem 6.9](#)); dynamics select the rank-one orbit ([Theorem 6.10](#)).

Axiom 2.2 (II: Quantum kinematics). States admit unitary evolution. In the semiclassical regime this evolution is realized through a Page–Wootters clock ϕ^0 and the Hamiltonian constraint of induced gravity ([subsection 8.4](#)).

2.2 Postulates P1–P6

	Postulate	Content
P1	Potential	E_6 -invariant $V(N, X^\#)$; rank-one minimum
P2	Vacuum	$\mathcal{M} = \text{EIII} = E_6/(\text{Spin}(10) \times U(1))$; proved from P1
P3	Soldering	Four clock fields $\rightarrow$ Lorentzian $(1, 3)$ via $H_2(\mathbb{C})$
P4	Matter	One chiral $\mathbf{16}$ per family
P5	Generations	$SU(3)_F$ on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$
P6	Yukawa	GUT Higgs sector; type-I seesaw

2.3 Logical dependencies

2.4 Principal parameters

Quantity	Value
Vacuum	$\mathcal{M} = E_6/(\text{Spin}(10) \times U(1))$, $\dim_{\mathbb{R}} = 32$
Coset tangent	$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$
Spectrum	28 massive scalars + 4 gauge modes
M_{Pl}^2	$\frac{28}{6} f^2 \mu^2 \log(\Lambda^2/\mu^2)/(16\pi^2)$
b_0	26 (one generation); 12 (three generations)
$\kappa_{\star}$	$96\pi^2/5$ (conjectural)

2.5 Summary

Theorem 2.3 (Embedding). *Under Axioms [Theorem 2.1](#)–[Theorem 2.2](#) and postulates P1–P6, UCFT defines a four-dimensional effective field theory with proved vacuum structure, induced gravitation, an asymptotically free gauge sector, and an embedded Standard Model. String-theoretic correspondences and the gravitational ultraviolet fixed point are conjectural.*

Proof. See [Table 1](#) and [Theorem 3.3](#). □

3 Introduction

3.1 Motivation

The present work rests on a three-tier foundation. At the deepest level, [section 1](#) posits a *graph description of the universe*—a weighted configuration graph with a Stone-spectral lift through the division algebras $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$ —and an exceptional cascade $0 \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \dots \rightarrow 0$ selected by BAMLP capacity balancing. The middle tier is UCFT proper: exceptional Lie algebras and Jordan algebras provide the arena for unification. The exceptional Jordan (Albert) algebra $J_3(\mathbb{O})$ —the algebra of 3×3 Hermitian octonionic matrices—is the unique finite-dimensional formally real Jordan algebra that is not special. It has dimension 27, automorphism group F_4 , and is preserved by the exceptional group E_6 through its cubic norm $N(X) = \det X$ (up to a real character). Under the maximal subgroup $H = \text{Spin}(10) \times U(1)$, the fundamental representation branches as

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \quad (4)$$

which contains a single chiral spinor $\mathbf{16}$ of $\text{Spin}(10)$ —one Standard Model generation together with a right-handed neutrino. The adjoint $\mathbf{78}$ decomposes as

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3} \quad (\text{under } H = \text{Spin}(10) \times U(1)), \quad (5)$$

exhibiting the coset tangent space $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ of real dimension 32. The appearance of a viable grand-unified gauge group together with the chiral matter content of one family motivates the present construction.

3.2 Axioms and scope

Universal Coset Field Theory (UCFT) is formulated on the compact real form of E_6 , with order parameter $\Phi \in \mathbf{27} = J_3(\mathbb{O})$. The theory is specified by two axioms and six additional postulates P1–P6, stated below. The axioms determine the algebraic arena and the kinematical framework; the postulates supply the potential, vacuum geometry, spacetime structure, matter content, and symmetry-breaking data required for contact with four-dimensional physics.

Axiom 3.1 (Order parameter). The order parameter is a field Φ valued in the matter representation $\mathbf{27} = J_3(\mathbb{O})$ of E_6 . The symmetric configuration $\Phi = 0$ is the unique E_6 -fixed point; the potential of postulate P1 renders it a strict local maximum ([Theorem 6.9](#)), so that the rank-one orbit is selected dynamically ([Theorem 6.10](#)).

Axiom 3.2 (Quantum kinematics). Physical states $\psi \in \mathcal{H}$ are acted on by a one-parameter unitary family $\{U(\tau)\}_{\tau \in \mathbb{R}}$ with $U(0) = \text{Id}$, $U(\tau + s) = U(\tau)U(s)$, and strong continuity.

In the main development, Axiom II is realized through emergent time: a Page–Wootters clock field φ^0 together with the Hamiltonian constraint $\hat{H}_{\text{phys}}|\Psi\rangle\rangle = 0$ of induced gravity, with $U(\tau)$ recovered on physical states in the semiclassical regime via a timelike Killing vector ([subsection 8.4](#)). The Standard Model and gravitation are not consequences of the axioms alone; they follow upon imposing postulates P1–P6.

3.3 Postulates P1–P6

P1 (Potential). An E_6 -invariant polynomial potential built from the cubic norm N and the sharp map $X^\#$, of the form

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \geq 2 \text{ even}, \quad (6)$$

Result	Content	Status
Vacuum selection	P1 $\Rightarrow$ P2: global minimum = rank-one EIII orbit	Theorem
Origin instability	Hess $V(0) \prec 0$, roll-down to $\mathcal{M}$	Theorem
Minkowski block	$H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$, $SL(2, \mathbb{O}) \cong \text{Spin}(1, 9)$	Theorem
Signature $(1, 3)$, $d = 4$	Soldering postulate P3	Postulate
Emergent time	Page–Wootters clock; OS reconstruction	Theorem / conjecture
Induced Planck mass	$M_{\text{Pl}}^2 = \frac{28}{6} f^2 \mu^2 \log(\Lambda^2/\mu^2)/(16\pi^2) > 0$	Theorem
Gauge running	Asymptotically free; $b_0(1) = 26$, $b_0(3) = 12$	Theorem
Gravitational fixed point	$\kappa_\star = 96\pi^2/5$	Conjecture
Standard Model content	One family per 16 ; anomaly cancellation	Theorem
Flavour sector	Three generations; Yukawa; $m_\nu \sim 0.05 \text{ eV}$	Postulate
Heterotic correspondence	$(c_L, c_R) = (16, 10)$, $h^1(V_{16}) = 3$	Correspondence

Table 2: Classification of the principal results.

whose global minimum is the rank-one idempotent orbit. Coercivity is established on the compact real form (section 6).

P2 (**Vacuum**). The vacuum manifold is

$$\mathcal{M} = E_6/(\text{Spin}(10) \times U(1)) = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32, \quad (7)$$

the rank-one primitive-idempotent orbit (the complex Cayley plane). This is proved from P1 as Theorem 6.7.

P3 (**Soldering**). A soldering map identifies four clock-field directions with a tangent frame carrying the Minkowski form inherited from N via $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$. The dimension $d = 4$ and signature $(1, 3)$ are part of this postulate (section 8).

P4 (**Matter**). One chiral **16** of $\text{Spin}(10)$ per family, with charges as in (4).

P5 (**Generations**). Three families via an $SU(3)_F$ horizontal symmetry on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$, corroborated by the heterotic index $h^1(\hat{Q}, V_{16}) = 3$.

P6 (**Yukawa sector**). Standard $\text{Spin}(10)$ Higgs and Yukawa multiplets (**45_H/54_H**, **126_H**, **10_H**) generating fermion masses and the type-I seesaw.

3.4 Status of results

Results are classified as follows.

Theorem. Proved from explicitly stated hypotheses.

Postulate. Additional structure beyond the axioms (P1–P6).

Conjecture / indicative. Motivated by explicit but uncontrolled calculations, or low-energy correspondences not established as dualities.

3.5 Main theorem

Theorem 3.3 (Conditional embedding). *Assume Axioms [Theorem 3.1](#) and [Theorem 3.2](#) on the compact real form of E_6 , together with postulates P1–P6. Then:*

- (i) ([Theorem 6.7](#)). *The global minimum of (6) is the orbit $\mathcal{M} = E_6/(\text{Spin}(10) \times U(1)) = \text{EIII}$, with 32 Goldstone modes spanning*

$$\mathbf{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (8)$$

and positive-definite coset metric $K|_{\mathbf{m}}$.

- (ii) (P3, [section 8](#)). *Soldering yields a four-dimensional Lorentzian manifold of signature $(-, +, +, +)$ with 28 physical massive scalars of mass $m^2 = \mu^2$.*
- (iii) ([section 11](#)). *One-loop fluctuations induce*

$$M_{\text{Pl}}^2 = \frac{28}{6} \frac{f^2 \mu^2 \log(\Lambda^2/\mu^2)}{16\pi^2} > 0. \quad (9)$$

- (iv) ([section 11](#), [section 12](#)). *The composite $\text{Spin}(10) \times U(1)$ gauge sector is asymptotically free with $\hat{g}_\star^2 = 0$ and*

$$b_0 = \frac{11}{3}C_A - \frac{4}{3}T n_F - \frac{1}{6}T n_S, \quad (10)$$

where $b_0 = 26$ for one generation and $b_0 = 12$ for three.

- (v) (P4–P6, [section 13](#)). *The model reduces to the Standard Model with three chiral families and neutrino masses via the type-I seesaw.*
- (vi) ([section 12](#)). *The gravitational coupling possesses a conjectural ultraviolet fixed point $\kappa_\star = 96\pi^2/5$.*

Proof. Parts (i), (iii), (iv), and the anomaly structure of (v) are established in the sections cited. Part (ii) combines the soldering postulate with the algebraic theorems of [section 8](#). The flavour content of (v) is postulated. Part (vi) rests on a one-loop truncation. $\square$

Remark 3.4 (Scope). (i) Lorentzian signature is not inherited from the coset metric $K|_{\mathbf{m}}$, which is positive-definite; it enters through P3. (ii) The vacuum is the rank-one EIII orbit (7), not the 26-dimensional E_6/F_4 orbit of $\text{diag}(1, 1, 1)$. (iii) The cosmological constant is a tunable parameter; the fine-tuning problem is not addressed here.

3.6 Organization

[section 2](#) gives an overview of the logical dependencies. [section 5](#) reviews the Albert algebra and E_6 structure. [section 6](#) proves vacuum selection; [section 7](#) develops the coset σ -model. [section 8](#) treats soldering and emergent time. [section 11](#) and [section 12](#) compute induced gravitation and renormalization-group flow. [section 13](#) carries out $\text{Spin}(10)$ descent to the Standard Model; [section 14](#) discusses string-theoretic correspondences. Appendices collect the underlying octonionic and representation-theoretic results.

We use signature $(-, +, +, +)$ throughout; $[f] = \text{mass}$, and $\hat{g}^2$, ξ , $\hat{y}^2$, κ , Λ_{cc} are dimensionless. The 32 coset Goldstone coordinates are termed *clock fields*.

4 Axioms and Added Postulates

This section fixes the foundation of Universal Coset Field Theory (UCFT). We state the single dynamical *axiom* that supplies the arena — an order parameter valued in the matter representation of E_6 , together with a quantization prescription — and we then enumerate, once and up front, the added postulates P1–P6 that convert this arena into a four-dimensional Lorentzian quantum field theory. UCFT is formulated as a *conditional embedding*: the axiom supplies the kinematical arena, P1–P6 supply the remaining structure, and every nontrivial claim below is classified as a **theorem** (proved from stated hypotheses), **postulate** (assumed: P1–P6), or **conjecture** (motivated, not controlled).

Conventions used throughout the manuscript are fixed here: the spacetime metric signature is $(-, +, +, +)$; the decay constant f carries mass dimension $[f] = \text{mass}$; the breaking/mass scale is μ , the UV cutoff is Λ , and the breaking-VEV (GUT) scale tied to f is M_{GUT} ; the couplings $\hat{g}^2$, ξ , $\hat{y}^2$, κ , Λ_{cc} are dimensionless. The 32 Goldstone directions of the symmetry-breaking coset are called the *clock fields*.

4.1 Axiom I: the order parameter and the E_6 /Albert arena

The arena of UCFT is the exceptional Jordan (Albert) algebra $J_3(\mathbb{O})$, the 27-dimensional formally real Jordan algebra of 3×3 Hermitian octonionic matrices,

$$J_3(\mathbb{O}) = \{X \in M_3(\mathbb{O}) : X = X^\dagger\}, \quad X \circ Y = \frac{1}{2}(XY + YX), \quad \dim_{\mathbb{R}} J_3(\mathbb{O}) = 27.$$

Its reduced structure group — the group preserving the cubic norm $N(X) = \det_{\mathbb{O}} X$ up to a real character — is the exceptional Lie group E_6 , and the 27 is its defining (matter) representation. We commit to *two real forms* (see [subsection 4.2](#)): the **compact** E_6 (equivalently $E_{6(-14)}$) for all dynamics, and the split-signature $E_{6(-26)}$ used *only* as the rigid classifier of the real orbits.

Axiom 4.1 (Order parameter). The fundamental order parameter of UCFT is a field Φ that is a section of an E_6 vector bundle valued in the matter representation **27** $\cong J_3(\mathbb{O})$. Locally, $\Phi : U \rightarrow J_3(\mathbb{O})$.

Quantization prescription (theorem-grade kinematics). The quantum theory is defined by the Euclidean functional integral over configurations of Φ with the E_6 -invariant weight $e^{-S_E[\Phi]}$, $S_E = \int (\frac{1}{2} K_{AB} \partial \Phi^A \partial \Phi^B + V(\Phi))$, treated as the generating functional of connected correlators. Its physical (Lorentzian, unitary, positive-energy) content is recovered not by any naive “ $e^0 \mapsto i e^0$ ” Wick trick but by Osterwalder–Schrader (OS) reconstruction along the soldered timelike clock field ϕ^0 ([subsection 4.5](#), and ??). The OS reconstruction theorem is a **theorem** given its hypotheses; that the UCFT σ -model *satisfies* reflection positivity is **indicative**, a property to be verified, not assumed.

Remark 4.2 (Axiom I and emergent time). Earlier formulations posited a second axiom asserting a global unitary $U(t) = e^{-iHt}$ with an external time parameter. In the present foundation that statement is **realized as a theorem**: time is emergent, recovered as a Page–Wootters clock on physical states subject to the induced-gravity Hamiltonian constraint ([Theorem 4.15](#)). UCFT therefore rests on the *single* dynamical Axiom 4.1 together with the quantization prescription; the axiom supplies the arena and P1–P6 supply the added structure required for contact with four-dimensional physics.

4.2 Real form, invariants, and the vacuum manifold

Two facts must not be conflated. The automorphism group of $J_3(\mathbb{O})$ (the group preserving the Jordan product, hence *both* $Q(X) = \text{Tr}(X^2)$ and $N(X)$) is the compact F_4 , dim 52. The reduced structure group preserving *only* N (up to a character) is the larger E_6 , dim 78; it does **not** preserve $Q = \text{Tr}(X^2)$. Consequently:

Theorem 4.3 (Invariants and orbit classification). *Only the cubic norm N is E_6 structure-group invariant, up to a real character $\lambda(g)$; the quadratic form $Q(X) = \text{Tr}(X^2)$ is F_4 -invariant only. Real E_6 -orbits in $J_3(\mathbb{O})$ are classified by $\text{rank } X \in \{0, 1, 2, 3\}$ together with $N(X)$ — not by the pair (Q, N) . Moreover*

$$\text{rank } X \leq 1 \iff X^\# = 0, \quad \text{rank } X \leq 2 \iff N(X) = 0, \quad \text{rank } X = 3 \iff N(X) \neq 0,$$

where $X^\# = \nabla N(X)$ is the E_6 -covariant sharp (adjoint) map.

Remark 4.4 (Real form committed to). For the dynamics we use the **compact** E_6 (equivalently $E_{6(-14)}$, whose maximal compact subgroup is exactly $H = \text{SO}(10) \times \text{U}(1)$). On the compact form the 27 is unitary, the trace form Q is the invariant metric, the norm character is identically 1, and the coset metric is *positive-definite* — giving a healthy kinetic term and a unitary gauge sector. Coercivity and boundedness-below of V are statements about the compact form (non-compact orbits cannot be coercive). The split form $E_{6(-26)}$ is invoked *only* in [Theorem 4.3](#) as the rigid classifier of the real 27 by rank and N .

Theorem 4.5 (Vacuum manifold = EIII). *The symmetry-breaking vacuum manifold of UCFT is the rank-1 primitive-idempotent orbit*

$$\mathcal{M} = \frac{E_6}{\text{SO}(10) \times \text{U}(1)} = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 78 - 46 = 32, \quad (11)$$

*the closed minimal E_6 -orbit in $\mathbb{P}(27_{\mathbb{C}})$ — the complex Cayley plane. This is the orbit of a rank-1 primitive idempotent, **not** the orbit of $\text{diag}(1, 1, 1)$ (whose stabiliser is F_4 and whose orbit E_6/F_4 has real dimension 26).*

The defining datum of the breaking is the branching of the adjoint and the fundamental under $H = \text{SO}(10) \times \text{U}(1)$:

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (12)$$

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}. \quad (13)$$

The coset tangent space is therefore

$$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3} \quad (32 \text{ real}), \quad (14)$$

a real-irreducible module of complex type (commutant = $\mathbb{C}$). The $\text{U}(1)$ charge forbids the symmetric $\mathbf{16} \times \mathbf{16}$ and $\overline{\mathbf{16}} \times \overline{\mathbf{16}}$ invariants, leaving the unique charge-zero cross-pairing; by Schur the invariant coset metric $K|_{\mathfrak{m}}$ is unique and *positive-definite*. Crucially, $\mathfrak{m}$ contains **no** $\text{SO}(10)$ singlet: it is *not* $2 \times \mathbf{10} + 4$, nor $\mathbf{10} + \overline{\mathbf{10}} + 4$, nor anything with a singlet. The 32 Goldstones spanning $\mathfrak{m}$ are the clock fields.

4.3 The added postulates P1–P6

We now state, once and explicitly, the added structure. These are **postulates**, not theorems; the manuscript cites them as P1–P6 wherever they are used.

Postulate 4.6 (Potential class). (P1) The dynamics for Φ are governed by an E_6 -invariant polynomial potential built from the cubic norm N and the sharp map $X^\#$,

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \geq 2 \text{ even}, \quad (15)$$

whose global minimum is the rank-1 idempotent orbit. The covariant term $\langle X^\#, X^\# \rangle \geq 0$ vanishes exactly on rank ≤ 1 ; the $\text{Tr}(X^2)$ terms are F_4 -invariant selectors fixing the nonzero scale and representative. On the compact real form every term of (15) is a genuine E_6 invariant.

Postulate 4.7 (Vacuum selection). (P2) The vacuum is the minimal orbit $\mathcal{M} = E_6/(\text{SO}(10) \times \text{U}(1)) = \text{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$. This is *proved* from P1 as the vacuum-selection theorem ([Theorem 6.7](#)); see also [Theorem 4.5](#).

Postulate 4.8 (Soldering and signature). (P3) A soldering map identifies four clock-field frame directions e^a_μ ($a = 0, 1, 2, 3$) with a tangent 4-plane of an emergent 4-manifold $\mathcal{M}^4$, with vierbein $e^a_\mu(x) = \langle \tau^a, \theta_m(\partial_\mu) \rangle$ the pull-back of the Maurer–Cartan form, and induced metric $g_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}$, $\eta = \text{diag}(-1, +1, +1, +1)$. The Minkowski form η is the restriction of the cubic norm N to the sub-block $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$, reduced to $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ by a vierbein condensate. The space-time dimension $d = 4$ and signature $(1, 3) = (-, +, +, +)$ enter *here*, as added structure — not as Killing-form, coset, or rank-lemma consequences. See ??.

Postulate 4.9 (Matter content and chirality). (P4) Each family contributes one chiral **16** of $\text{SO}(10)$, with the $\text{U}(1)$ charges of (13) (the family lives in $\mathbf{16}_{+1} \subset \mathbf{27}$); there are no mirror partners.

Postulate 4.10 (Family symmetry). (P5) There are three generations, realized (primary route) by an $SU(3)_F$ horizontal symmetry acting on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$, yielding three chiral **16**s with no mirrors. This is corroborated (a correspondence, not a second mechanism) by the heterotic count $h^1(\widehat{Q}, V_{16}) = 3$. See [subsection 13.3](#).

Postulate 4.11 (GUT-Higgs / Yukawa sector). (P6) Fermion masses arise from a real $\text{SO}(10)$ Higgs/Yukawa sector (**45_H** or **54_H**, and **126_H**, with electroweak **10_H**), generating Dirac masses and the type-I seesaw. The naive coupling $-y \bar{\psi} \phi^a T_a \psi$ with $\phi \in \{\mathbf{16}, \overline{\mathbf{16}}\}$ is forbidden: $\mathbf{16} \times \mathbf{16} = \mathbf{10} + \mathbf{120} + \mathbf{126}$ and $\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$ contain no H -singlet from such a scalar, and H is unbroken. See [subsection 13.4](#).

Remark 4.12 (Epistemic status of the postulates). Of these, P2 is *proved* from P1 ([Theorem 6.7](#)); P3–P6 are *added structure*. The gravitational fixed point of ?? is *conjectural*, and the Standard Model is *embedded* under P4–P6.

4.4 From Axiom I to a broken vacuum

The **27** is a non-trivial irreducible E_6 representation, so the *only* vector fixed by the whole structure group is the origin $\Phi = 0$. Maximal symmetry is therefore not a stable ground state: the potential P1 makes $\Phi = 0$ a strict local maximum.

Theorem 4.13 (The symmetric origin is a strict maximum). *For V as in (15),*

$$\text{Hess } V(0) = -4\beta c_2 \mathbb{K} \prec 0,$$

i.e. all 27 eigenvalues equal $-4\beta c_2 < 0$, so $\Phi = 0$ is a strict local maximum. Since $V(0) = \beta c_2^2$ strictly exceeds the value $W(q_)$ attained on the rank-1 orbit, the broken vacuum $\mathcal{M} = \text{EIII}$ beats the origin both locally and globally.*

Proof sketch. The covariant term contributes $\langle X^\#, X^\# \rangle = O(\|X\|^4)$, so its Hessian vanishes at the origin; only the selector $\beta[\text{Tr}(X^2) - c_2]^2$ contributes at quadratic order, giving $\text{Hess } V(0) = -4\beta c_2 \mathbb{K}$. The global statement is the vacuum-selection theorem [Theorem 6.7](#): $V = \alpha \langle X^\#, X^\# \rangle + W(\text{Tr}(X^2))$ with W strictly convex and $W(q_*) < W(0) = \beta c_2^2$, so every global minimizer has $X^\# = 0$ (rank ≤ 1) and $\text{Tr}(X^2) = q_* > 0$ (rank exactly 1). $\square$

Remark 4.14 (Forward reference to roll-down). Because $\Phi = 0$ is linearly unstable (growth rate $4\beta c_2$) and V is a strict Lyapunov function that is coercive on the compact real form, generic initial data flow under the equations of motion to the broken vacuum $\mathcal{M}$. The roll-down dynamics, the tree-level vanishing of $\text{Hess } V|_{\mathfrak{m}}$ (the 32 exact Goldstone/clock fields), and the single common radiative mass fixed by Schur on the irreducible $\mathfrak{m}$, are established in [Theorem 6.9](#) and [Theorem 6.10](#), with the spectrum analysed in [subsection 11.1](#).

4.5 Reconciliation of axiomatic and emergent time

The legacy ‘‘Axiom II’’ posited a global unitary $U(t) = e^{-iHt}$ acting along an external time. UCFT has no external time: time is one of the soldered clock fields (P3). We reconcile the two pictures by realizing the unitary-evolution statement as a theorem on physical states.

Theorem 4.15 (Emergent time as a Page–Wootters clock). *Induced gravity supplies a diffeomorphism/Hamiltonian constraint $\hat{H}_{\text{phys}}|\Psi\rangle = 0$ on physical states. Designate the soldered timelike clock field ϕ^0 a Page–Wootters clock. Conditioning,*

$$|\psi(\tau)\rangle = \langle\langle \phi^0 = \tau | \Psi \rangle\rangle, \quad i \partial_\tau |\psi(\tau)\rangle = \hat{H}_{\text{sys}} |\psi(\tau)\rangle,$$

so the one-parameter unitary group $U(\tau) = e^{-i\hat{H}_{\text{sys}}\tau}$ is recovered on physical states in the semiclassical regime, with the identification $t \leftrightarrow x^0$ via $H = \int T_{00}$ along the timelike Killing direction. Global hyperbolicity is an assumption.

Remark 4.16 (Euclidean to Lorentzian). The passage to Lorentzian, unitary, positive-energy dynamics is via Osterwalder–Schrader reconstruction along ϕ^0 (reflection positivity), *not* by the ill-defined ‘‘multiply e^0 by i ’’ Wick prescription. The metric $\eta^{(1,3)}$ is already the soldered (P3) Lorentzian form before any continuation; OS reconstruction then supplies unitary, positive-energy evolution. As noted under the quantization prescription, OS reconstruction is a **theorem** given its hypotheses, while reflection positivity of the UCFT σ -model is **indicative**. See ??.

Part II

Albert Algebra, Vacuum Selection, and Coset Geometry

5 The Albert Algebra and Exceptional Structure

This section assembles the algebraic foundation on which the whole construction rests: the exceptional Jordan (Albert) algebra $J_3(\mathbb{O})$, its trace and cubic invariants, the sharp map, the notion of rank, and the four real forms of the exceptional groups that act on it. We isolate, with care, *which* invariant is preserved by *which* group, since the entire vacuum-selection argument (section 6) and the Lorentzian soldering (??) depend on getting this exactly right. The heavy octonionic identities (associator gymnastics, the explicit cubic norm in components, the Freudenthal cross product) are deferred to ??; here we state the structural facts and prove the ones that are short. Throughout, the metric signature is $(-, +, +, +)$, $[f] = \text{mass}$, and the dimensionless couplings are $\hat{g}^2, \xi, \hat{y}^2, \kappa$.

5.1 The exceptional Jordan algebra $J_3(\mathbb{O})$

Let $\mathbb{O}$ denote the (real, division) octonions, with conjugation $x \mapsto \bar{x}$, norm $n(x) = x\bar{x} \in \mathbb{R}_{\geq 0}$, and real part $\text{Re } x = \frac{1}{2}(x + \bar{x})$. Octonionic multiplication is nonassociative; this nonassociativity is exactly what forces the *exceptional* structure below and obstructs any matrix-algebra shortcut.

Definition 5.1 (Albert algebra). The *Albert algebra* is the space of 3×3 octonionic Hermitian matrices

$$J_3(\mathbb{O}) = \left\{ X = X^\dagger \in M_3(\mathbb{O}) \right\} = \left\{ \begin{pmatrix} \xi_1 & z_3 & \bar{z}_2 \\ \bar{z}_3 & \xi_2 & z_1 \\ z_2 & \bar{z}_1 & \xi_3 \end{pmatrix} : \xi_i \in \mathbb{R}, z_i \in \mathbb{O} \right\}, \quad (16)$$

equipped with the commutative *Jordan product*

$$X \circ Y = \frac{1}{2}(XY + YX). \quad (17)$$

The three real diagonal entries and three octonionic off-diagonal entries give

$$\dim_{\mathbb{R}} J_3(\mathbb{O}) = 3 \cdot 1 + 3 \cdot 8 = 27. \quad (18)$$

Theorem 5.2 (Formally real Jordan algebra). $(J_3(\mathbb{O}), \circ)$ is a 27-dimensional, commutative, formally real (Euclidean) Jordan algebra: the Jordan identity $X \circ (Y \circ X^{\circ 2}) = (X \circ Y) \circ X^{\circ 2}$ holds, and $\sum_i X_i \circ X_i = 0$ implies every $X_i = 0$. It is the unique exceptional simple formally real Jordan algebra (it is not special, i.e. not a Jordan subalgebra of any associative algebra under the symmetrised product).

Proof sketch. Commutativity is manifest from (17). The Jordan identity is a power-associativity statement that, by linearisation, reduces to relations involving at most two octonions at a time; since any two octonions generate an associative (indeed commutative) subalgebra $\cong \mathbb{C}$ or $\mathbb{H}$ (Artin's theorem), the identity holds despite global nonassociativity. Formal reality follows from positivity of the trace form (20) below. Exceptionality (non-speciality) is the Albert–Jacobson theorem; see ?? and [2, 10, 12]. $\square$

5.2 Trace form, cubic norm, and the sharp map

Three invariant objects organise the algebra: a quadratic trace form, a cubic norm, and a quadratic-in- X *sharp* (adjoint) map. Their distinct invariance groups are the crux of the whole construction.

Linear and quadratic traces. Define the linear trace and the symmetric *trace form*

$$\mathrm{tr}(X) = \xi_1 + \xi_2 + \xi_3, \quad (19)$$

$$\langle X, Y \rangle = \mathrm{Tr}(X \circ Y), \quad Q(X) \equiv \mathrm{Tr}(X^2) = \langle X, X \rangle. \quad (20)$$

In components,

$$Q(X) = \mathrm{Tr}(X^2) = \sum_i \xi_i^2 + 2 \sum_i n(z_i) \geq 0, \quad (21)$$

so Q is positive-definite: $J_3(\mathbb{O})$ is Euclidean, and Q is the natural $O(27)$ -type inner product. This positivity is what makes the dynamical coset metric definite later (section 7).

Cubic norm. The *cubic norm* (the octonionic “determinant”) is

$$N(X) = \det_{\mathbb{O}}(X) = \xi_1 \xi_2 \xi_3 - \sum_i \xi_i n(z_i) + 2 \mathrm{Re}(z_1 z_2 z_3). \quad (22)$$

The bracketing of the cubic term $z_1 z_2 z_3$ is unambiguous under $\mathrm{Re}(\cdot)$ despite nonassociativity (a standard octonionic identity; ??). Unlike Q , the cubic norm is *odd* under $X \mapsto -X$ and is genuinely indefinite; its restriction to the $H_2(\mathbb{O})$ sub-block supplies the Minkowski form used in soldering (??).

Sharp map. The *sharp* (adjoint) map $X \mapsto X^\#$ is the gradient of N with respect to the trace form,

$$\langle X^\#, Y \rangle = \mathrm{D} N(X)[Y] \iff X^\# = \nabla N(X), \quad (23)$$

equivalently the quadratic map characterised by

$$X \circ X^\# = N(X) \mathbf{1}, \quad (X^\#)^\# = N(X) X. \quad (24)$$

In a *Jordan frame* $X = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$,

$$X^\# = \mathrm{diag}(\lambda_2 \lambda_3, \lambda_3 \lambda_1, \lambda_1 \lambda_2), \quad \langle X^\#, X^\# \rangle = (\lambda_2 \lambda_3)^2 + (\lambda_3 \lambda_1)^2 + (\lambda_1 \lambda_2)^2 \geq 0. \quad (25)$$

The combination $\langle X^\#, X^\# \rangle$ is a sum of squares, hence non-negative; this is the quantity whose vanishing characterises the vacuum orbit.

5.3 Rank, idempotents, and the rank-one criterion

Definition 5.3 (Rank). The *rank* of $X \in J_3(\mathbb{O})$ is the degree of its minimal polynomial; equivalently, in a Jordan frame it is the number of nonzero eigenvalues λ_i . Thus $\mathrm{rank}(X) \in \{0, 1, 2, 3\}$, with $\mathrm{rank}(0) = 0$. A *primitive idempotent* is a rank-one E with $E \circ E = E$ (equivalently $E^2 = E$, $\mathrm{tr} E = 1$).

Theorem 5.4 (Rank-one criterion). *For all $X \in J_3(\mathbb{O})$,*

$$\mathrm{rank}(X) \leq 1 \iff X^\# = 0. \quad (26)$$

Moreover $\mathrm{rank}(X) \leq 2 \Leftrightarrow N(X) = 0$ (with $X^\# \neq 0$), and $\mathrm{rank}(X) = 3 \Leftrightarrow N(X) \neq 0$.

Proof. Every X is conjugate by $F_4 = \text{Aut}(J_3(\mathbb{O}))$ (Theorem 5.6) to a Jordan frame $\text{diag}(\lambda_1, \lambda_2, \lambda_3)$ (spectral theorem for Euclidean Jordan algebras). By (25), $X^\# = 0$ iff all three pairwise products $\lambda_i \lambda_j$ vanish, i.e. at most one λ_i is nonzero, i.e. $\text{rank}(X) \leq 1$. The norm statements follow from $N(X) = \lambda_1 \lambda_2 \lambda_3$ in a frame: $N = 0$ iff some $\lambda_i = 0$ ($\text{rank} \leq 2$), and $N \neq 0$ iff all nonzero ($\text{rank} = 3$). Since F_4 preserves both N and $X^\#$, the criteria are frame-independent. $\square$

Remark 5.5 (Scope). Theorem 5.4 is a *theorem* of Jordan algebra, used verbatim in the vacuum-selection argument: the E_6 -covariant term $\langle X^\#, X^\# \rangle$ in the potential V vanishes *exactly* on the $\text{rank} \leq 1$ cone (section 6). The selection of the rank-one orbit as the global minimum is proved there; here we only supply the algebraic criterion.

5.4 The four real forms and what each preserves

The single most consequential fact of this section is that *different groups preserve different invariants*. Conflating them (in particular, asserting that the structure group preserves the trace form Q , or that (Q, N) separates orbits) is the root error against which the entire construction is hardened.

Theorem 5.6 (Four real forms). *The following groups act on $J_3(\mathbb{O})$ and the real 27:*

1. $F_4 = \text{Aut}(J_3(\mathbb{O}), \circ)$ is the **compact** exceptional group, $\dim F_4 = 52$. It preserves the full Jordan product, hence both the trace form $Q = \text{Tr}(X^2)$ and the cubic norm N , and acts as $F_4 \subset \text{SO}(27)$ (it is the stabiliser of the identity $\mathbf{1}$).
2. The **reduced structure group** $E_{6(-26)}$ is the **non-compact** real form, $\dim E_{6(-26)} = 78$. It preserves the cubic norm N up to a real positive character $\lambda(g)$,

$$N(g \cdot X) = \lambda(g) N(X), \quad \lambda : E_{6(-26)} \rightarrow \mathbb{R}_{>0}, \quad (27)$$

and does not preserve Q . (If it did, it would lie in $\text{SO}(27)$ and be compact, a contradiction.)

3. The **compact** real form, here denoted compact E_6 , $\dim E_6 = 78$, acts unitarily on the complexified $27_{\mathbb{C}}$; on it the norm character is identically 1 (a continuous homomorphism from a compact connected group to $\mathbb{R}_{>0}$ is trivial), and Q is the invariant Hermitian form.
4. The real form $E_{6(-14)}$, $\dim = 78$, has **maximal compact subgroup** $\text{Spin}(10) \times \text{U}(1)$.

Proof sketch. 1 An automorphism of the Jordan product preserves powers, hence the trace form (via $Q(X) = \text{Tr } X^2$) and the norm (via Cayley–Hamilton in the algebra); compactness and $\dim 52$ are classical [18]. 2 The reduced structure group is by definition the norm-similarity group; (27) is its defining property, and non-compactness follows because a non-trivial similarity character exists. 3 Triviality of the character on a compact group is the homomorphism argument given. 4 is the standard real-form classification; the maximal compact of $E_{6(-14)}$ is $\text{Spin}(10) \times \text{U}(1)$ [18, 17, 8]. $\square$

Postulate 5.7 (Real form for dynamics). For the *dynamics* of UCFT we commit to the **compact** real form (equivalently $E_{6(-14)}$, whose maximal compact is exactly $H = \text{Spin}(10) \times \text{U}(1)$). Then H is compact, the coset metric on E_6/H is positive-definite, the kinetic term is healthy, and the gauge sector is unitary. The non-compact form $E_{6(-26)}$ is used **only** as the rigid orbit-classifier of the real 27. This choice is part of the postulate set (it is the form on which the potential V is coercive; non-compact orbits cannot be coercive, since the norm character (27) is then unbounded). Cf. P1 and the vacuum theorem in section 6.

Theorem 5.8 (Orbit classification by rank and norm). *The orbits of $E_{6(-26)}$ on the real $27 = J_3(\mathbb{O})$ are classified by the pair $(\text{rank}(X), N(X))$, and not by $(Q(X), N(X))$. Explicitly, the nonzero orbit strata are*

$$\underbrace{\text{rank} = 1}_{X^\# = 0} \subset \underbrace{\text{rank} = 2}_{X^\# \neq 0, N = 0} \subset \underbrace{\text{rank} = 3}_{N \neq 0}, \quad (28)$$

*the rank-3 stratum further refined by the sign / value of N . The trace form Q is **not** an E_6 -invariant (only an F_4 -invariant), so any “ (Q, N) separates orbits” statement is false.*

Proof. $E_{6(-26)}$ preserves N up to the character (27) and acts transitively on each rank stratum of fixed N -class (Springer; ??). The rank is an algebraic invariant (Theorem 5.3), preserved because the sharp map is E_6 -covariant: $(g \cdot X)^\# = \lambda(g)(g^\#) \cdot X^\#$ in the appropriate contragredient action, so $X^\# = 0$ is a group-invariant condition (and likewise $N = 0$). That Q is not invariant is Theorem 5.62: a structure-group element with $\lambda(g) \neq 1$ generically rescales $\text{Tr}(X^2)$. $\square$

Corollary 5.9 (Closed minimal orbit). *The unique closed nonzero E_6 -orbit in $\mathbb{P}(27_{\mathbb{C}})$ is the rank-one (primitive-idempotent) orbit, the complex Cayley plane*

$$\mathcal{M} = E_6 / (\text{Spin}(10) \times \text{U}(1)) = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32. \quad (29)$$

*This is the orbit of a rank-one primitive idempotent, characterised by $X^\# = 0$ (Theorem 5.4); it is **not** the orbit of $\text{diag}(1, 1, 1)$ (which is rank 3, has $\langle X^\#, X^\# \rangle = 3 > 0$, stabiliser F_4 , orbit E_6/F_4 of real dimension 26).*

Proof. The closed orbit of a reductive group in a projective space is the orbit of a highest-weight line; for the 27 of E_6 this is the rank-one line, with stabiliser the parabolic whose reductive part fixes the H -decomposition (30). Over the compact form the isotropy is $H = \text{Spin}(10) \times \text{U}(1)$, and $\dim_{\mathbb{R}} E_6 - \dim_{\mathbb{R}} H = 78 - 46 = 32$. The real rank-one idempotents alone form $F_4/\text{Spin}(9) \cong \mathbb{O}\mathbb{P}^2$ of dimension 16; the full complex line under E_6 sweeps the larger 32-dimensional EIII [8, 18]. $\square$

Remark 5.10 (Consistency constraints). Three statements are *false* and are never used: (i) that the structure group preserves $Q = \text{Tr}(X^2)$ (only F_4 does); (ii) that (Q, N) separates E_6 -orbits (orbits are classified by rank and N , Theorem 5.8); (iii) that the 32-dimensional vacuum is the orbit of $\text{diag}(1, 1, 1)$ with stabiliser F_4 (that orbit is E_6/F_4 , dimension 26; the vacuum is the rank-one EIII orbit, Theorem 5.9).

5.5 Branchings under $H = \text{Spin}(10) \times \text{U}(1)$

The isotropy group $H = \text{Spin}(10) \times \text{U}(1)$ of the vacuum orbit (29) is exactly the gauge group of the construction. The relevant representations branch as follows; these are the canonical decompositions used everywhere downstream.

Theorem 5.11 (Defining branchings). *Under $H = \text{Spin}(10) \times \text{U}(1)$ (subscripts are the $\text{U}(1)$ charges),*

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3} \quad (\text{adjoint}), \quad (30)$$

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1} \quad (\text{fundamental} / \text{matter}). \quad (31)$$

Consequently the coset tangent space is

$$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3} \quad (32 \text{ real dimensions}), \quad (32)$$

*which contains **no** $\text{Spin}(10)$ singlet.*

Proof. The decompositions (30)–(31) are the standard $E_6 \downarrow \text{Spin}(10) \times \text{U}(1)$ branchings [17]; the $\text{U}(1)$ charges are fixed by requiring the cubic E_6 -invariant $\mathbf{27}^3$ and the adjoint commutators to be charge-neutral. The coset directions are the non- H generators in (30), namely $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, of real dimension $16 + 16 = 32 = \dim_{\mathbb{R}} \mathcal{M}$ (Theorem 5.9). Dimension and charge counting exclude any singlet in $\mathfrak{m}$. $\square$

Theorem 5.12 (Irreducibility and the definite coset metric). *The coset tangent module $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is real-irreducible of complex type (its commutant is $\mathbb{C}$). The H -invariant metric on $\mathfrak{m}$ is therefore **unique up to scale** and, on the compact real form (Theorem 5.7), **positive-definite**.*

Proof. The $\text{U}(1)$ charge ± 3 forbids the symmetric self-pairings $\mathbf{16}_{-3} \otimes \mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3} \otimes \overline{\mathbf{16}}_{+3}$ (charges $\mp 6 \neq 0$), leaving only the charge-0 cross-pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3}$, which is the restriction of the Killing form. Hence the invariant bilinear is one-dimensional (Schur, complex type), so unique up to scale. On the compact form the Killing form is definite; restricted to $\mathfrak{m}$ it is positive-definite, giving the Euclidean target-space metric $K|_{\mathfrak{m}}$ used in section 7. $\square$

Remark 5.13 (No Lorentzian content in the coset metric). The coset metric $K|_{\mathfrak{m}}$ is positive-definite, signature $(32, 0)$: it carries *zero* Lorentzian content. In particular there is no $(4, 28)$ Killing/coset signature, and the physical $(1, 3)$ Minkowski signature does *not* fall out of the coset geometry. Lorentzian signature enters only through the soldering postulate P3, which uses the indefinite cubic norm N restricted to $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ on a *different* module (??). The two forms (definite $K|_{\mathfrak{m}}$, indefinite N) live on distinct spaces; there is no tension.

5.6 Summary of canonical structure

For reference, Table 3 collects the canonical algebraic data established above; every downstream section uses these values.

Object	Canonical value / property
$\dim_{\mathbb{R}} J_3(\mathbb{O})$	27
Trace form $Q = \text{Tr}(X^2)$	positive-definite; F_4 -invariant <i>only</i>
Cubic norm $N = \det_{\mathbb{O}} X$	indefinite; E_6 -invariant up to character $\lambda(g)$
Sharp map $X^\# = \nabla N$	E_6 -covariant; $X^\# = 0 \Leftrightarrow \text{rank} \leq 1$
$F_4 = \text{Aut}(J_3(\mathbb{O}))$	compact, dim 52; preserves Q and N
$E_{6(-26)}$ (orbit classifier)	non-compact, dim 78; preserves N only
compact $E_6 / E_{6(-14)}$ (dynamics)	dim 78; max. compact $\text{Spin}(10) \times \text{U}(1)$
Vacuum manifold $\mathcal{M}$	$E_6 / (\text{Spin}(10) \times \text{U}(1)) = \text{EIII}$, dim 32
Adjoint branching	$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$
Fundamental branching	$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$
Coset tangent $\mathfrak{m}$	$\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ (32 real); no singlet
Coset metric $K _{\mathfrak{m}}$	positive-definite (Euclidean), unique up to scale

Table 3: Canonical algebraic data of the Albert algebra and its exceptional structure, as used throughout UCFT.

These facts are purely algebraic theorems, with the single *postulate* being the commitment to the compact real form for the dynamics (Theorem 5.7, part of P1). The vacuum-selection theorem that promotes (29) from “a distinguished orbit” to “the global minimum of V ” is proved in section 6; the Lorentzian soldering that uses N to install signature $(-, +, +, +)$ is in ??; the gauge/clock-field dynamics on $\mathfrak{m}$ are in section 7.

6 The Invariant Potential and Vacuum Selection

The order parameter of UCFT is a field Φ valued in the matter representation $27 = J_3(\mathbb{O})$ of E_6 (Axiom I). This section presents the E_6 -invariant potential V of postulate P1 and proves the vacuum-selection statement P2: the global-minimum locus of V is exactly the rank-1 primitive-idempotent orbit $\mathcal{M} = \text{EIII} = E_6/(\text{Spin}(10) \times U(1))$ of real dimension 32. We also show that the symmetric origin $\Phi = 0$ is a strict local *maximum*, which supplies the roll-down dynamics. The potential class is the added postulate; that the chosen member of the class selects the rank-1 orbit is a **theorem**. Throughout, the dynamics live on the *compact* real form E_6 (equivalently $E_{6(-14)}$, maximal compact $H = \text{Spin}(10) \times U(1)$); the non-compact $E_{6(-26)}$ enters only as the rigid orbit classifier (rank and N only). The vacuum geometry and its tangent space are used in ?? (signature) and section 11 (spectrum, induced gravity).

6.1 Albert-algebra setup and the two real forms

Let $\mathbb{O}$ be the real octonions with conjugation $x \mapsto \bar{x}$ and norm $n(x) = x\bar{x}$. The real Albert algebra is the 27-dimensional Euclidean (formally real) Jordan algebra

$$J_3(\mathbb{O}) = \left\{ X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix} : \xi_i \in \mathbb{R}, x_i \in \mathbb{O} \right\}, \quad X \circ Y = \frac{1}{2}(XY + YX). \quad (33)$$

Its two basic invariants are the *trace form* (a Euclidean, positive-definite bilinear form) and the *cubic norm*:

$$\langle X, Y \rangle = \text{Tr}(X \circ Y), \quad Q(X) := \text{Tr}(X^2) = \langle X, X \rangle \geq 0, \quad N(X) = \det X \quad (\text{cubic}). \quad (34)$$

With the linear trace $\text{tr}(X) = \xi_1 + \xi_2 + \xi_3$ and identity $E = \text{diag}(1, 1, 1)$, every X obeys the cubic characteristic identity $X^{\circ 3} - \text{tr}(X) X^{\circ 2} + \sigma_2(X) X - N(X) E = 0$ with $\sigma_2(X) = \frac{1}{2}(\text{tr}(X)^2 - Q(X))$. The Jordan eigenvalues $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ satisfy $\text{tr}(X) = \sum_i \lambda_i$, $Q(X) = \sum_i \lambda_i^2$, $N(X) = \lambda_1 \lambda_2 \lambda_3$.

The *sharp map* (Jordan adjoint) $X \mapsto X^\#$ is the unique quadratic map with

$$X^\# \circ X = N(X) E, \quad (X^\#)^\# = N(X) X, \quad X^\# = \nabla N(X), \quad (35)$$

and is the matrix 2×2-cofactor (adjugate): in a Jordan frame $\text{diag}(\lambda_1, \lambda_2, \lambda_3)^\# = \text{diag}(\lambda_2 \lambda_3, \lambda_3 \lambda_1, \lambda_1 \lambda_2)$.

The two E_6 ’s (a locked distinction). The reduced structure group is the non-compact $E_{6(-26)}$ (dim 78); it preserves N only up to a scalar character, $N(gX) = \lambda(g)N(X)$, and covariantly transforms the sharp map. It does *not* preserve $Q = \text{Tr}(X^2)$; indeed $\text{Aut}(J_3(\mathbb{O})) = F_4$ (compact, dim 52) is exactly the subgroup preserving both Q and N . We use $E_{6(-26)}$ solely to classify orbits of the real 27 by **rank and N** . The *dynamics*, by contrast, live on the **compact** E_6 : there $H = \text{Spin}(10) \times U(1)$ is compact, so the coset metric is positive-definite and the gauge sector

unitary; on the compact form $Q = \langle X, X \rangle$ is the invariant trace form and the norm character is identically 1. Orbits are classified by rank and N (*not* by (Q, N)):

$$\text{rank}(X) \leq 1 \iff X^\# = 0, \quad \text{rank}(X) \leq 2 \iff N(X) = 0, \quad \text{rank}(X) = 3 \iff N(X) \neq 0. \quad (36)$$

6.2 The potential class (Postulate P1)

Postulate 6.1 (Potential class, P1). The order parameter Φ is governed by the E_6 -invariant polynomial potential

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta (\text{Tr}(X^2) - c_2)^2 + \gamma (\text{Tr}(X^2))^k, \quad \alpha, \beta, \gamma > 0, \quad k \geq 2 \text{ even}, \quad (37)$$

built from the cubic norm (through the sharp map $X^\# = \nabla N$) and the trace form Q . On the compact real form every term is a genuine E_6 -invariant.

The first term is genuinely E_6 -covariant and vanishes precisely on $\text{rank} \leq 1$; the Q -terms are F_4 -invariant selectors that fix the scale and the representative. Postulate P1 asserts that one is *allowed* to write such a potential; the theorems below establish *what it selects*.

6.3 The vacuum-selection theorem (P1 $\Rightarrow$ P2)

Lemma 6.2 (Zero locus of the sharp map). *For $X \in J_3(\mathbb{O})$, $X^\# = 0 \iff \text{rank}(X) \leq 1$. The normalized ($\text{tr} = 1$) rank-1 elements are exactly the primitive idempotents, forming the single F_4 -orbit $F_4/\text{Spin}(9) \cong \mathbb{O}\mathbb{P}^2$ (real dim 16); the projective E_6 -orbit of a rank-1 line in $\mathbb{P}(27_{\mathbb{C}})$ is the complex Cayley plane $\text{EIII} = E_6/(\text{Spin}(10) \times U(1))$, real dim 32.*

Proof. In a Jordan frame $X = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, the sharp map gives $X^\# = \text{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$, which vanishes iff at most one $\lambda_i \neq 0$, i.e. $\text{rank}(X) \leq 1$; F_4 -equivariance transports this to all of $J_3(\mathbb{O})$. The orbit statements are the standard identifications of the real rank-1 idempotents ($\mathbb{O}\mathbb{P}^2$, dim 16) and the minimal E_6 -orbit / complex Cayley plane (EIII, dim 32). $\square$

Theorem 6.3 (Positivity of the sharp term). *For all $X \in J_3(\mathbb{O})$, $\langle X^\#, X^\# \rangle \geq 0$, with equality iff $X^\# = 0$ iff $\text{rank}(X) \leq 1$. Explicitly*

$$\langle X^\#, X^\# \rangle = (\lambda_2\lambda_3)^2 + (\lambda_3\lambda_1)^2 + (\lambda_1\lambda_2)^2 = \sigma_2(X)^2 - 2 \text{tr}(X) N(X) \quad (\text{a sum of squares}). \quad (38)$$

Proof. The trace form is positive-definite on the Euclidean Jordan algebra, so $\langle Y, Y \rangle \geq 0$ with equality iff $Y = 0$; take $Y = X^\#$ and apply [Theorem 6.2](#). The frame expression (38) is manifestly a sum of squares. $\square$

Theorem 6.4 (Compact- E_6 invariance of the sharp term). *The scalar $\langle X^\#, X^\# \rangle$ is invariant under the compact real form E_6 (and $E_{6(-14)}$): $\langle (gX)^\#, (gX)^\# \rangle = \langle X^\#, X^\# \rangle$ for all $g \in E_6^{\text{compact}}$.*

Proof. On the compact form the 27 is unitary, with invariant positive-definite form whose real part is the trace form; hence the compact E_6 preserves $\langle \cdot, \cdot \rangle$. As a degree-4 relative invariant, $\langle X^\#, X^\# \rangle$ transforms by a power of the norm character $\lambda(g)$; but a continuous homomorphism from a compact connected group to $\mathbb{R}_{>0}$ is identically 1, so $\lambda \equiv 1$ and the relative invariant is a genuine invariant. (Under $E_{6(-26)}$, λ is unbounded and $\langle \cdot, \cdot \rangle$ is not preserved; this is precisely why the dynamics use the compact form.) $\square$

Status of Q . $Q = \text{Tr}(X^2)$ is F_4 -invariant only and is *not* preserved by the non-compact structure group; we never claim otherwise, nor that (Q, N) separates E_6 -orbits. On the compact form, however, $Q = \langle X, X \rangle$ is the invariant trace form, so V in (37) is a genuine compact- E_6 invariant potential.

Lemma 6.5 (The selector has a unique positive minimizer). *Set $W(q) = \beta(q - c_2)^2 + \gamma q^k$ for $q \geq 0$, $\beta, \gamma > 0$, $k \geq 2$ even. There is a $c_2^* > 0$ such that for $c_2 > c_2^*$, W is strictly convex on $[0, \infty)$ and attains its global minimum at a unique interior point $q_* \in (0, c_2)$ with $W'(q_*) = 0$, $W''(q_*) > 0$, and $W(q_*) < W(0) = \beta c_2^2$. Moreover*

$$q_* = c_2 - \frac{\gamma k}{2\beta} c_2^{k-1} + O(c_2^{2k-3}), \quad q_* \xrightarrow[\gamma \rightarrow 0^+]{\quad} c_2. \quad (39)$$

Proof. $W(0) = \beta c_2^2 > 0$ and $W(q) \rightarrow +\infty$, so a minimum exists on a compact set. Since $W''(q) = 2\beta + \gamma k(k-1)q^{k-2} > 0$ for all $q \geq 0$ ($k \geq 2$ even), W is strictly convex; with $W'(0) = -2\beta c_2 < 0$ it has a unique stationary point $q_* > 0$, the global minimizer, and $W(q_*) < W(0)$. As $W'(c_2) = \gamma k c_2^{k-1} > 0$, the root lies below c_2 ; the expansion follows by solving $2\beta(q_* - c_2) = -\gamma k q_*^{k-1}$ perturbatively. $\square$

The selector thus *fixes the scale*: among rank-1 elements $X = \lambda P$ (with $Q = \lambda^2$, $N = 0$, $X^\# = 0$), the minimizers have $\lambda^2 = q_*$, a fixed nonzero norm; in particular the origin ($q = 0$, $W(0) = \beta c_2^2 > W(q_*)$) is excluded.

Lemma 6.6 (Coercivity requires the compact form). *On $J_3(\mathbb{O})$ with the compact- E_6 Euclidean norm $\|X\|^2 = Q(X)$, the potential V is coercive and bounded below: $V(X) \geq \gamma \|X\|^{2k} \rightarrow +\infty$. On a non-compact real-form orbit, where Q is not preserved and not a proper exhaustion, V need not be coercive.*

Proof. $V \geq \gamma Q^k = \gamma \|X\|^{2k}$ from (37) (α -term and the square are ≥ 0), giving coercivity on the compact form. Under $E_{6(-26)}$ the character $\lambda(g)$ is unbounded, so a fixed X with $N(X) \neq 0$ can be driven to $Q \rightarrow 0$ along a fixed N -level set; Q is not proper there and sublevel sets are unbounded. $\square$

Theorem 6.7 (Global-minimum locus is the rank-1 EIII orbit; P2). *With $\alpha, \beta, \gamma > 0$, $k \geq 2$ even, and $c_2 > c_2^*$ (Theorem 6.5), every global minimizer X_* of V on $J_3(\mathbb{O})$ satisfies $X_*^\# = 0$ and $\text{Tr}(X_*^2) = q_*$; equivalently*

$$\arg \min_{X \in J_3(\mathbb{O})} V(X) = \{X = \lambda P : P \text{ primitive idempotent}, \lambda^2 = q_*\}, \quad (40)$$

whose projective E_6 -orbit is the vacuum manifold

$$\mathcal{M} = \text{EIII} = \frac{E_6}{\text{Spin}(10) \times U(1)}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32. \quad (41)$$

This proves postulate **P2** from Theorem 6.1.

Proof. By Theorem 6.6 a global minimum is attained. For any X ,

$$V(X) = \underbrace{\alpha \langle X^\#, X^\# \rangle}_{\geq 0} + W(Q(X)) \geq W(Q(X)) \geq W(q_*),$$

using Theorem 6.3 and Theorem 6.5. Equality in the first step forces $X^\# = 0$, i.e. $\text{rank}(X) \leq 1$ (Theorem 6.2); equality in the second forces $Q(X) = q_* > 0$, excluding $X = 0$ and forcing rank exactly 1. Both hold iff $X = \lambda P$ with $\lambda^2 = q_*$, and every such X achieves $V = W(q_*)$. The projective orbit of these rank-1 idempotents is EIII (Theorem 6.2). $\square$

Why rank 2 and rank 3 strictly lose. For $\text{rank}(X) \geq 2$, [Theorem 6.2](#) gives $X^\# \neq 0$, so $\langle X^\#, X^\# \rangle > 0$ strictly ([Theorem 6.3](#)) and $V(X) > W(q_\star)$ by a deficit at least $\alpha \langle X^\#, X^\# \rangle > 0$. In particular $\text{diag}(1, 1, 1)$ (rank 3, $\langle X^\#, X^\# \rangle = 3 > 0$, orbit E_6/F_4 of real dimension 26) is *not* a minimizer. The vacuum is the rank-1 idempotent EIII orbit (dim 32), *not* the E_6/F_4 orbit of $\text{diag}(1, 1, 1)$ (dim 26).

Remark 6.8 (Coset tangent and metric). The tangent space to $\mathcal{M}$ at the base idempotent is the coset module

$$\mathfrak{m} = 16_{-3} \oplus \overline{16}_{+3} \quad (32 \text{ real}), \quad (42)$$

read off from the adjoint branching $78 = 45_0 \oplus 1_0 \oplus 16_{-3} \oplus \overline{16}_{+3}$ under $H = \text{Spin}(10) \times U(1)$. It is real-irreducible of complex type (commutant $\mathbb{C}$) and contains *no* $\text{SO}(10)$ singlet; the $U(1)$ charge forbids the symmetric 16×16 and $\overline{16} \times \overline{16}$ invariants, leaving a unique charge-0 cross-pairing. The induced coset metric $K|_{\mathfrak{m}}$ is therefore unique and positive-definite (Schur), and serves as the kinetic/target-space metric for the 32 clock fields (Goldstones). It carries *no* Lorentzian content: the $(1, 3)$ signature is added structure (P3), soldered from the cubic norm via $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$, not a Killing-form consequence; see ??.

6.4 The symmetric origin is a strict local maximum

Theorem 6.9 (Origin is a strict local maximum). *At $X = 0$,*

$$\nabla V(0) = 0, \quad \text{Hess } V(0) = -4\beta c_2 \mathbb{K} \prec 0 \quad (\text{negative-definite, } \beta, c_2 > 0), \quad (43)$$

so $X = 0$ is a strict local maximum, all 27 Hessian eigenvalues equal to $-4\beta c_2 < 0$. Since $V(0) = \beta c_2^2 > W(q_\star)$, the rank-1 orbit beats the origin locally and globally.

Proof. V has no linear part. The $\alpha \langle X^\#, X^\# \rangle$ term is homogeneous of degree 4 and the γQ^k term of degree $2k \geq 4$, so both have vanishing Hessian at 0. Only the quadratic piece $-2\beta c_2 Q$ of $\beta(Q - c_2)^2$ contributes; with Q having Hessian $2\mathbb{K}$ in the trace metric this gives $\text{Hess } V(0) = -4\beta c_2 \mathbb{K}$, negative-definite. $\square$

Theorem 6.10 (Roll-down to $\mathcal{M}$). *Under gradient flow $\dot{X} = -\nabla V(X)$ (or the EOM with damping), the origin is linearly unstable with growth rate $4\beta c_2 > 0$ in all 27 directions; V is a strict Lyapunov function, coercivity ([Theorem 6.6](#)) confines trajectories to compact sublevel sets, and by LaSalle the ω -limit set lies in $\{\nabla V = 0\}$. The unique stable critical set is the global minimum $\mathcal{M}$ ([Theorem 6.7](#)); the origin (maximum, [Theorem 6.9](#)) and the rank-2/rank-3 saddles ($X^\# \neq 0$) are all unstable. Hence generic initial data roll down to $\mathcal{M} = \text{EIII}$.*

Proof. Linearization at 0 uses $\text{Hess } V(0) \prec 0$ ([Theorem 6.9](#)), giving positive growth in every direction. $\dot{V} = -\|\nabla V\|^2 \leq 0$ vanishes only at critical points; LaSalle plus coercivity confine ω -limits to the critical set, of which only $\mathcal{M}$ is Lyapunov-stable. The basin of $\mathcal{M}$ is open and dense. $\square$

Tree-level Goldstone consistency. On $\mathcal{M}$, V is E_6 -invariant and constant along the orbit, so $\text{Hess } V|_{\mathfrak{m}} = 0$ at tree level: the 32 clock fields are exact Goldstones. Radiative (Coleman–Weinberg) corrections are H -invariant; by Schur on the real-irreducible module $\mathfrak{m}$ they give *one* common mass on all 32 directions. Of these, 28 are physical massive scalars (single common mass) and 4 are the gauge/soldering modes eaten by diffeomorphism and local Lorentz; this $28+4$ split and the common radiative mass are carried into the spectrum and induced-gravity analysis of [section 11](#). The origin's 27 negative modes and the vacuum's flat coset directions are not in tension: they live at different points of field space.

6.5 Summary of the logical chain

step	statement	type
Theorem 6.2	$X^\# = 0 \iff \text{rank}(X) \leq 1$; rank-1 orbit = EIII (dim 32)	Theorem
Theorem 6.3	$\langle X^\#, X^\# \rangle \geq 0, = 0 \iff \text{rank} \leq 1$	Theorem
Theorem 6.4	$\langle X^\#, X^\# \rangle$ is compact- E_6 invariant	Theorem
Theorem 6.5	selector W strictly convex, unique $q_\star > 0$, $W(q_\star) < W(0)$	Theorem
Theorem 6.6	coercivity holds on the compact form, fails on non-compact orbits	Theorem
Theorem 6.7	global-min locus = rank-1 EIII orbit, dim 32 (P2)	Theorem
Theorem 6.9	$\text{Hess } V(0) = -4\beta c_2 \mathbb{K} \prec 0$: origin strict max	Theorem
Theorem 6.10	roll-down: generic flows attract to $\mathcal{M}$	Theorem
Theorem 6.1	the potential class is added structure	Postulate

Given the postulated potential class P1, [Theorem 6.7](#) proves P2: the unique global-minimum locus of V is the rank-1 primitive-idempotent orbit $\mathcal{M} = \text{EIII} = E_6/(\text{Spin}(10) \times U(1))$, real dimension 32. The sharp term enforces $\text{rank} \leq 1$; the Q -selector fixes the nonzero scale and excludes the origin; the compact real form supplies coercivity and a definite invariant metric; and the origin is a strict local maximum driving the roll-down. No rank-2/rank-3 point — in particular none on the E_6/F_4 orbit of $\text{diag}(1, 1, 1)$ (dim 26) — can be a minimizer.

7 Coset Geometry and the Clock Fields

The vacuum-selection theorem (??) establishes that the global minimum of the E_6 -invariant potential of Postulate P1 is the closed rank-1 primitive-idempotent orbit in $\mathbb{P}(\mathbf{27}_{\mathbb{C}})$, namely the Hermitian symmetric space

$$\mathcal{M} = \frac{E_6}{\text{Spin}(10) \times U(1)} = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32. \quad (44)$$

Here and throughout the *dynamics* are carried by the *compact* real form of E_6 (equivalently $E_{6(-14)}$, whose maximal compact subgroup is exactly $H = \text{Spin}(10) \times U(1)$); the non-compact $E_{6(-26)}$ enters only as the rigid classifier of orbits in the real **27** by rank and cubic norm N (??). Because H is compact, the isotropy representation carries an invariant *positive-definite* metric, which we will identify as the kinetic/target-space metric of the σ -model.

This section is geometric and tree-level. We (i) identify the tangent space $\mathfrak{m}$ of $\mathcal{M}$ and prove, from the $U(1)$ charge structure of a Hermitian symmetric space, that the invariant metric $K|_{\mathfrak{m}}$ is unique and positive-definite; (ii) build the nonlinear σ -model from the Maurer–Cartan form, with its 32 Goldstone *clock fields*; and (iii) record that at tree level all 32 clock fields are exact massless Goldstone modes. The Lorentzian frame of four of them (Postulate P3) is supplied in ??; their one-loop spectrum (Schur, 28 massive + 4 gauge/soldering) in ??.

7.1 The tangent space $\mathfrak{m}$

The orbit (44) is defined by the Cartan involution ϑ whose $(+1)$ -eigenspace is $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$ and whose (-1) -eigenspace $\mathfrak{m}$ is the tangent space at the base point. The branching of the adjoint **78** of E_6 under $H = \text{Spin}(10) \times U(1)$ is the involution datum

$$\mathbf{78} = \underbrace{\mathbf{45}_0 \oplus \mathbf{1}_0}_{\mathfrak{h}} \oplus \underbrace{\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}}_{\mathfrak{m}}, \quad (45)$$

so that

$$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad \dim_{\mathbb{R}} \mathfrak{m} = 32. \quad (46)$$

The two summands are the chiral spinor and antispinor of $\text{Spin}(10)$, carrying opposite $U(1)$ charges ∓ 3 . For comparison, the matter representation branches as

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \quad (47)$$

the $\mathbf{16}_{+1}$ of which will house one chiral family (??).

Remark 7.1. $\mathfrak{m}$ contains *no* $\text{Spin}(10)$ singlet. In particular $\mathcal{M}$ is *not* the orbit E_6/F_4 of $\text{diag}(1, 1, 1)$ (real dimension 26), and $\mathfrak{m}$ admits no decomposition of the spurious form $2 \times \mathbf{10} \oplus \mathbf{4}$ or $\mathbf{10} \oplus \overline{\mathbf{10}} \oplus \mathbf{4}$. The four Lorentzian frame directions of ?? are *not* an H -invariant subspace of $\mathfrak{m}$; they are selected by the soldering map σ (Postulate P3), not by the branching (46).

The decisive structural fact is the type of $\mathfrak{m}$ as a real H -representation.

Theorem 7.2 (Real-irreducibility of $\mathfrak{m}$). *As a real representation of $H = \text{Spin}(10) \times U(1)$, the tangent space $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is irreducible of complex type: its commutant algebra is $\mathbb{C}$.*

Proof. The complexified module $\mathfrak{m}_{\mathbb{C}}$ is the sum of the two distinct complex-irreducible H -modules $\mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3}$, which are interchanged by complex conjugation (the $U(1)$ charge flips sign and $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$). A real-irreducible module whose complexification is $V \oplus \overline{V}$ with $V \not\cong \overline{V}$ is irreducible of *complex type*; its commutant is $\mathbb{C}$, acting as the complex structure J that rotates the $\mathbf{16}_{-3}$ phase by $e^{i\theta}$ and the $\overline{\mathbf{16}}_{+3}$ phase by $e^{-i\theta}$. (This J is precisely the action of the $\mathfrak{u}(1)$ generator $\mathbf{1}_0$ in (45); it is the integrable almost-complex structure that makes $\mathcal{M} = \text{EIII}$ Hermitian symmetric.) No real H -invariant subspace of $\mathfrak{m}$ other than 0 and $\mathfrak{m}$ exists, since any such subspace would complexify to a proper nonzero H -submodule of $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ stable under conjugation, hence to $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ itself. $\square$

Remark 7.3. [Theorem 7.2](#) replaces, with a cleaner argument, the older “single weight orbit” claim of `lem:Irrep`: irreducibility follows directly from the Hermitian-symmetric-space structure (a single $\mathbb{C}$ -line $\mathfrak{m}^{(1,0)} = \mathbf{16}_{-3}$ and its conjugate), without inspecting the root diagram.

7.2 The invariant metric

Theorem 7.4 (Unique positive-definite coset metric). *Up to an overall positive constant f^2 , there is a unique E_6 -invariant symmetric bilinear form on $\mathfrak{m}$,*

$$\mathfrak{g}_{\alpha\beta} = f^2 K_{\alpha\beta}, \quad K_{\alpha\beta} = (K_{E_6})|_{\mathfrak{m}}, \quad (48)$$

the restriction to $\mathfrak{m}$ of the Killing form of (compact) E_6 . For $f^2 > 0$ this metric is positive-definite (Euclidean signature $(32, 0)$); it is the kinetic/target-space metric of the σ -model. Here $[f] = \text{mass}$.

Proof. Existence and the charge selection rule. An H -invariant symmetric form on $\mathfrak{m}$ is an element of $(\text{Sym}^2 \mathfrak{m}^*)^H$. Complexifying, $\text{Sym}^2 \mathfrak{m}_{\mathbb{C}}^*$ decomposes through

$$\text{Sym}^2(\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}) = \underbrace{\text{Sym}^2 \mathbf{16}_{-3}}_{U(1) \text{ charge } -6} \oplus \underbrace{(\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3})}_{\text{charge } 0} \oplus \underbrace{\text{Sym}^2 \overline{\mathbf{16}}_{+3}}_{\text{charge } +6}. \quad (49)$$

The first and third summands carry nonzero $U(1)$ charge (∓ 6) and therefore contain *no* H -singlet: the symmetric $\mathbf{16} \times \mathbf{16}$ and $\overline{\mathbf{16}} \times \overline{\mathbf{16}}$ invariants are forbidden by the $U(1)$ charge. Only the charge-0 cross-pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3} \supset \mathbf{1}_0$ contributes an H -singlet, and it does so exactly once (the $\text{Spin}(10)$

contraction $\langle \cdot, \cdot \rangle$ is unique up to scale). Hence $\dim(\text{Sym}^2 \mathfrak{m}^*)^H = 1$. This unique invariant is the Killing restriction $K|_{\mathfrak{m}}$.

Uniqueness (Schur). By [Theorem 7.2](#), $\mathfrak{m}$ is real-irreducible, so Schur's lemma gives at most a one-dimensional space of invariant symmetric forms; the count above shows it is exactly one-dimensional. Any two are proportional, $\mathfrak{g}_{\alpha\beta} = c K_{\alpha\beta}$.

Definiteness. On the compact real form of E_6 the Killing form is negative-definite, so $-K_{E_6}$ is a positive-definite invariant inner product on $\mathfrak{e}_6$; its restriction to $\mathfrak{m}$ is therefore positive-definite. Fixing the sign and scale by the kinetic normalization, $c = f^2 > 0$, gives [\(48\)](#). $\square$

Remark 7.5. [Theorem 7.4](#) forbids any (4, 28) or (1, 3) “Killing/coset signature.” The compact- E_6 Killing form on $\mathfrak{m}$ has signature (32, 0) and zero Lorentzian content. Lorentzian signature enters *only* through the soldering Postulate P3 (??), rooted in the cubic norm N on $J_3(\mathbb{O})$ via $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$, not in the coset geometry.

7.3 The Maurer–Cartan form and the clock-field σ -model

Following the coset construction of nonlinear realizations, choose a local representative $g(x) \in E_6$ of the map $x \mapsto \pi(x) \in \mathcal{M}$. The 32 broken generators $X_\alpha \in \mathfrak{m}$ furnish local coordinates $\phi^\alpha(x)$ on $\mathcal{M}$; these are the *clock fields*. The left-invariant Maurer–Cartan one-form

$$\theta_\mu(x) = g^{-1}(x) \partial_\mu g(x) \in \mathfrak{e}_6 \quad (50)$$

decomposes into unbroken ($\mathfrak{h}$) and broken ($\mathfrak{m}$) parts,

$$\theta_\mu = \underbrace{A_\mu^i H_i}_{\mathfrak{h}\text{-connection}} + \underbrace{e_\mu^\alpha X_\alpha}_{\mathfrak{m}\text{-vielbein}}, \quad i = 1, \dots, 46, \quad \alpha = 1, \dots, 32. \quad (51)$$

Under a local H transformation $g(x) \mapsto g(x)h(x)^{-1}$, the $\mathfrak{h}$ -part A_μ transforms as a $\text{Spin}(10) \times U(1)$ gauge connection and the $\mathfrak{m}$ -part e_μ^α transforms covariantly; the composite gauge sector built from A_μ is developed in ??, and the soldering of four components of e_μ^α to a spacetime vierbein in ??.

The H -covariant derivative of the clock fields and the field strength of the composite connection are

$$D_\mu \phi^\alpha = \partial_\mu \phi^\alpha + A_\mu^i (t_i)^\alpha_\beta \phi^\beta, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \quad (52)$$

with $(t_i)^\alpha_\beta$ the matrices of $\mathfrak{h}$ acting on $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$. The most general two-derivative action consistent with global E_6 , local H , and (after Postulate P3) local Lorentz invariance is fixed up to the couplings to be the gauged nonlinear σ -model

$$S_\sigma[\phi, A] = \int d^4x \sqrt{-g} \frac{1}{2} g^{\mu\nu} \mathfrak{g}_{\alpha\beta} D_\mu \phi^\alpha D_\nu \phi^\beta - \frac{1}{4} \hat{g}^{-2} \text{Tr}(F_{\mu\nu} F^{\mu\nu}), \quad (53)$$

with the target metric $\mathfrak{g}_{\alpha\beta} = f^2 K_{\alpha\beta}$ of [Theorem 7.4](#) and signature convention $(-, +, +, +)$. The matter (Weyl-fermion) and Yukawa sectors are added in ?? (Postulates P4, P6); the gauge coupling $\hat{g}^2$, the dimensionless $\xi, \hat{y}^2, \kappa$, and the running of f are studied in ??.

Lemma 7.6 (Healthy kinetic term). *For $f^2 > 0$ the clock-field kinetic term in [\(53\)](#) is positive-definite, and the classical σ -model energy is bounded below.*

Proof. Immediate from [Theorem 7.4](#): $\mathfrak{g}_{\alpha\beta}$ is positive-definite on $\mathfrak{m}$, so for the spatial gradient and time derivative the energy density $\frac{1}{2} \mathfrak{g}_{\alpha\beta} (\dot{\phi}^\alpha \dot{\phi}^\beta + \partial_i \phi^\alpha \partial_i \phi^\beta) \geq 0$, and the gauge term enters at most quadratically. (This is where the *compact* real form is essential; a non-compact form would render $K|_{\mathfrak{m}}$ indefinite.) $\square$

7.4 Tree-level massless Goldstones

Theorem 7.7 (Tree-level Goldstone theorem). *At tree level the potential V of Postulate P1 is E_6 -invariant, and $\mathcal{M}$ is an orbit of broken vacua. Hence the Hessian of V vanishes along the coset directions,*

$$\text{Hess } V_{\text{tree}}|_{\mathfrak{m}} = 0, \quad (54)$$

and all 32 clock fields ϕ^α are exact massless Goldstone bosons.

Proof. The vacuum $X_0 \in \mathcal{M}$ minimizes V (?). For any broken generator $X_\alpha \in \mathfrak{m}$ the curve $t \mapsto e^{tX_\alpha} \cdot X_0$ lies in the minimizing orbit, so V is constant along it; differentiating twice gives $X_\alpha^\top \text{Hess } V(X_0) X_\beta = 0$ for all $X_\alpha, X_\beta \in \mathfrak{m}$. Thus the 32 coset directions are exact null eigenvectors of the tree Hessian, i.e. massless Goldstone modes (Goldstone’s theorem). The 46 unbroken directions in $\mathfrak{h}$ produce the massless $\text{Spin}(10) \times U(1)$ gauge bosons of ??. $\square$

Remark 7.8 (What lifts the masses, and what stays massless). The tree degeneracy (54) is lifted radiatively. Because the one-loop Coleman–Weinberg potential is H -invariant and $\mathfrak{m}$ is a *single* real-irreducible H -module (Theorem 7.2), Schur’s lemma forces a *single common eigenvalue* on all of $\mathfrak{m}$: one common radiative mass, with no $\text{Spin}(10)$ -singlet subspace to split off. Of the 32 clock fields, four are then identified by the soldering map (Postulate P3, ??) with the spacetime frame and are protected by gauge symmetry (diffeomorphism + local Lorentz, Ward identity $\nabla_\mu(\delta\Gamma/\delta e^a_\mu) = 0$); they are eaten, not Hessian zeros. This yields the locked count $32 = 28$ (massive, common mass) + 4 (gauge/soldering), with the canonical mass $m^2 = \mu^2$ ($[f] = \text{mass}$; the dim-4 Hessian eigenvalue is $f^2\mu^2$). The spectrum is established in ??; we do *not* claim a four-dimensional H -invariant Goldstone subspace here.

Remark 7.9 (Topology of the clock-field manifold). $\mathcal{M} = \text{EIII}$ is simply connected as a compact Hermitian symmetric space, so $\pi_1(\mathcal{M})$ is trivial; in particular $\pi_1(\mathcal{M}) = \mathbb{Z}$ is *false* of EIII itself. The vortex/string configurations of ?? therefore arise from the *gauged* ($U(1)$ -quotient) configuration space rather than from $\pi_1(\mathcal{M})$, and are presented there as a low-energy correspondence.

Status of the claims. Theorem 7.2, Theorem 7.4, Theorem 7.6, and Theorem 7.7 are **theorems**, proved from the locked branching (45) and the compact real form. The four-direction soldering invoked in Theorem 7.1 and Theorem 7.8 is **Postulate P3** (added structure, ??); the matter and Yukawa content is **Postulates P4, P6** (?). The coset geometry and its unique positive-definite metric are derived; $d = 4$, signature $(1, 3)$, and the Standard-Model content are added postulates.

Part III

Spacetime, Gauge Sector, and Mass Spectrum

8 Emergent Spacetime, Soldering, and Lorentzian Signature

The dynamics of UCFT live on the compact real form of E_6 , whose vacuum manifold is the rank-1 primitive-idempotent orbit $\mathcal{M} = \text{EIII} = E_6/(\text{Spin}(10) \times U(1))$, $\dim_{\mathbb{R}} \mathcal{M} = 32$ (Theorem 6.7). The induced metric on $\mathcal{M}$ — the kinetic/target-space metric of the coset σ -model — is *positive-definite*: the maximal compact subgroup $H = \text{Spin}(10) \times U(1)$ acts on the reductive complement $\mathfrak{m} = \mathbf{16}_{-3} \oplus \mathbf{\overline{16}}_{+3}$, a real-irreducible module of complex type whose Killing restriction $K|_{\mathfrak{m}}$ is the unique (Schur) H -invariant form, of signature $(32, 0)$. There is *no* Lorentzian content in the coset geometry.

This section establishes how a four-dimensional *Lorentzian* spacetime is nonetheless *soldered* onto this Euclidean arena. The logical separation, which we maintain throughout, is the following. The positive-definite object is the coset metric $K|_{\mathfrak{m}}$ on the 32-dimensional internal tangent space $\mathfrak{m}$. The indefinite object is the cubic norm N on the matter representation $\mathbf{27} = J_3(\mathbb{O})$, whose 2×2 octonion-Hermitian sub-block $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ carries a Minkowski quadratic form. These are two distinct forms on two distinct spaces; no single form is asked to be both definite and indefinite. The soldering map (Postulate P3) is the bridge that imports the indefinite $(1, 3)$ structure onto a tangent 4-plane of the world manifold, leaving $K|_{\mathfrak{m}}$ untouched.

We classify every nontrivial claim as a **theorem** (proved from stated hypotheses), **postulate** (added structure, P1–P6), or **conjecture** (motivated, not controlled). The structural constraint recorded in [Theorem 8.12](#) is that $d = 4$ and signature $(1, 3)$ are added structure (P3), not consequences of the Killing form.

8.1 The Minkowski block inside the Albert algebra

Let $\mathbb{O}$ be the real division octonions, with conjugation $x \mapsto \bar{x}$, norm $n(x) = x\bar{x} \in \mathbb{R}_{\geq 0}$, and Euclidean inner product $\langle x, y \rangle = \frac{1}{2}(x\bar{y} + y\bar{x})$ on $\mathbb{O} \cong \mathbb{R}^8$. The space of 2×2 octonion-Hermitian matrices is

$$H_2(\mathbb{O}) = \left\{ X = \begin{pmatrix} a & x \\ \bar{x} & b \end{pmatrix} : a, b \in \mathbb{R}, x \in \mathbb{O} \right\}, \quad \dim_{\mathbb{R}} H_2(\mathbb{O}) = 10, \quad (55)$$

with the (well-defined, despite non-associativity) determinant $\det X = ab - x\bar{x} = ab - n(x)$.

Theorem 8.1 (Minkowski identification of $H_2(\mathbb{O})$). *In light-cone coordinates $a = X^0 + X^9$, $b = X^0 - X^9$, and $x = (X^1, \dots, X^8)$ in an orthonormal octonion basis,*

$$\det X = (X^0)^2 - (X^9)^2 - \sum_{i=1}^8 (X^i)^2 = -\eta_{MN} X^M X^N, \quad \eta = \text{diag}(-1, \underbrace{+1, \dots, +1}_9), \quad (56)$$

so that $(H_2(\mathbb{O}), -\det) \cong \mathbb{R}^{1,9}$.

Proof. $\det X = ab - n(x) = (X^0 + X^9)(X^0 - X^9) - \sum_i (X^i)^2 = (X^0)^2 - (X^9)^2 - \sum_i (X^i)^2$. By Sylvester's law of inertia the diagonal entries contribute one $+$ (X^0) and one $-$ (X^9), and $-n(x)$ contributes eight $-$'s; hence $-\det$ has one timelike and nine spacelike directions in the convention $(-, +, \dots, +)$. $\square$

Theorem 8.2 (Octonionic exceptional isomorphisms). *There are Lie-group isomorphisms, the four members of the classical family $\text{SL}(2, \mathbb{A}) \cong \text{Spin}(n+1, 1)$ for the normed division algebras $\mathbb{A} = \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ of real dimension $n = 1, 2, 4, 8$:*

$$\text{SL}(2, \mathbb{R}) \cong \text{Spin}(1, 2), \quad \text{SL}(2, \mathbb{C}) \cong \text{Spin}(1, 3), \quad \text{SL}(2, \mathbb{H}) \cong \text{Spin}(1, 5), \quad \text{SL}(2, \mathbb{O}) \cong \text{Spin}(1, 9). \quad (57)$$

The defining 10-dimensional representation of $\text{Spin}(1, 9)$ on $\mathbb{R}^{1,9}$ is realized on $(H_2(\mathbb{O}), -\det)$; the two chiral $\mathbf{16}$ spinors are realized on $\mathbb{O}^2 \cong \mathbb{R}^{16}$.

Proof. Classical (Sudbery; Kugo–Townsend; Manogue–Schray; Baez). For $\mathbb{A} = \mathbb{O}$ the group $\text{SL}(2, \mathbb{O})$ is defined as $\text{Spin}(1, 9)$ with its $\mathbb{R}^{1,9}$ action presented by (55)–(56), non-associativity precluding a literal matrix group. $\square$

Theorem 8.3 (The physical Lorentz 4-plane). *The chain of composition-algebra inclusions $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O}$ induces $\text{Spin}(1,1) \subset \text{Spin}(1,3) \subset \text{Spin}(1,5) \subset \text{Spin}(1,9)$. In particular $\text{SL}(2,\mathbb{C}) = \text{Spin}(1,3) \subset \text{SL}(2,\mathbb{O}) = \text{Spin}(1,9)$ via the sub-block $H_2(\mathbb{C}) = \{[a, z], [\bar{z}, b] : a, b \in \mathbb{R}, z \in \mathbb{C}\} \subset H_2(\mathbb{O})$, $\dim_{\mathbb{R}} H_2(\mathbb{C}) = 4$, with $\det|_{H_2(\mathbb{C})} = ab - |z|^2$ the (1,3) Minkowski form in the convention $(-, +, +, +)$.*

Proof. The same light-cone change of variables as [Theorem 8.1](#), restricted to $\mathbb{C}$, gives $\det = (X^0)^2 - (X^1)^2 - (X^2)^2 - (X^3)^2$. The $\text{SL}(2,\mathbb{C})$ determinant-covariance is the standard four-dimensional statement. $\square$

The ambient indefinite structure is the octonionic $\mathbb{R}^{1,9}$; the *physical* Lorentz signature (1,3) is a soldered 4-plane inside it ([subsection 8.3](#)).

8.2 Source of indefiniteness: the cubic norm, not the trace form

The Albert algebra $J_3(\mathbb{O})$ of 3×3 octonion-Hermitian matrices, $\dim_{\mathbb{R}} J_3(\mathbb{O}) = 27$, carries two basic polynomial invariants: the quadratic trace form $Q(X) = \text{Tr}(X^2)$ (a sum of real squares and positive-definite octonion norms, hence *positive-definite*) and the cubic norm $N(X) = \det X$ (a homogeneous cubic). Their invariance groups differ crucially.

Theorem 8.4 (Separation of invariance groups). *$F_4 = \text{Aut}(J_3(\mathbb{O}))$ (compact, dim 52) preserves the Jordan product, hence both Q and N ; since Q is positive-definite, $F_4 \subset \text{SO}((27))$. The reduced structure group $E_{6(-26)}$ (dim 78) preserves N only up to a real character $\lambda(g)$ and does not preserve Q : were it to, it would lie in $\text{SO}((27))$ and be compact, contradicting its non-compactness (maximal compact F_4 , 26 non-compact directions move Q).*

Proof. Automorphisms fix **1**, commute with $\circ$, hence fix the characteristic coefficients Tr , $Q = \text{Tr}(X^2)$, $N = \det$; Euclidean signature forces $F_4 \subset \text{SO}((27))$. The reduced structure group is by definition the linear group fixing N up to scalar; $\dim E_{6(-26)} = 78 > 52 = \dim F_4$ with non-compactness gives the second claim. $\square$

This excludes the incorrect claim that the structure group preserves Q or that (Q, N) separates orbits: only N is structure-group invariant (up to a character), and real E_6 -orbits on the **27** are classified by $\text{rank} \in \{0, 1, 2, 3\}$ and the value of N , via the sharp map $X^\#$ ($\text{rank } X \leq 1 \Leftrightarrow X^\# = 0$; $\text{rank } X \leq 2 \Leftrightarrow N(X) = 0$), cf. ??.

Theorem 8.5 (N is genuinely indefinite, and contains the Minkowski block). *N takes strictly positive, strictly negative, and zero values on $J_3(\mathbb{O}) \cong \mathbb{R}^{27}$. Restricting N to the upper-left 2×2 block $Y = [[\xi_1, z_3], [\bar{z}_3, \xi_2]] \in H_2(\mathbb{O})$ with the third pivot set to $\xi_3 = 1$ and $z_1 = z_2 = 0$,*

$$N(X)|_{\text{block}} = \xi_1 \xi_2 - n(z_3) = \det Y, \quad (58)$$

so the Minkowski form $-\det = -\eta^{(1,9)}$ of [Theorem 8.1](#) is the restriction of the cubic norm N to an $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ sub-block.

Proof. On $X = \text{diag}(\xi_1, \xi_2, \xi_3)$, $N = \xi_1 \xi_2 \xi_3$ takes values $+1, -1, 0$ at $(1, 1, 1), (-1, 1, 1), (0, 1, 1)$; an odd-degree form is necessarily indefinite. The restriction (58) is the explicit evaluation of $N(X) = \xi_1 \xi_2 \xi_3 + 2\Re(z_1 z_2 z_3) - \sum_i \xi_i n(z_i)$ at the stated pivot. $\square$

Theorem 8.6 (Definite coset metric and indefinite norm are consistent). *The positive-definite coset metric $K|_{\mathfrak{m}}$ and the indefinite spacetime form are mutually consistent because they live on different modules:*

<i>Object</i>	<i>Lives on</i>	<i>Signature</i>	<i>Invariance</i>
<i>Coset metric</i> $K _{\mathfrak{m}}$	$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ (32)	(32, 0) <i>definite</i>	H (unique, Schur)
<i>Trace form</i> $Q = \text{Tr}(X^2)$	$J_3(\mathbb{O}) = \mathbb{R}^{27}$	(27, 0) <i>definite</i>	F_4 (not E_6)
<i>Cubic norm</i> $N = \det$	$J_3(\mathbb{O}) = \mathbb{R}^{27}$	<i>indefinite</i>	E_6 up to character
<i>Minkowski</i> $-\det$	$H_2(\mathbb{O}) \subset J_3(\mathbb{O})$	(1, 9)	$\text{SL}(2, \mathbb{O}) = \text{Spin}(1, 9)$
<i>Soldered</i> η	<i>4-plane of</i> $T_x \mathcal{M}^4$	(1, 3), $(-, +, +, +)$	$\text{SO}((1, 3))$ (postulated, P3)

Proof. $K|_{\mathfrak{m}}$ is positive-definite by the compact-real-form choice (compact E_6 has negative-definite Killing form; the induced metric on $\mathfrak{m}$ is its sign-flip), unique by Schur since $\mathfrak{m}$ is real-irreducible of complex type with commutant $\mathbb{C}$ (??). The indefinite N lives on the **27**, a representation distinct from the adjoint complement $\mathfrak{m}$; hence “ $K|_{\mathfrak{m}}$ Euclidean” and “ N indefinite” are statements about different data on different spaces. The spacetime signature is imported from $N|_{H_2(\mathbb{O})}$ via P3, and is never asked of $K|_{\mathfrak{m}}$. $\square$

This excludes the incorrect (4, 28) Killing/coset signature: Lorentzian structure does *not* follow from the coset geometry alone.

8.3 The soldering postulate P3

We now state P3. It is a **Postulate** throughout: *added* structure, not a theorem of the coset geometry.

Theorem-level inputs. From the dynamics we have: (T-a) the vacuum manifold $\mathcal{M} = \text{EIII} = E_6/(\text{Spin}(10) \times \text{U}(1))$, $\dim_{\mathbb{R}} = 32$, with reductive decomposition $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$, $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$, $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, and Maurer–Cartan form $\theta = \theta_{\mathfrak{h}} + \theta_{\mathfrak{m}}$ valued in $\mathfrak{e}_6$; (T-b) the matter rep **27** = $J_3(\mathbb{O})$ carrying the indefinite N with its $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ and $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ sub-blocks ([Theorem 8.1](#)–[Theorem 8.3](#), [Theorem 8.5](#)); (T-c) the branching **27** = $\mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$ under H .

The 32 clock fields are the coordinates along $\mathfrak{m}$ (Goldstone directions of EIII). Of these, 28 are physical massive scalars (single common radiative mass by Schur on the irreducible $\mathfrak{m}$, ??) and 4 are gauge/soldering modes eaten by world-volume diffeomorphism plus local Lorentz. The 4 frame modes are *gauge-protected*, not Hessian zeros: $\mathfrak{m}$ is irreducible and contains no $\text{SO}((10))$ singlet, so no 4-dimensional H -invariant subspace of $\mathfrak{m}$ exists. The old rank-four lemma ([1em:Rank4](#)) is hereby relabeled: $d = 4$ is fixed by P3, and its circular $r > 4$ branch is deleted.

Postulate 8.7 (P3a, soldering / frame identification). There exists a soldering map σ identifying four distinguished clock-field frame directions, written e_{μ}^a ($a = 0, 1, 2, 3$; μ a world index), with a tangent 4-plane $\Pi_x \subset T_x \mathcal{M}^4$ of an emergent four-dimensional world manifold $\mathcal{M}^4$. The vierbein is the pull-back of the Maurer–Cartan form along these four directions,

$$e_{\mu}^a(x) = \langle \tau^a, \theta_{\mathfrak{m}}(\partial_{\mu}) \rangle, \quad (59)$$

where $\{\tau^a\}_{a=0}^3$ is the soldering frame — the image of the $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ basis under the embedding of [Theorem 8.10](#). The vierbein *emerges dynamically*.

Postulate 8.8 (P3b, induced Minkowski form). The induced metric on Π_x is the Minkowski form inherited from the cubic-norm / $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ structure:

$$g_{\mu\nu}(x) = e_{\mu}^a(x) e_{\nu}^b(x) \eta_{ab}, \quad \eta_{ab} = \text{diag}(-1, +1, +1, +1), \quad (60)$$

with η the restriction of $-\det = -N$ to the $H_2(\mathbb{C})$ sub-block ([Theorem 8.3](#), [Theorem 8.5](#)).

Postulate 8.9 (P3c, dimension and signature are added). The numbers $d = 4$ and signature $(1, 3) = (-, +, +, +)$ are *part of P3*: added structure, *not* consequences of the Killing form, the coset signature, or any rank lemma.

Postulate 8.10 (P3d, vierbein condensate / frame reduction). A vierbein condensate (a VEV of the soldering bilinear) selects the $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ sub-block of $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ and breaks the ambient frame symmetry

$$\text{Spin}(1, 9) = \text{SL}(2, \mathbb{O}) \longrightarrow \text{Spin}(1, 3) \times \text{Spin}(6) = \text{SL}(2, \mathbb{C}) \times \text{SU}((4), \quad (61)$$

where the orthogonal $\mathbb{R}^6$ complement of $\mathbb{R}^{1,3}$ in $\mathbb{R}^{1,9}$ is acted on by internal $\text{Spin}(6) \cong \text{SU}((4)$. The residual $\text{Spin}(1, 3) = \text{SL}(2, \mathbb{C})$ is the local Lorentz group.

Theorem 8.11 (Internal consistency of P3). *Given [Theorem 8.7](#)–[Theorem 8.10](#), the metric $g_{\mu\nu}$ of (60) is a non-degenerate symmetric rank-2 tensor of Lorentzian signature $(1, 3)$ on $\mathcal{M}^4$, invariant under the residual local Lorentz group $\text{SO}((1, 3)$; the four frame modes e_μ^a carry exactly the gauge degrees of freedom eaten by diffeomorphisms plus local Lorentz, consistent with the count $32 = 28 \text{ physical} + 4 \text{ gauge}$.*

Proof. Non-degeneracy and signature are inherited from $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ ([Theorem 8.3](#)) for generic condensate $\det(e_\mu^a) \neq 0$; $\text{SO}((1, 3)$ -invariance is $\eta_{ab} = \Lambda^c_a \Lambda^d_b \eta_{cd}$ for $\Lambda \in \text{SO}((1, 3)$. The gauge count is the Ward identity $\nabla_\mu(\delta\Gamma/\delta e_\mu^a) = 0$: the four frame directions are protected by diffeomorphism plus local Lorentz, not by a Hessian zero. $\square$

What P3 does not claim. P3 does not make $K|_{\mathfrak{m}}$ Lorentzian (it stays positive-definite); it does not use a $(4, 28)$ Lorentzian split of $\mathfrak{m}$ (the 4 are gauge, the 28 are Euclidean target coordinates); and it does not build the 4-plane from $\nabla I_2, \nabla I_3$ singlets (there are none in $\mathfrak{m}$, and $\nabla Q|_{X_0} = 2X_0 \notin \mathfrak{m}$). The 4-plane is the soldered image of $H_2(\mathbb{C}) \subset \mathbf{27}$, not a submodule of $\mathfrak{m}$.

Theorem 8.12 (No Killing-form derivation of dimension or signature). *There is no derivation of $d = 4$ or signature $(1, 3)$ from the Killing form of E_6 , from the coset metric $K|_{\mathfrak{m}}$, or from any rank lemma. The Killing form of compact E_6 restricted to $\mathfrak{m}$ is positive-definite of signature $(32, 0)$, with zero Lorentzian content. Any $(4, 28)$ or $(1, 3)$ Killing/coset signature claim is false. Lorentzian signature enters only through P3, rooted in the indefinite cubic norm $N / H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$.*

Proof. Immediate from [Theorem 8.6](#): $K|_{\mathfrak{m}}$ is definite, and the indefinite datum lives on the disjoint module $\mathbf{27}$. $\square$

8.4 Emergent time: Page–Wootters clock and Osterwalder–Schrader

Lorentzian time is recovered by a relational-clock plus reconstruction-theorem mechanism. We use *no* “multiply e^0 by i ” Wick trick.

Postulate 8.13 (Timelike clock selection). Among the 32 clock fields, the soldering condensate ([Theorem 8.10](#)) selects a distinguished timelike direction whose clock field we call ϕ^0 : the trace/ X^0 direction of the soldered $H_2(\mathbb{C}) \subset J_3(\mathbb{O})$ block (the unique η -timelike frame leg e_μ^0). Its three spacelike companions are e_μ^i , $i = 1, 2, 3$. The selection of *one* timelike direction is the $(1, 3)$ content of P3.

Promote ϕ^0 to a **Page–Wootters clock**: enlarge the kinematical Hilbert space to $\mathcal{H} = \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{sys}}$ and impose the induced-gravity Hamiltonian constraint

$$\hat{H}_{\text{phys}} |\Psi\rangle = 0, \quad \hat{H}_{\text{phys}} = \hat{H}_{\text{clock}} + \hat{H}_{\text{sys}} (+ \text{ interaction}), \quad (62)$$

on physical states $|\Psi\rangle$. Conditioning on clock readings $\phi^0 = \tau$ defines the relational state $|\psi(\tau)\rangle_{\text{sys}} = \langle \phi^0 = \tau | \Psi \rangle$.

Theorem 8.14 (Page–Wootters relational evolution). *With a good clock ($\hat{H}_{\text{clock}} = \hat{p}_{\phi^0}$ canonically conjugate to ϕ^0), the conditioned state satisfies the Schrödinger equation in the relational time τ ,*

$$i \partial_\tau |\psi(\tau)\rangle_{\text{sys}} = \hat{H}_{\text{sys}} |\psi(\tau)\rangle_{\text{sys}}, \quad (63)$$

and the unitary group $U(\tau) = e^{-i\hat{H}_{\text{sys}}\tau}$ is recovered on physical states in the semiclassical/background regime where the soldered $g_{\mu\nu}$ admits a timelike Killing vector and $\hat{H}_{\text{sys}} = \int T_{00}$ along it.

Proof. Standard Page–Wootters reduction: under the constraint (62) with a conjugate clock, the conditioned state obeys (63). The identification $t \leftrightarrow x^0$ is $\hat{H} = \int T_{00}$ along the timelike Killing direction. Global hyperbolicity is flagged as an assumption. $\square$

This demotes the original Axiom II (an externally postulated $U(t) = e^{-iHt}$) to a theorem: time is emergent.

Euclidean to Lorentzian via Osterwalder–Schrader. The compact-form σ -model with positive-definite $K|_{\text{m}}$ defines a Euclidean functional measure. Lorentzian QFT is obtained by Osterwalder–Schrader (OS) reconstruction along the selected time ϕ^0 , *not* by rotating $e^0 \rightarrow i e^0$. Let $\tau = \phi^0$ be the OS time, and assume the Euclidean Schwinger functions satisfy the OS axioms, in particular *reflection positivity* under $\theta : \tau \mapsto -\tau$ about a ϕ^0 -slice,

$$\sum_{i,j} \bar{c}_i c_j S(\theta F_i \cdot F_j) \geq 0, \quad F_i \text{ supported in } \{\tau > 0\}. \quad (64)$$

Theorem 8.15 (OS reconstruction $\Rightarrow$ Lorentzian Wightman theory). *Given the OS axioms (Euclidean invariance, reflection positivity (64), symmetry, clustering), the OS reconstruction theorem produces a Hilbert space $\mathcal{H}$, a positive self-adjoint Hamiltonian $\hat{H} \geq 0$ generating τ -translations as the contraction semigroup $e^{-\hat{H}\tau}$, and — by analytic continuation of the Schwinger functions in the clock-time argument $\tau \rightarrow it$ — Lorentzian Wightman functions on $(\mathcal{M}^4, \eta^{(1,3)})$ with unitary evolution $e^{-i\hat{H}t}$ satisfying the spectrum condition and Lorentz covariance under the reconstructed $\text{SO}((1,3))$.*

Proof. Osterwalder–Schrader (1973, 1975): reflection positivity defines a physical inner product on $\{F : \tau > 0\}/(\text{null})$; $\hat{H}$ is the positive self-adjoint generator of $e^{-\hat{H}\tau}$; the Schwinger functions, analytic in a tube, continue to Wightman distributions. $\square$

Indicative 8.16 (Reflection positivity of the UCFT σ -model). The OS theorem (Theorem 8.15) is rigorous given its hypotheses. That the UCFT σ -model *satisfies* reflection positivity (64) along ϕ^0 is the precise property to be checked — it is the rigorous replacement for the Wick trick, and is tagged **Indicative**.

The analytic continuation $\tau \rightarrow it$ acts on the Schwinger distributions in the clock-time argument (justified by reflection positivity and the resulting analyticity tube), *not* on a frame field. The Lorentzian metric $\eta^{(1,3)}$ is the soldered form of P3, already Lorentzian before any continuation; OS supplies the unitary, positive-energy dynamics, with ϕ^0 serving as the OS reflection time. The two mechanisms dovetail: [Theorem 8.14](#) gives the relational unitary $U(\tau)$ on physical states, and [Theorem 8.15](#) gives the positive-energy, Lorentz-covariant Wightman theory.

8.5 Master statement

Postulate 8.17 (P3, hardened soldering — master statement). Given (i) the compact- E_6 dynamics with vacuum $\text{EIII} = E_6/(\text{Spin}(10) \times \text{U}(1))$ and positive-definite coset metric $K|_{\mathfrak{m}}$ (Theorem-level inputs), and (ii) the matter rep $\mathbf{27} = J_3(\mathbb{O})$ with its indefinite cubic norm N containing the $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ ($\text{SL}(2, \mathbb{O}) = \text{Spin}(1, 9)$) and $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ ($\text{SL}(2, \mathbb{C}) = \text{Spin}(1, 3)$) sub-blocks ([Theorem 8.1–Theorem 8.3](#), [Theorem 8.5](#)), we postulate a soldering map σ ([Theorem 8.7](#)) identifying four clock-field frame directions e_μ^a (pull-backs of the Maurer–Cartan form) with a tangent 4-plane of an emergent four-dimensional world manifold, carrying $g_{\mu\nu} = e_\mu^a e_\nu^b \eta_{ab}$, $\eta = \text{diag}(-1, +1, +1, +1)$ ([Theorem 8.8](#)), with $d = 4$ and signature $(1, 3)$ as added structure ([Theorem 8.9](#)), realized by a vierbein condensate breaking $\text{Spin}(1, 9) \rightarrow \text{Spin}(1, 3) \times \text{Spin}(6)$ ([Theorem 8.10](#)). The timelike clock ϕ^0 is a Page–Wootters clock and Lorentzian unitary dynamics follows from Osterwalder–Schrader reconstruction along ϕ^0 ([Theorem 8.15](#)), not by Wick rotation of the frame.

The vierbein e_μ^a of (59), carrying simultaneously a world index μ and a soldering index a , is the fundamental gravitational field; its σ -model dynamics and the induced Einstein–Hilbert term are developed in [section 11](#). The low-energy field content on the soldered spacetime — the $46 = \dim \text{SO}((10)) + 1$ massless gauge bosons, the chiral $\mathbf{16}_{+1}$ fermions, and the 28 massive coset scalars — transforms under the unbroken $H = \text{Spin}(10) \times \text{U}(1)$ and is catalogued in [section 13](#).

9 The composite $\text{Spin}(10) \times \text{U}(1)$ gauge sector

Having fixed the vacuum manifold as the rank-one primitive-idempotent orbit

$$\mathcal{M} = E_6/(\text{Spin}(10) \times \text{U}(1)) = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32,$$

([Theorem 6.7](#)), we now show that the *same* geometric data — the Maurer–Cartan form of E_6 pulled back along a coset map — generate a composite $H = \text{Spin}(10) \times \text{U}(1)$ Yang–Mills theory. The gauge bosons are not fundamental: they are the unbroken-subalgebra components of the canonical connection on the homogeneous bundle $E_6 \rightarrow \mathcal{M}$. We establish (i) the composite connection and the $\mathbf{45}_0 \oplus \mathbf{1}_0$ gauge content, (ii) the induced Yang–Mills term with the *corrected* coefficient computed from the massive multiplet $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, (iii) that H remains *unbroken* at the vacuum so that the $\dim \mathbf{45} + \dim \mathbf{1} = 46$ gauge bosons stay massless, and (iv) Green–Schwarz cancellation of the residual mixed anomaly. Throughout we work on the *compact* real form of E_6 , so that H is compact, the coset metric is positive-definite, and the gauge sector is unitary (?). Signature is $(-, +, +, +)$ and $[f] = \text{mass}$.

9.1 The composite connection from the Maurer–Cartan form

The Lie algebra splits, H -equivariantly, as the reductive sum

$$\mathfrak{e}_6 = \underbrace{\mathfrak{so}_{10} \oplus \mathfrak{u}_1}_{\mathfrak{h}} \oplus \underbrace{\mathfrak{m}}_{\text{coset}}, \quad [\mathfrak{h}, \mathfrak{m}] \subseteq \mathfrak{m}, \quad (65)$$

with the branching of the adjoint under $H = \text{Spin}(10) \times \text{U}(1)$ fixed by the involution datum

$$\mathbf{78} = \underbrace{\mathbf{45}_0 \oplus \mathbf{1}_0}_{\mathfrak{h}} \oplus \underbrace{\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}}_{\mathfrak{m}}. \quad (66)$$

Here the subscripts are the $\text{U}(1)$ charges; $\mathfrak{m}$ is real-irreducible of complex type and contains *no* $\text{Spin}(10)$ singlet (??). The 46 generators of $\mathfrak{h}$ are exactly the composite gauge degrees of freedom.

Let $g(x) \in \text{E}_6$ be a smooth local section of the principal H -bundle, representing the coset map $\pi_-(x) \in \mathcal{M}$. The Maurer–Cartan form decomposes along (65),

$$g^{-1}\partial_\mu g = A_\mu(x) + e_\mu(x), \quad A_\mu \in \mathfrak{so}_{10} \oplus \mathfrak{u}_1, \quad e_\mu \in \mathfrak{m}, \quad (67)$$

where A_μ is the $\mathfrak{h}$ -part (the canonical connection) and e_μ is the $\mathfrak{m}$ -part (the soldering one-form of the clock fields, used in ??).

Theorem 9.1 (Composite connection; THEOREM). *Under a change of local section $g \mapsto gh^{-1}(x)$ with $h(x) \in H$, the $\mathfrak{h}$ -component of (67) transforms as a genuine H gauge potential,*

$$\mathcal{A}_\mu := A_\mu \mapsto h \mathcal{A}_\mu h^{-1} - (\partial_\mu h) h^{-1}, \quad (68)$$

while $e_\mu \mapsto h e_\mu h^{-1}$ transforms homogeneously (in the $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$). Its curvature

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + [\mathcal{A}_\mu, \mathcal{A}_\nu] = -P_{\mathfrak{h}}(g^{-1}[\partial_\mu, \partial_\nu]g) + [e_\mu, e_\nu]_{\mathfrak{h}} \quad (69)$$

is the $\mathfrak{h}$ -valued field strength of a $\text{Spin}(10) \times \text{U}(1)$ gauge theory.

Proof. Equation (68) is the standard transformation of the canonical (reductive) connection on a homogeneous bundle: writing $g \mapsto gh^{-1}$ gives $g^{-1}\partial g \mapsto h(g^{-1}\partial g)h^{-1} - (\partial h)h^{-1}$, and projecting onto $\mathfrak{h}$ (which is H -stable by (65)) yields (68); the $\mathfrak{m}$ -projection transforms homogeneously since $[\mathfrak{h}, \mathfrak{m}] \subseteq \mathfrak{m}$. The Maurer–Cartan equation $d(g^{-1}dg) + \frac{1}{2}[g^{-1}dg, g^{-1}dg] = 0$, projected onto $\mathfrak{h}$, gives (69) (the Gauss–Codazzi relation: the $\mathfrak{h}$ -curvature receives the soldering contribution $[e_\mu, e_\nu]_{\mathfrak{h}}$). Both are intrinsic to the reductive split. $\square$

The composite gauge content is therefore precisely $\mathbf{45}_0 \oplus \mathbf{1}_0$: the $\mathbf{45}$ adjoint of $\text{Spin}(10)$ and the $\text{U}(1)$ singlet, 46 vector fields in all.

9.2 Induced Yang–Mills term — corrected coefficient

Fluctuating the massive clock fields around the vacuum and integrating them out at one loop produces the functional determinant $[\det(-D^2 + m^2)]^{-1/2}$ with $D_\mu = \partial_\mu + \mathcal{A}_\mu$ and $m^2 = f^2\mu^2$ (subsection 11.1). The Seeley–DeWitt a_4 coefficient supplies a local Yang–Mills counterterm,

$$S_{\text{YM}} = -\frac{1}{4g^2} \int_X \text{Tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) d^4x, \quad (70)$$

whose logarithmic coefficient is fixed by the total Dynkin index of the fields running in the loop.

Remark 9.2 (The corrected loop multiplet; THEOREM). The integrated-out matter is the *massive coset multiplet* $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ (dim 32), the chiral spinor pair of $\text{Spin}(10)$, *not* a spurious $\mathbf{10} \oplus \overline{\mathbf{10}} \oplus \mathbf{4}$. With the $\text{Spin}(10)$ Dynkin indices $T(\mathbf{16}) = 2$ for each spinor (and the $\text{U}(1)$ singlet $\mathbf{1}$ not contributing to the $\text{Spin}(10)$ trace), the total index is

$$\sum_{R \subset \mathfrak{m}} \text{mult}(R) T(R) = T(\mathbf{16}) + T(\overline{\mathbf{16}}) = 2 + 2 = 4, \quad (71)$$

the value used everywhere. The previously quoted $\frac{5}{2}$, obtained from the (forbidden) decomposition $\mathbf{10} \oplus \overline{\mathbf{10}} \oplus \mathbf{4}$ with indices $1, 1, \frac{1}{2}$, is wrong: that multiplet is not the coset tangent (which carries no Spin(10) singlet, ??).

Theorem 9.3 (Induced gauge coupling; THEOREM). *Integrating out $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ at one loop induces (70) with*

$$g^{-2} = \frac{4}{1} \frac{f^2}{48\pi^2} \log \frac{\Lambda^2}{\mu^2} + \mathcal{O}(\Lambda^0) = \frac{f^2}{12\pi^2} \log \frac{\Lambda^2}{\mu^2} + \mathcal{O}(\Lambda^0), \quad (72)$$

with Λ the UV cutoff and μ the breaking scale. In particular $g^{-2} > 0$ for $\Lambda > \mu$, so the induced gauge sector is healthy.

Proof sketch. The proper-time representation of $\Gamma_{1L} = \frac{1}{2} \text{Tr} \log(-D^2 + m^2)$, expanded with the heat-kernel coefficient $a_4(R) = \frac{1}{12} \int_X \text{Tr}_R(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu})$ and the cutoff $s \geq \Lambda^{-2}$, gives a logarithmically divergent $\text{Tr}(\mathcal{F}\mathcal{F})$ coefficient proportional to the total index (71). Collecting the factors $\frac{1}{2} \cdot \frac{1}{12} \cdot \frac{1}{(4\pi)^2}$ and the index 4 reproduces (72); the full computation, including normalization of $\text{Tr}(\mathcal{F}\mathcal{F})$ to the Spin(10) vector, is deferred to ??.

The renormalization-group running of g^2 — gauge *asymptotic freedom*, with $\beta_{g^2} = 2g^2 + (b_0/48\pi^2)g^4$, $b_0 = 26$ for one generation and $b_0 = 12$ for three (Theorem 13.7) — follows from this same induced action and is treated alongside the gravitational flow; there is no interacting gauge fixed point.

9.3 H is unbroken at the vacuum

Theorem 9.4 (Unbroken gauge group; THEOREM). *At the vacuum $\langle \Phi \rangle \in \mathcal{M}$ the stabilizer is exactly $H = \text{Spin}(10) \times \text{U}(1)$. Consequently none of the 46 gauge bosons of $\mathbf{45}_0 \oplus \mathbf{1}_0$ acquires a mass: the gauge sector remains massless and H is unbroken.*

Proof. By construction $\mathcal{M} = \text{E}_6/H$ is the orbit of a rank-one primitive idempotent, whose isotropy group is H (Theorem 6.7). A vector boson associated with a generator $T \in \mathfrak{e}_6$ becomes massive only if T moves the vacuum, i.e. if $T \in \mathfrak{m}$; the would-be Goldstone–Higgs mechanism eats a coset scalar for each broken generator. The unbroken generators $\mathfrak{h}$ annihilate $\langle \Phi \rangle$ and the corresponding 46 vectors stay massless. There is no Higgsing of $\mathfrak{h}$: the 32 coset Goldstones (“clock fields”) are not in $\mathfrak{h}$, and the scalar potential is H -invariant, so the one-loop scalar mass matrix is H -invariant and commutes with the $\mathfrak{h}$ -action. Schur’s lemma on the irreducible $\mathfrak{m}$ gives a single common scalar mass with no $\mathfrak{h}$ admixture (subsection 11.1), preserving H .

Remark 9.5 (Ward identity; THEOREM). Masslessness is protected to all orders by the H Ward identity. With the BRST-gauge-fixed effective action Γ , current conservation $D_\mu \langle J_a^\mu \rangle = 0$ for a ranging over $\mathfrak{h}$ forces the transverse vacuum polarization $\Pi_{ab}(p)$ to vanish at $p^2 = 0$; no H -invariant scalar condensate in $\mathfrak{m}$ exists to generate a mass term (there is no Spin(10) singlet in $\mathfrak{m}$), so the 46 gauge bosons remain exactly massless. This is consistent with the count $32 = 28$ massive scalars + 4 gauge/soldering: the 4 frame modes are eaten by diffeomorphism + local Lorentz, not by H (??).

9.4 Green–Schwarz anomaly cancellation

The chiral matter is one family in the $\text{SO}(10)$ generation, the $\mathbf{16}_{+1}$ together with the remaining components of the fundamental,

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1} \quad (\text{under } H), \quad (73)$$

with $\text{U}(1)$ charges as shown. The cubic norm N on $J_3(\mathbb{O})$ furnishes a closed, E_6 -invariant 3-form Ω_3 on $\mathcal{M}$; let B be a two-form on X with $dB = \pi_-^* \Omega_3$. The Chern–Simons transgression of (67) produces the Green–Schwarz coupling

$$S_{GS} = \int_X B \wedge \frac{\text{Tr}(\mathcal{F} \wedge \mathcal{F})}{8\pi^2}, \quad dB = \frac{1}{2\pi} F_1, \quad (74)$$

where F_1 is the $\text{U}(1)$ field strength.

Theorem 9.6 (Anomaly cancellation; THEOREM). *The combined gauge variation $\delta_H(\Gamma_{1L} + S_{YM} + S_{GS})$ vanishes. The only nonvanishing one-loop gauge anomaly is the mixed $\text{U}(1)$ – $\text{Spin}(10)^2$ anomaly, with coefficient -1 , and it is cancelled by the inflow from S_{GS} ; the cubic $\text{U}(1)^3$ anomaly vanishes identically.*

Proof sketch. For a chiral Weyl fermion in rep R of $\text{Spin}(10)$ with $\text{U}(1)$ charge q , the relevant six-form is $I_6(R, q) = q A_R \frac{F_1}{2\pi} \wedge \frac{\text{Tr}(\mathcal{F} \wedge \mathcal{F})}{8\pi^2}$, with $A_{\mathbf{16}} = A_{\mathbf{10}} = 1$, $A_{\mathbf{1}} = 0$. Summing over (73),

$$(+1)A_{\mathbf{16}} + (-2)A_{\mathbf{10}} + (+4)A_{\mathbf{1}} = 1 - 2 + 0 = -1, \quad (75)$$

giving $I_6^{\text{tot}} = -\frac{F_1}{2\pi} \wedge \frac{\text{Tr}(\mathcal{F} \wedge \mathcal{F})}{8\pi^2}$. The cubic abelian anomaly cancels,

$$16 \cdot (+1)^3 + 10 \cdot (-2)^3 + 1 \cdot (+4)^3 = 16 - 80 + 64 = 0. \quad (76)$$

Under an abelian gauge shift $B \mapsto B + \Lambda_1/2\pi$ the variation of (74) is $\delta_\Lambda S_{GS} = \int_X \frac{\Lambda_1}{2\pi} \wedge \frac{\text{Tr}(\mathcal{F} \wedge \mathcal{F})}{8\pi^2} = -\int_X I_6^{\text{tot}} = -\delta_\Lambda \Gamma_{1L}$, exactly cancelling the mixed anomaly. The descent equations and the full arithmetic appear in section E. $\square$

Theorem 9.7 (Emergent gauge sector; THEOREM). *The functional $S_{\text{tot}} = S_{\text{eff}} + S_{YM} + S_{GS}$ describes a local, unitary, anomaly-free $\text{Spin}(10) \times \text{U}(1)$ Yang–Mills theory coupled to the clock-field σ -model, with the dynamically induced coupling (72). Locality and unitarity follow from those of the scalar σ -model (??) together with compactness of H and standard BRST gauge fixing; anomaly freedom is Theorem 9.6.*

Status. The composite connection (Theorem 9.1), the unbroken- H statement (Theorem 9.4), the corrected total index (Theorem 9.2), and the anomaly cancellation (Theorem 9.6) are *theorems* given the vacuum $\mathcal{M} = \text{EIII}$ and the compact real form. The chiral matter assignment (73) with no mirror partners is *postulate* P4. The detailed Yang–Mills coefficient is in ?? and the full anomaly arithmetic in section E.

10 Radiative Masses and the Physical Spectrum

The vacuum manifold of UCFT is the rank-one primitive-idempotent orbit

$$\mathcal{M} = E_6/(\text{Spin}(10) \times U(1)) = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32,$$

established as the global minimum of the E_6 -invariant potential in ?? (vacuum-selection theorem; cf. the locked foundation, Hardened Result A). The 32 coset Goldstones — the *clock fields* — span the tangent module

$$\mathfrak{m} = T_{X_0}\mathcal{M} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad \dim_{\mathbb{R}} \mathfrak{m} = 32, \quad (77)$$

under the unbroken $H = \text{Spin}(10) \times U(1)$ (?). This section determines their physical mass spectrum. The central structural fact, used throughout, is that (77) is a *single* real-irreducible H -module: the $U(1)$ charge forbids the symmetric $\mathbf{16} \times \mathbf{16}$ and $\overline{\mathbf{16}} \times \overline{\mathbf{16}}$ invariants, leaving only the charge-zero cross pairing, so the commutant of H on $\mathfrak{m}$ is $\mathbb{C}$ (real-irreducible of complex type) and $\mathfrak{m}$ contains *no* $\text{Spin}(10)$ singlet.

We work on the compact real form (equivalently $E_{6(-14)}$, whose maximal compact subgroup is precisely H), so the coset metric $K|_{\mathfrak{m}}$ is positive-definite and the gauge sector is unitary. Throughout, $[f] = \text{mass}$ is the decay constant, μ the breaking/mass scale, and Λ the UV cutoff; the signature is $(-, +, +, +)$.

10.1 Tree level: thirty-two exact Goldstones

Theorem 10.1 (Tree-level Goldstone degeneracy). *At any vacuum $X_0 \in \mathcal{M}$ the Hessian of the E_6 -invariant tree potential V vanishes identically on the coset tangent space,*

$$\text{Hess}_{X_0} V|_{\mathfrak{m}} = 0, \quad (78)$$

so that all 32 clock fields are exact Goldstone bosons.

Proof. V is constant along the E_6 -orbit $\mathcal{M}$ of the minimizer (?), so its directional derivatives along the broken generators, which span $\mathfrak{m}$, vanish to all orders in the flat directions; (78) is Goldstone's theorem applied to the spontaneously broken $E_6 \rightarrow H$. The unbroken directions $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ produce no scalars. $\square$

The degeneracy (78) is exact only at tree level; the physical spectrum is fixed by the one-loop effective potential.

10.2 One loop: a single common radiative mass

The Coleman–Weinberg effective potential V_{eff} is built from the H -invariant fluctuation operator about X_0 and is therefore itself H -invariant. Its Hessian on $\mathfrak{m}$ is consequently an H -intertwiner. We exploit irreducibility *positively*: it forces a single eigenvalue, rather than splitting off any singlet subspace.

Theorem 10.2 (Common radiative mass via Schur). *The one-loop Hessian on the coset is a multiple of the identity,*

$$\text{Hess}_{X_0} V_{\text{eff}}|_{\mathfrak{m}} = m_{\mathfrak{m}}^2 \mathbf{1}_{32}, \quad m_{\mathfrak{m}}^2 \in \mathbb{R}_{\geq 0}, \quad (79)$$

i.e. there is one common radiative mass on all 32 coset directions.

Proof. $\text{Hess}_{X_0} V_{\text{eff}}|_{\mathfrak{m}}$ is a symmetric H -equivariant endomorphism of $\mathfrak{m}$. By (77), $\mathfrak{m}$ is a single real-irreducible H -module of complex type, with commutant $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$. A symmetric (real, self-adjoint with respect to the positive-definite $K|_{\mathfrak{m}}$) intertwiner must lie in the real part of the commutant, hence is a real scalar multiple of the identity; this is Schur's lemma for the complex-type case restricted to self-adjoint operators. Thus the Hessian equals $m_{\mathfrak{m}}^2 \mathbf{1}_{32}$. Stability of the vacuum (??) and positivity of $K|_{\mathfrak{m}}$ give $m_{\mathfrak{m}}^2 \geq 0$. $\square$

Remark 10.3 (No internal singlet subspace). Because $\mathfrak{m}$ carries *no* $\text{Spin}(10)$ singlet, there is no H -invariant four-dimensional subspace of $\mathfrak{m}$ on which the Hessian could independently vanish, and in particular no subspace spanned by gradients of independent invariants $\nabla I_2, \nabla I_3$. Any decomposition of $\mathfrak{m}$ as $2 \times \mathbf{10} \oplus \mathbf{4}$, $\mathbf{10} \oplus \overline{\mathbf{10}} \oplus \mathbf{4}$, or with any singlet factor is representation-theoretically false. The earlier “ $4 \oplus 28$ ” Hessian splitting of $\mathfrak{m}$ into an internally massless quartet and a massive 28-plet is therefore *not* the mechanism; the four light directions are gauge artefacts, treated next.

10.3 The four frame directions are gauge, not Hessian zeros

Four of the 32 clock fields are the soldering/frame directions identified by Postulate P3 (??): the soldering map σ aligns four clock-field frame directions e^a_μ ($a = 0, 1, 2, 3$) with a tangent 4-plane of the emergent spacetime M^4 , providing the composite vierbein $e^a_\mu(x) = \langle \tau^a, \theta_{\mathfrak{m}}(\partial_\mu) \rangle$. These four directions are not lifted by the Hessian; they are removed by *gauge* symmetry.

Postulate 10.4 (Frame gauge protection, P3). The four soldered frame directions are gauge degrees of freedom of the emergent geometry: they are the modes eaten under world-volume diffeomorphisms and local Lorentz transformations $\text{SO}(1, 3)$. They satisfy the Ward identity

$$\nabla_\mu \left(\frac{\delta \Gamma}{\delta e^a_\mu} \right) = 0, \quad (80)$$

which removes them from the physical spectrum (cf. ??).

Theorem 10.5 (Physical count). *Of the 32 real clock fields, four are eaten as gauge/soldering modes under diffeomorphism + local Lorentz, and the remaining 28 are physical massive scalars carrying the single common mass of Theorem 10.2:*

$$\underbrace{32}_{\dim_{\mathbb{R}} \mathfrak{m}} = \underbrace{28}_{\text{physical massive scalars}} + \underbrace{4}_{\text{gauge / soldering (eaten)}}. \quad (81)$$

The unbroken $H = \text{Spin}(10) \times U(1)$ is not affected: its $46 = \dim(\mathbf{45}) + 1$ gauge bosons remain massless.

Proof. By Theorem 10.2 the one-loop Hessian is $m_{\mathfrak{m}}^2 \mathbf{1}_{32}$, so *no* internal direction is a Hessian zero (Theorem 10.3). The reduction of the physical count is purely gauge-theoretic: by Postulate P3 (Theorem 10.4, ??) the four soldered frame directions are gauge artefacts removed by diffeomorphism and local Lorentz invariance through (80), leaving $32 - 4 = 28$ propagating physical scalars. Since the vacuum preserves H , no H gauge boson is eaten and the 46 adjoint gauge bosons stay massless. $\square$

Remark 10.6 (Logical separation). It is essential to distinguish the two reductions. Schur’s lemma (Theorem 10.2) uses *irreducibility* to fix ONE common mass on all 32 directions; it never produces a massless internal subspace. The 28 vs. 4 split (Theorem 10.5) is a *gauge* statement (Theorem 10.4), not a statement about the internal Hessian. The two are independent: the same 28 massive scalars all share the one common mass, and the four removed modes were never internal zeros.

10.4 Canonical normalization of the mass

Theorem 10.7 (Canonical physical mass). *With $[f] = \text{mass}$, the dimension-four Hessian eigenvalue on the coset is $f^2 \mu^2$, and the canonically normalized physical mass of each of the 28 massive scalars is*

$$m^2 = \mu^2. \quad (82)$$

Proof. The σ -model kinetic term carries the dimensionful prefactor f^2 (section E), so the clock fields π^α are not canonically normalized; the canonical fields are $\phi^\alpha = f \pi^\alpha$. The E_6 -invariant effective potential contributes a Hessian whose dimension-four eigenvalue on $\mathfrak{m}$ is $m_{\mathfrak{m}}^2 = f^2 \mu^2$ in terms of the non-canonical fields. Re-expressing in the canonical field ϕ^α divides the mass-squared by f^2 , giving $m^2 = f^2 \mu^2 / f^2 = \mu^2$. The spurious extra factor of f^2 is thereby removed. $\square$

The residual symmetry of the full 27-dimensional scalar block is H , not E_6 , so off the coset the mass matrix is block-diagonal along the fundamental branching $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$,

$$\mathcal{M}_{27} = \text{diag}(\mu_1 \mathbf{1}_1, \mu_{10} \mathbf{1}_{10}, \mu_{16} \mathbf{1}_{16}), \quad (83)$$

with one common eigenvalue per H -irreducible factor (Schur, applied factorwise).

10.5 Consequence: the loop sum uses twenty-eight scalars

The physical count (81) and the canonical mass (82) fix the input to every downstream loop sum: *only* the 28 massive scalars, each with $m^2 = \mu^2$ (i.e. dimension-four eigenvalue $f^2 \mu^2$, signature factor $s = +1$), enter. In particular the Sakharov-induced Planck mass (??; cf. Hardened Result C) is built from precisely these 28 scalars,

$$M_{\text{Pl}}^2 = \frac{1}{16\pi^2} \frac{1}{6} \sum_{\text{massive } i} s_i m_i^2 \log \frac{\Lambda^2}{m_i^2} = \frac{28}{6} \frac{f^2 \mu^2 \log(\Lambda^2/\mu^2)}{16\pi^2} > 0, \quad (84)$$

with prefactor $28/6$ (not 32), keeping the conformal coefficient $1/6$, the scale μ^2 , and the log. Massless fields do not produce the $m^2 \log$ term, since the heat-kernel a_2 (R -coefficient) is mass-independent; hence the four eaten frame modes do not contribute. Every factor in (84) is positive for $\Lambda > \mu$, so the induced Newton constant has the correct sign.

Remark 10.8 (Status). Theorem 10.1, Theorem 10.2, Theorem 10.5, and Theorem 10.7 are *theorems* given the locked branchings, the irreducibility of (77), and Postulate P3. The reduction $32 \rightarrow 28$ rests on the soldering postulate P3 (added structure), labelled as Theorem 10.4. The downstream use of 28 in (84) and in the gauge β -function scalar count $n_S = 28$ is fixed once and for all by Theorem 10.5.

Part IV

Induced Gravity, Renormalization, and Standard Model Descent

11 Sakharov-Induced Gravitation

Gravity in UCFT is not a fundamental input. It is the Sakharov mechanism [15]: integrating out the quantum fluctuations of the matter sector over the curved emergent background M^4 (the soldered 4-manifold of Postulate P3, ??) generates an Einstein–Hilbert term whose coefficient is the induced Planck mass. The curvature dependence is carried entirely by the second Seeley–DeWitt heat-kernel coefficient a_2 , and the only fields that feed the running R -coefficient are the *massive* ones. In UCFT the relevant massive multiplet is the set of 28 clock fields of common mass $m^2 = f^2 \mu^2$ (subsection 11.1), so the induced Newton constant is fixed, positive, and computable.

We work throughout in signature $(-, +, +, +)$, with $[f] = \text{mass}$ and the dimensionless couplings $\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\text{cc}}$.

This section establishes three things: (i) the induced Planck mass $M_{\text{Pl}}^2 = \frac{28}{6} f^2 \mu^2 \log(\Lambda^2/\mu^2)/(16\pi^2) > 0$ (*theorem*, given the spectrum and P3); (ii) that the cosmological constant is *not* solved — the a_0 coefficient forces an uncanceled $\sim \Lambda^4$ vacuum energy, handled only by a tuned counterterm Λ_{cc} ; and (iii) the place of the gravitational coupling κ in the renormalization-group flow, whose ultraviolet-attractive fixed point $\kappa_\star = 96\pi^2/5$ is conjectural. The full heat-kernel coefficient table and the proper-time computation are deferred to [section E](#).

11.1 The massive multiplet and the ghost operator

Recall ([section 6](#), Theorem T4) that the vacuum manifold is the rank-1 primitive-idempotent orbit

$$\mathcal{M} = \frac{E_6}{\text{Spin}(10) \times U(1)} = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32,$$

with coset tangent module $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, a single real-irreducible $H = \text{Spin}(10) \times U(1)$ representation of complex type (commutant $\mathbb{C}$), containing *no* $\text{Spin}(10)$ singlet ([section 7](#)). At tree level $\text{Hess } V_{\text{tree}}|_{\mathfrak{m}} = 0$: all 32 clock fields are exact Goldstones. The radiative spectrum is fixed by symmetry alone.

Theorem 11.1 (Common radiative mass; $28 + 4$ split). *The one-loop Coleman–Weinberg effective potential is H -invariant. Since $\mathfrak{m}$ is a single real-irreducible H -module, Schur’s lemma forces the radiative Hessian to be a multiple of the identity on all 32 directions,*

$$\text{Hess } V_{1\text{-loop}}|_{\mathfrak{m}} = f^2 \mu^2 \mathbf{1}_{32},$$

i.e. a single common eigenvalue. Of these, four directions are the frame/soldering modes that are gauge degrees of freedom (world-volume diffeomorphism + local Lorentz, Ward identity $\nabla_\mu(\delta\Gamma/\delta e_\mu^a) = 0$); they are eaten by the vierbein and local-Lorentz connection and carry no physical mass. The remaining

$$32 = 28 \text{ physical massive scalars (single common mass)} + 4 \text{ gauge/soldering (eaten)}$$

are the 28 physical clock fields. With $[f] = \text{mass}$, the dim-4 Hessian eigenvalue is $f^2 \mu^2$; the canonically normalized physical mass is

$$m^2 = \mu^2.$$

The unbroken H leaves the $46 = \dim(\mathbf{45}) + 1$ gauge bosons massless.

Proof. H -invariance of Γ and Schur’s lemma on the irreducible $\mathfrak{m}$ give a scalar Hessian; this is the corrected version of the spectrum analysis ([section E](#)). The 4 frame modes are not Hessian zeros but gauge-protected directions, as established by the soldering Ward identity of ?? (Theorem 3.1 there). Canonical normalization divides the dim-4 eigenvalue $f^2 \mu^2$ by f^2 . $\square$

Remark 11.2 (No singlet subspace; against the legacy construction). There is no four-dimensional H -invariant subspace of $\mathfrak{m}$ built from $\nabla I_2, \nabla I_3$ singlets: $\mathfrak{m}$ is irreducible and has no $\text{Spin}(10)$ singlet, so a singlet-projector construction is algebraically impossible. The mass is one common eigenvalue from Schur, and the four light modes are gauge-protected, not radiative zeros.

The Euclidean fluctuation operator for the massive multiplet, used in the heat-kernel expansion, is

$$\Delta_m = -D^2 + m^2, \quad m^2 = f^2 \mu^2 \quad (\text{Hessian normalization}), \quad (85)$$

with D the H -covariant derivative on $\mathbf{m}$. The Faddeev–Popov ghosts that accompany the local $\text{Spin}(10) \times U(1)$ gauge fixing are Lorentz scalars in the adjoint of H , and their fluctuation operator is the *adjoint covariant Laplacian*,

$$\Delta_{\text{ghost}} = -(D_{\text{adj}})^2, \quad D_{\text{adj}} = d + [A_{\text{adj}}, \cdot], \quad (86)$$

acting on $\text{adj}(H)$. The ghosts are massless and contribute to the subtracted quadratic divergence but, being massless, do *not* contribute to the curvature (a_2) running — see [Theorem 11.3](#) and [section E](#); no quoted heat-kernel number changes.

11.2 Induced Planck mass from the a_2 coefficient

We evaluate $\frac{1}{2} \text{Tr} \log \Delta_m$ in the proper-time (Schwinger) representation on the slowly varying curved background M^4 . The heat-kernel expansion of a real scalar of mass m is

$$\text{Tr} e^{-s\Delta_m} = \frac{1}{(4\pi s)^2} e^{-sm^2} \int_{M^4} \sqrt{-g} \left[a_0 + a_1 s + a_2 s^2 + \cdots \right], \quad (87)$$

with the Seeley–DeWitt coefficients ([section E](#))

$$a_0 = 1, \quad a_1 = \frac{1}{6} R, \quad a_2 = \frac{1}{180} (R_{\mu\nu\rho\sigma}^2 - R_{\mu\nu}^2) + \frac{1}{72} R^2 + \frac{1}{6} \square \left(\frac{R}{5} \right) + \cdots \quad (88)$$

The Einstein–Hilbert term comes from the curvature-linear coefficient $a_1 = \frac{1}{6} R$; its coefficient in the effective action is the $n = 1$ proper-time moment. Because the a_n are *mass-independent* geometric invariants, the entire mass dependence sits in the e^{-sm^2} factor, and the curvature-linear term integrates to a logarithm cut off at the scale μ . Crucially this is finite and non-zero *only* for $m \neq 0$: a massless field gives $e^{-sm^2} = 1$ and the R -coefficient reduces to a pure (subtracted) quadratic divergence $\sim \Lambda^2$ with no $m^2 \log$ running. Massless fields (Goldstones, gauge bosons, ghosts) therefore do not contribute to M_{Pl}^2 .

Theorem 11.3 (Induced Planck mass; Sakharov). *Given the soldered background of Postulate P3 (??) and the 28 massive clock fields of [Theorem 11.1](#), the one-loop effective action contains an Einstein–Hilbert term*

$$\Gamma^{(1)} \supset \frac{M_{\text{Pl}}^2}{2} \int_{M^4} \sqrt{-g} R, \quad (89)$$

with induced Planck mass

$$M_{\text{Pl}}^2 = \frac{1}{16\pi^2} \cdot \frac{1}{6} \sum_{\text{massive } i} s_i m_i^2 \log \frac{\Lambda^2}{m_i^2} = \frac{28}{6} \frac{f^2 \mu^2 \log(\Lambda^2/\mu^2)}{16\pi^2} > 0, \quad (90)$$

where $s_i = +1$ for scalars, Λ is the ultraviolet cutoff, and the 28 scalars all carry $m^2 = f^2 \mu^2$.

Proof. The curvature-linear term in $\frac{1}{2} \text{Tr} \log \Delta_m$ is, by (87)–(88), the $n = 1$ proper-time moment of $a_1 = \frac{1}{6} R$. Writing the proper-time integral with the cutoff Λ in the lower limit and the mass scale in the renormalization condition, the (a priori 0/0-singular) $n = 1$ term evaluates explicitly to

$$[n = 1 \text{ term}] = a_1 \cdot \frac{1}{2} \log \frac{\Lambda^2}{\mu^2} = \frac{1}{6} R \cdot \frac{1}{2} \log \frac{\Lambda^2}{\mu^2}, \quad (91)$$

per real scalar of mass $m^2 = \mu^2$ in canonical normalization (the $\frac{1}{6}$ is the conformal Seeley–DeWitt coefficient). Multiplying by the loop measure $1/(16\pi^2)$ and summing $s_i = +1$ over the 28 massive clock fields, each with $m_i^2 = f^2\mu^2$, gives the prefactor $28/6$ and the common $\log(\Lambda^2/\mu^2)$, hence (90). The detailed proper-time bookkeeping is in [section E](#). $\square$

Proposition 11.4 (Positivity of the induced Newton constant). $M_{\text{Pl}}^2 > 0$ for $\Lambda > \mu$.

Proof. Every factor in (90) is strictly positive: $\frac{28}{6} = 4.6667 > 0$, $1/(16\pi^2) = 0.0063326 > 0$, $f^2\mu^2 > 0$ (since $[f] = \text{mass}$ and μ real), and $\log(\Lambda^2/\mu^2) > 0$ for $\Lambda > \mu$. The 28 scalars ($s = +1$) dominate any massive-fermion negative ($s = -1$) contribution, so the induced Newton constant $G_N = 1/(8\pi M_{\text{Pl}}^2)$ has the correct attractive sign. A worked representative ($f = \mu = 1$ in μ -units, $\Lambda/\mu = 100$, so $\log(\Lambda^2/\mu^2) = \log 10^4 = 9.2103$) gives $M_{\text{Pl}}^2 = 4.6667 \times 9.2103 \times 0.0063326 = 0.27218 \mu^2 > 0$. $\square$

Remark 11.5 (Running of M_{Pl}^2 from the same expression). The static value and the running are derived from the *one* expression (90), so there is no 32-versus-28 or normalization mismatch. Differentiating with respect to the renormalization scale,

$$\beta_{M_{\text{Pl}}^2} = \mu \frac{dM_{\text{Pl}}^2}{d\mu} = -\frac{28}{6} \frac{2f^2\mu^2}{16\pi^2} + \frac{28}{6} \frac{2f^2\mu^2 \log(\Lambda^2/\mu^2)}{16\pi^2}, \quad (92)$$

so $M_{\text{Pl}}^2(\mu)$ grows (logarithmically dominated) and remains positive throughout the regime $\mu \ll \Lambda$. The prefactor $28/6$ is the same number that appears statically.

Remark 11.6 (Why $28/6$ and not 32). The legacy value used $N_s = 32$ Goldstones and dropped the conformal $\frac{1}{6}$. That is wrong on two counts: only the 28 *physical massive* clock fields run the R -coefficient (the 4 frame modes are eaten and massless; massless fields give no $m^2 \log$), and the per-scalar heat-kernel coefficient is $a_1 = \frac{1}{6}R$, so the $\frac{1}{6}$ and the μ^2 must be kept. The correct, manuscript-wide prefactor is $28/6$.

11.3 Cosmological constant: inherited, not solved

The curvature-independent coefficient $a_0 = 1$ in (87) produces a vacuum-energy (cosmological-constant) term whose proper-time $n = -1$ moment is quartically divergent,

$$\Gamma^{(1)} \supset \int_{M^4} \sqrt{-g} \rho_{\text{vac}}, \quad \rho_{\text{vac}} \sim \frac{1}{16\pi^2} \sum_i s_i \Lambda^4 \sim \frac{28}{32\pi^2} \Lambda^4, \quad (93)$$

with *no* supersymmetric cancellation in UCFT. This $\sim \Lambda^4$ vacuum energy is not removed by any symmetry of the construction.

Postulate 11.7 (Cosmological-constant counterterm). A counterterm Λ_{cc} is added to cancel (93) down to the observed value. In the renormalization-group language Λ_{cc} is a tuned *relevant* coupling (canonical eigenvalue -4 of the stability matrix, [subsection 11.4](#)).

Remark 11.8 (Status). UCFT *inherits*, and does *not* solve, the cosmological-constant problem: the magnitude of Λ_{cc} is fixed by hand, exactly as in the Standard Model coupled to gravity. No claim of naturalness is made.

11.4 Place in the renormalization-group flow

The gravitational coupling κ (dimensionless Newton coupling) joins the gauge–Yukawa flow. The gauge sector is asymptotically *free* (Gaussian fixed point $\hat{g}_\star^2 = 0$; full one- and two-loop analysis in [section 12](#)); here we record only the gravitational beta function and the combined fixed point, with the truncation caveat below.

Conjecture 11.9 (Gravitational fixed point). *The single-scalar-loop gravitational beta function is*

$$\beta_\kappa = 2\kappa - \frac{5}{48\pi^2} \kappa^2, \quad (94)$$

with a nontrivial ultraviolet-attractive fixed point

$$\kappa_\star = \frac{96\pi^2}{5} = 189.4964, \quad \left. \frac{d\beta_\kappa}{d\kappa} \right|_{\kappa_\star} = 2 - \frac{2 \cdot 5}{48\pi^2} \kappa_\star = -2. \quad (95)$$

The negative sign in (94) is essential: κ flows away from 0 toward $\kappa_\star$ (gravity does not decouple), consistent with the anomalous dimension $\eta_M = -(5/48\pi^2)\kappa$.

Remark 11.10 (Caveat). $\kappa_\star = 96\pi^2/5 \approx 189 \gg 1$ lies outside the controlled weak-coupling regime, so this fixed point is *indicative*, not rigorous. We label the gravitational asymptotic-safety statement a conjecture for this reason.

Collecting the couplings $(\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\text{cc}})$ and using the asymptotic-safety convention $\theta_i = -\text{eig}(M)$ (UV-attractive $\iff \text{eig}(M) > 0$; $\#\text{relevant} = \#\{\text{eig}(M) < 0\}$), the representative fixed point self-consistent with the gauge asymptotic freedom of [section 12](#) and with (95) is

$$(\xi_\star, \hat{g}_\star^2, \hat{y}_\star^2, \kappa_\star, \Lambda_{\text{cc},\star}) = (0.05, 0, 0, 96\pi^2/5, 0.01). \quad (96)$$

At this point $\hat{g}_\star^2 = \hat{y}_\star^2 = 0$ are Gaussian (asymptotic freedom), so the stability matrix $M = \partial\beta_i/\partial g_j$ is effectively triangular with diagonal entries

$$\frac{\partial\beta_\xi}{\partial\xi} = +2, \quad \frac{\partial\beta_{\hat{g}^2}}{\partial\hat{g}^2} = +2, \quad \frac{\partial\beta_{\hat{y}^2}}{\partial\hat{y}^2} = +2, \quad \frac{\partial\beta_\kappa}{\partial\kappa} = -2, \quad \frac{\partial\beta_{\Lambda_{\text{cc}}}}{\partial\Lambda_{\text{cc}}} = -4, \quad (97)$$

so that

$$\text{eig}(M) = \{+2, +2, +2, -2, -4\}, \quad \theta = -\text{eig}(M) = \{-2, -2, -2, +2, +4\}. \quad (98)$$

The two relevant directions ($\#\{\text{eig}(M) < 0\} = 2$) are κ (eigenvalue -2 , $\theta = +2$, the gravitational fixed point) and Λ_{cc} (eigenvalue -4 , $\theta = +4$, the tuned cosmological-constant counterterm of [Theorem 11.7](#)). The positive central $(\hat{g}^2, \hat{y}^2)$ eigenvalues are the correct signature of asymptotic freedom under this convention. Ultraviolet completeness of UCFT is therefore asymptotic freedom of the gauge–Yukawa sector *plus* the (indicative) gravitational fixed point *plus* irrelevant higher-curvature operators — *not* a non-Gaussian gauge fixed point.

Remark 11.11 (Higher-curvature operators). The a_2 coefficient (88) supplies curvature-squared terms $c_1 R^2 + c_2 R_{\mu\nu} R^{\mu\nu}$ with $|c_{1,2}| \sim 1/(16\pi^2)$, suppressed by μ^2/Λ^2 relative to $M_{\text{Pl}}^2 R$ and hence irrelevant in the regime $R \ll M_{\text{Pl}}^2$. These are the irrelevant operators of the ultraviolet-completeness statement above.

12 Functional Renormalization and Ultraviolet Behaviour

We now track the coset field theory under continuous coarse-graining. Throughout we work in signature $(-, +, +, +)$, with $[f] = \text{mass}$ the decay constant and $\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\text{cc}}$ dimensionless. After introducing the functional renormalization-group (FRG) machinery and the dimensionless couplings we record the gauge, gravitational and scalar β -functions, state the explicit numerical fixed point and its recomputed stability spectrum, and recast the claim of ultraviolet (UV) completeness in a truncation-conditional form. The gauge–Yukawa sector is *asymptotically free*; the gravitational sector exhibits an *asymptotically safe* interacting fixed point that we flag as *indicative*. All loop machinery (Wetterich kernel, threshold integrals, heat-kernel coefficients, the 27×27 scalar block) is deferred to [section E](#); the induced Planck mass is established in [??](#).

12.1 Setup: Wetterich equation and Litim regulator

The scale-dependent effective average action Γ_k obeys the exact Wetterich flow

$$k \partial_k \Gamma_k = \frac{1}{2} \text{STr} \left[\left(\Gamma_k^{(2)} + R_k \right)^{-1} k \partial_k R_k \right], \quad (99)$$

where $\Gamma_k^{(2)}$ is the second functional derivative, STr is the supertrace over all fields (with the Faddeev–Popov ghosts of the diffeomorphism and local-Lorentz gauge symmetries entering as Lorentz scalars in $\text{adj}(H)$, $\Delta_{\text{ghost}} = -(D_{\text{adj}})^2$), and R_k is an infrared cutoff. We adopt Litim’s optimized regulator

$$R_k(p^2) = Z_k (k^2 - p^2) \theta(k^2 - p^2), \quad (100)$$

which renders the threshold integrals algebraic while preserving locality; all UV statements below are regulator independent. The natural dimensionless couplings are

$$\xi = \frac{k^2}{f^2}, \quad \hat{g}^2 = \frac{k^2 g^2}{f^2}, \quad \hat{y}^2 = \frac{k^2 y^2}{f^2}, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2(k)}, \quad (101)$$

together with the tuned vacuum-energy coupling Λ_{cc} . Projecting (99) onto the coefficients of $\sqrt{-g} R$, $\sqrt{-g} \bar{\psi} \gamma^\mu D_\mu \psi$, the gauge invariant $\sqrt{-g} \text{tr} F_{\mu\nu} F^{\mu\nu}$, and the cosmological term $\sqrt{-g}$ yields the β -functions quoted below.

Remark 12.1. We evaluate functional determinants on a finite four-torus $\mathbb{T}^4$ and take the volume to infinity at the end, removing infrared regulators without affecting UV behaviour. The mass spectrum entering the loops is the locked $32 = 28 + 4$ split (??): the 28 massive coset “clock fields” (one common mass $m^2 = \mu^2$, by Schur on the irreducible H -module $\mathfrak{m} = \mathbf{16}_{-3} \oplus \mathbf{\bar{16}}_{+3}$) run in every loop sum, while the 4 frame/soldering modes are gauge protected (eaten by diffeomorphism + local Lorentz) and do not.

12.2 Gauge sector: asymptotic freedom

The residual gauge symmetry on the vacuum manifold is $H = \text{Spin}(10) \times U(1)$ (??); the gauge running is governed by the $\text{SO}(10)$ group theory, *not* by E_6 . The locked Casimirs are

$$C_A(\text{SO}(10)) = 8, \quad T(\mathbf{16}) = 2 \text{ (the spinor is index-2)}, \quad C_2(\mathbf{16}) = \frac{45}{8}. \quad (102)$$

Theorem 12.2 (Gauge β -function and asymptotic freedom). *With $[f] = \text{mass}$ and $\hat{g}^2 = k^2 g^2 / f^2$ of engineering dimension 2, the gauge coupling obeys*

$$\beta_{\hat{g}^2} = 2 \hat{g}^2 + \frac{b_0}{48\pi^2} \hat{g}^4, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} T n_F - \frac{1}{6} T n_S. \quad (103)$$

The canonical $+2\hat{g}^2$ term forces the only UV fixed point to be the Gaussian one, $\hat{g}_*^2 = 0$. The gauge sector is therefore asymptotically free, and there is no interacting gauge fixed point.

Proof. The loop coefficient is the standard one-loop result for a gauge algebra with n_F Weyl fermions and n_S real scalars, evaluated with the SO(10) data (102); the derivation through the Litim threshold integrals is given in section E. The canonical scaling term $+2\hat{g}^2$ follows from the engineering dimension of $\hat{g}^2$. Setting $\beta_{\hat{g}^2} = 0$ gives $\hat{g}_*^2(2 + \frac{b_0}{48\pi^2}\hat{g}_*^2) = 0$; the nonzero root $\hat{g}_*^2 = -96\pi^2/b_0$ is negative for $b_0 > 0$ and hence unphysical, leaving $\hat{g}_*^2 = 0$ as the unique admissible fixed point. Linearizing, $\partial_{\hat{g}^2}\beta_{\hat{g}^2}|_0 = +2 > 0$, so the Gaussian point is UV attractive in this direction: $\hat{g}^2 \rightarrow 0$ logarithmically as $k \rightarrow \infty$, i.e. asymptotic freedom. $\square$

Remark 12.3 (The integer b_0). For *one* SO(10) generation ($n_F = 1$, with the scalar Higgs/Yukawa content fixed by $n_S = 2$),

$$b_0 = \frac{11}{3}(8) - \frac{4}{3}(2)(1) - \frac{1}{6}(2)(2) = \frac{88}{3} - \frac{8}{3} - \frac{2}{3} = \frac{78}{3} = 26. \quad (104)$$

This is the canonical anchor at the matching scale with minimal GUT scalars ($n_S = 2$). For the physical *three*-generation theory, $n_F = 3$ (three chiral **16** representations) and $n_S = 28$ (the massive coset scalars used in all loop sums), and direct evaluation gives

$$b_0(3 \text{ gen}) = \frac{11}{3}(8) - \frac{4}{3}(2)(3) - \frac{1}{6}(2)(28) = \frac{88 - 24 - 28}{3} = 12. \quad (105)$$

We use $b_0 = 12$ for three-generation running everywhere downstream. The $+2\hat{g}^2$ scaling term, not the sign of b_0 , governs the Gaussian UV fixed point $\hat{g}_*^2 = 0$, so the gauge sector remains asymptotically free.

12.3 Gravitational sector: asymptotic safety (indicative)

Integrating out the 28 massive clock fields generates the Sakharov flow of the induced Planck mass (??); the *same* expression $M_{\text{Pl}}^2 = (28/6)f^2\mu^2 \log(\Lambda^2/\mu^2)/(16\pi^2) > 0$ controls both the static value and the running. In terms of $\kappa = k^2/M_{\text{Pl}}^2(k)$ the result is the following.

Conjecture 12.4 (Gravitational fixed point; INDICATIVE). *The gravitational coupling runs as*

$$\beta_\kappa = 2\kappa - \frac{5}{48\pi^2}\kappa^2, \quad (106)$$

matching the anomalous dimension $\eta_M = -(5/48\pi^2)\kappa$. Besides the Gaussian point it possesses a nontrivial UV-attractive fixed point

$$\kappa_* = \frac{96\pi^2}{5} = 189.4964, \quad \left. \frac{\partial\beta_\kappa}{\partial\kappa} \right|_{\kappa_*} = -2. \quad (107)$$

The minus sign in (106) is essential: κ flows away from 0 toward κ_ in the UV (gravity does not decouple). Because $\kappa_* \approx 189 \gg 1$ lies outside the controlled weak-coupling regime, this single-scalar-loop fixed point is INDICATIVE, not rigorous.*

Computation. Setting $\beta_\kappa = 0$ with $\kappa \neq 0$ gives $2 = (5/48\pi^2)\kappa_*$, hence $\kappa_* = 96\pi^2/5$. Differentiating, $\partial_\kappa\beta_\kappa = 2 - 2(5/48\pi^2)\kappa$; at κ_* this is $2 - 2(5/48\pi^2)(96\pi^2/5) = 2 - 4 = -2$. The sign of the slope being negative means κ_* is UV attractive along the κ direction. The status is downgraded to indicative because $\kappa_* \gg 1$. $\square$

Remark 12.5 (AS sign convention). We use $\theta_i = -\text{eig}(M)$ for the critical exponents, where $M = \partial\beta_i/\partial g_j$ is the stability matrix. A direction is UV attractive (relevant) iff $\text{eig}(M) < 0$, i.e. $\theta > 0$, so the number of relevant directions equals $\#\{\text{eig}(M) < 0\}$. By this convention the negative slope -2 at κ_* in (107) corresponds to a *relevant* eigendirection with $\theta_\kappa = +2$.

12.4 Scalar block and Yukawa direction

The classically marginal coset-scalar interactions live in a 27×27 mass/coupling block whose multiplicities follow the fundamental branching $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$ under the *residual* symmetry H (not E_6):

$$M = \text{diag}(\mu_1 \mathbb{1}_1, \mu_{10} \mathbb{1}_{10}, \mu_{16} \mathbb{1}_{16}), \quad (108)$$

i.e. block multiplicities $1 + 10 + 16$. The three computed block eigenvalues $\mu_1, \mu_{10}, \mu_{16}$ are negative (section E), so the scalar block introduces *no* additional relevant directions at the fixed point. The Yukawa coupling $\hat{y}^2$, defined through the real $\mathbf{10}_H/\mathbf{126}_H$ Higgs sector of ?? (Postulate P6; the naive $\mathbf{16}$ -scalar Yukawa is forbidden, having no H -singlet), runs with a β -function of the schematic form $\beta_{\hat{y}^2} = 2\hat{y}^2 + (\text{loop})\hat{y}^4 + (\text{loop})\hat{g}^2\hat{y}^2$, whose only admissible fixed point at $\hat{g}_*^2 = 0$ is again Gaussian, $\hat{y}_*^2 = 0$ (Yukawa asymptotic freedom).

12.5 Explicit fixed point and stability spectrum

Collecting the directions $(\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\text{cc}})$, the canonical scaling terms and the loop results above give

$$\begin{aligned} \beta_\xi &= 2\xi, & \beta_{\hat{g}^2} &= 2\hat{g}^2 + \frac{b_0}{48\pi^2}\hat{g}^4, & \beta_{\hat{y}^2} &= 2\hat{y}^2 + \dots, \\ \beta_\kappa &= 2\kappa - \frac{5}{48\pi^2}\kappa^2, & \beta_{\Lambda_{\text{cc}}} &= -4\Lambda_{\text{cc}} + \dots \end{aligned} \quad (109)$$

Theorem 12.6 (Numerical fixed point and eigenvalues). *The system (109) admits the fixed point*

$$(\xi_*, \hat{g}_*^2, \hat{y}_*^2, \kappa_*, \Lambda_{\text{cc},*}) = (0.05, 0, 0, \frac{96\pi^2}{5} = 189.4964, 0.01), \quad (110)$$

where ξ_* and $\Lambda_{\text{cc},*}$ are representative tuned values. At this point the stability matrix $M = \partial\beta_i/\partial g_j$ is effectively triangular (every loop correction to a β -function is proportional to a coupling that vanishes at $\hat{g}_*^2 = \hat{y}_*^2 = 0$), with diagonal

$$\partial_\xi\beta_\xi = +2, \quad \partial_{\hat{g}^2}\beta_{\hat{g}^2} = +2, \quad \partial_{\hat{y}^2}\beta_{\hat{y}^2} = +2, \quad \partial_\kappa\beta_\kappa = -2, \quad \partial_{\Lambda_{\text{cc}}}\beta_{\Lambda_{\text{cc}}} = -4. \quad (111)$$

Hence

$$\text{eig}(M) = \{+2, +2, +2, -2, -4\}, \quad \theta = -\text{eig}(M) = \{-2, -2, -2, +2, +4\}, \quad (112)$$

so there are exactly two relevant directions, κ ($\theta = +2$) and Λ_{cc} ($\theta = +4$); the $(\xi, \hat{g}^2, \hat{y}^2)$ directions are irrelevant, with $\hat{g}^2$ and $\hat{y}^2$ asymptotically free. The interacting entry κ_* is *INDICATIVE* (Theorem 12.4).

Proof. The Gaussian gauge and Yukawa values are forced by Theorem 12.2 and subsection 12.4; κ_* is (107); $\xi_*, \Lambda_{\text{cc},*}$ are tuned. The diagonal entries (111) are the engineering-dimension scaling coefficients (+2 for the dimension-2 couplings $\xi, \hat{g}^2, \hat{y}^2$; -4 for the dimension-4 relevant Λ_{cc}) plus the gravitational slope -2 from Theorem 12.4. Triangularity makes these the eigenvalues; applying the convention of Theorem 12.5 gives (112). $\square$

Remark 12.7. The positive central eigenvalues $(\hat{g}^2, \hat{y}^2) = +2$ are *correct* under the AS convention: they encode gauge and Yukawa asymptotic freedom, not a pathology. The legacy quartet $\{-2, -1.3, -0.9, -0.2\}$ (which presumed an interacting gauge fixed point and all-relevant directions) is not reproducible and is replaced by (112).

12.6 Ultraviolet completeness, recast

We deliberately drop the earlier higher-curvature “irrelevance” sketch: the previous argument assigned a dimensionally inconsistent scaling $\Delta = -2 + \mathcal{O}(\kappa_*)$ to curvature-squared operators while simultaneously asserting $\kappa \rightarrow 0$, which contradicts Theorem 12.4. In the present, corrected picture UV completeness rests on three pillars.

Theorem 12.8 (UV completeness, conditional on the truncation). *Within the truncation defined by the coset-field action (??) and the β -system (109), UCFT possesses a finite-dimensional UV-critical surface, with only the two relevant directions κ and Λ_{cc} , provided the following hold:*

- (i) Gauge–Yukawa asymptotic freedom (*Theorem 12.2, subsection 12.4*): $\hat{g}_*^2 = \hat{y}_*^2 = 0$, UV attractive, providing a Gaussian completion of the matter sector;
- (ii) a gravitational fixed point (*Theorem 12.4*): the interacting $\kappa_* = 96\pi^2/5$, with a single relevant direction $\theta_\kappa = +2$ — INDICATIVE;
- (iii) irrelevant higher operators: every operator beyond those in (109), including the 27×27 scalar block (108) (three negative block eigenvalues), has $\theta < 0$ in this truncation.

The continuum limit is then defined by tuning the two relevant couplings.

Proof. By Theorem 12.6 the stability spectrum is $\{+2, +2, +2, -2, -4\}$, giving exactly two relevant directions $(\kappa, \Lambda_{\text{cc}})$ under Theorem 12.5. The matter directions are UV attractive (i); the gravitational direction is the sole interacting one (ii); the marginal scalar block and the remaining higher operators carry $\theta < 0$ (iii, section E). A finite-dimensional UV-critical surface with finitely many relevant couplings is precisely the asymptotic-safety/freedom criterion for a predictive continuum limit. $\square$

Remark 12.9 (Truncation caveat). Theorem 12.8 is conditional on the truncation and on the indicative status of Theorem 12.4: the gravitational fixed point sits at $\kappa_* \approx 189 \gg 1$, outside controlled weak coupling, so it is a strong-coupling extrapolation rather than a rigorous result. The cosmological constant enters as a tuned relevant direction Λ_{cc} ($\theta = +4$): UCFT inherits, and does not solve, the cosmological-constant problem. Thus UV completeness here is a well-posed property of the stated truncation, not an unconditional theorem.

13 Standard Model Descent, Generations, and Fermion Masses

This section develops the Standard Model descent within UCFT. We start from the vacuum-selection theorem (section 6), which fixes the residual gauge group $H = \text{Spin}(10) \times U(1)$ and the matter kinematics, and we spell out the symmetry-breaking cascade down to $SU(3)_c \times U(1)_{\text{em}}$. The descent chain, the branchings, and the anomaly arithmetic are *theorems* given the residual group and the assigned charges; chirality (P4), the three-generation horizontal symmetry (P5), and the GUT–Higgs/Yukawa sector (P6) are *added postulates*; the heterotic generation match and the listed phenomenology are *conjectural correspondences*. subsection 13.7 summarizes the epistemic status. Conventions are fixed globally: metric signature $(-, +, +, +)$, $[f] = \text{mass}$, the 32 coset Goldstones are the *clock fields*, and M_{GUT} denotes the breaking-VEV scale tied to the decay constant f .

13.1 The explicit symmetry-breaking chain

Postulate 13.1 (GUT–Higgs sector, P6 part I). The descent below $H = \text{Spin}(10) \times U(1)$ proceeds through the real $SO(10)$ Higgs multiplets 45_H or 54_H , 126_H , and 10_H , whose VEVs realize the cascade of [Theorem 13.2](#). These multiplets are added structure, not consequences of the two axioms.

Theorem 13.2 (Descent chain). *With the residual group $H = \text{Spin}(10) \times U(1)$ fixed by the proved rank-1 idempotent vacuum $M = E_6/(\text{Spin}(10) \times U(1)) = \text{EIII}$ (??), and granting the Higgs content of [Theorem 13.1](#), the gauge symmetry descends as*

$$\begin{aligned}
 E_6 & \xrightarrow{\text{P1: rank-1 idempotent vacuum } M=\text{EIII}} \text{Spin}(10) \times U(1) \\
 & \xrightarrow{45_H \text{ or } 54_H} SU(5) \times U(1)_X \\
 & \xrightarrow{126_H \text{ (breaks } B-L \text{ and rank)}} SU(3)_c \times SU(2)_L \times U(1)_Y \\
 & \xrightarrow{10_H \text{ (electroweak, } v=246 \text{ GeV)}} SU(3)_c \times U(1)_{\text{em}}.
 \end{aligned} \tag{113}$$

Proof sketch. Step 0 ($E_6 \rightarrow \text{Spin}(10) \times U(1)$) is not a Higgs VEV in the GUT sense but the vacuum-selection step: the E_6 -invariant potential V of P1 has its global minimum on the rank-1 primitive-idempotent orbit, whose stabilizer is exactly $H = \text{Spin}(10) \times U(1)$ (??). Steps 1–3 are standard $SO(10)$ symmetry breaking by real Higgs reps: $\langle 45_H \rangle$ (or $\langle 54_H \rangle$) reduces $SO(10)$ to its $SU(5)$ subgroup; the SM-singlet, $B-L = 2$ component of 126_H acquires a VEV $v_R \sim M_{\text{GUT}}$, breaking $B-L$ and lowering the rank by one (removing the extra Abelian factor and supplying M_R , [Theorem 13.13](#)); $\langle 10_H \rangle = v = 246 \text{ GeV}$ is ordinary electroweak breaking. Each embedding is a standard maximal/regular subgroup chain; no step uses a forbidden $\{16, 16\text{bar}\}$ Higgs. $\square$

Theorem 13.3 (Hypercharge embedding and GUT normalization). *Hypercharge is the standard $SU(5)$ -embedded Cartan combination. Acting on the $SU(5)$ fundamental 5,*

$$T_Y = \frac{1}{\sqrt{60}} \text{diag}(-2, -2, -2, 3, 3), \quad Y = \text{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right) \tag{114}$$

on $5\text{bar} = (d^c, l)$. In the canonical group-theoretic (GUT) normalization the unified $U(1)$ coupling obeys

$$\alpha_1 = \frac{5}{3} \alpha_Y, \quad \Longleftrightarrow \quad g_1^2 = \frac{5}{3} g_Y^2. \tag{115}$$

The orthogonal $U(1)_X$ (the $SO(10)/SU(5)$ Abelian factor, orthogonal to Y) is broken at the 126_H scale as a heavy Z' , or kept light only if separately anomaly-checked. A scale remark that we use repeatedly downstream: M_{GUT} is the *breaking-VEV scale tied to f* , not the imported MSSM $2 \times 10^{16} \text{ GeV}$ crossing. Below M_{GUT} one runs the *three* Standard-Model β -functions, not a single b_0 .

13.2 One family from the 16

Theorem 13.4 ($16 = \text{one SM family} + \nu^c$). *Under $SO(10) \rightarrow SU(5) \times U(1)_X$ the chiral spinor decomposes as*

$$16 \longrightarrow 10_{+1} + \bar{5}_{-3} + 1_{+5}, \tag{116}$$

and further under $SU(5) \rightarrow SU(3)_c \times SU(2)_L \times U(1)_Y$ (with $Q = T_3 + Y$) the content is exactly one Standard-Model family plus a singlet:

$$\begin{aligned}
 10 & \supset q = (3, 2)_{+1/6}, \quad u^c = (\bar{3}, 1)_{-2/3}, \quad e^c = (1, 1)_{+1}, \\
 \bar{5} & \supset d^c = (\bar{3}, 1)_{+1/3}, \quad l = (1, 2)_{-1/2}, \\
 1 & = \nu^c = (1, 1)_0 \text{ (right-handed neutrino, } SU(5) \text{ singlet)}.
 \end{aligned} \tag{117}$$

Thus $10 + \bar{5} + 1 = 16$ Weyl fermions $= (q, u^c, d^c, l, e^c) + \nu^c$: one SM family plus a right-handed neutrino, with no mirror partners.

The chiral family sits inside 16_{+1} of the matter 27 of E_6 , whose canonical UCFT branching (??) is

$$27 = 1_{+4} + 10_{-2} + 16_{+1}, \quad (118)$$

where the 1_{+4} is an extra SM-singlet modulus available as a right-handed-neutrino modulus. That a single 16 rather than a vector-like $16 \oplus \bar{16}$ appears per family is the content of the chirality postulate:

Postulate 13.5 (Matter content and chirality, P4). Each family is one chiral 16 of Spin(10) with the $U(1)$ charge of (118); there are no mirror partners.

13.3 Three generations: $SU(3)_F$ on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$

Postulate 13.6 (Family symmetry, P5). The matter carries an $SU(3)_F$ horizontal (family) symmetry acting on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$; the three copies in the $\mathbb{C}^3$ factor are three chiral 16s of Spin(10), with no mirror partners. Three generations are *not* a consequence of the two axioms; they are added structure.

Theorem 13.7 (Recomputed n_F and three-generation b_0). *With one chiral **16** per family (P4, P5), the fermion count n_F in the one-loop Spin(10) β -function is the number of chiral **16** representations (one per generation), not the Weyl-fermion dimension:*

$$n_F = \# \text{generations} = 3 \quad (\text{three chiral } \mathbf{16}s, \text{ each weighted at } T(\mathbf{16}) = 2). \quad (119)$$

The gauge β -function (canonical +2 term restored; asymptotically free, no interacting fixed point) is

$$\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2} \hat{g}^4, \quad b_0 = \frac{11}{3}C_A - \frac{4}{3}T n_F - \frac{1}{6}T n_S, \quad (120)$$

with the locked Casimirs $C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $C_2(\mathbf{16}) = 45/8$. At the GUT matching scale with the full UCFT scalar content, $n_S = 28$ massive coset scalars enter the loop. Direct evaluation gives

$$\boxed{b_0(3 \text{ gen}) = \frac{11}{3}(8) - \frac{4}{3}(2)(3) - \frac{1}{6}(2)(28) = \frac{88-24-28}{3} = 12.} \quad (121)$$

The canonical one-generation anchor $b_0(1 \text{ gen}) = 26$ uses $n_F = 1$ and $n_S = 2$ (minimal GUT Higgs/Yukawa scalars at that scale):

$$b_0(1 \text{ gen}) = \frac{11}{3}(8) - \frac{4}{3}(2)(1) - \frac{1}{6}(2)(2) = \frac{78}{3} = 26. \quad (122)$$

We use $b_0 = 12$ for three-generation running below M_{GUT} and in the AS stability check of ??.

Remark 13.8. The gauge sector is asymptotically free: the canonical $+2\hat{g}^2$ term in (120) drives a Gaussian UV point $\hat{g}_*^2 = 0$; there is no interacting gauge fixed point. More fermions *decrease* b_0 in (120), so $b_0(3) < b_0(1)$. UV completeness is asymptotic freedom plus the (indicative) gravitational fixed point $\kappa_* = 96\pi^2/5$ (??) plus irrelevant higher operators.

Postulate 13.9 (Hierarchy and mixing from flavon VEVs, P5 part II). $SU(3)_F$ is broken sequentially ($SU(3)_F \rightarrow SU(2)_F \rightarrow 1$) by flavon VEVs, generating the inter-generation Yukawa hierarchy and the CKM (quark) and PMNS (lepton) mixing matrices as ratios of flavon VEVs to the family scale.

Indicative 13.10 (Heterotic corroboration). The heterotic $\mathbb{Z}_5$ quintic $\hat{Q} = Q/\mathbb{Z}_5$ compactification with the tangent-bundle $SU(3)$ embedding gives the same generation count,

$$h^1(\hat{Q}, V_{16}) = 3, \quad h^1(\hat{Q}, V_{10}) = 14, \quad h^1(\hat{Q}, \mathbf{1}) = 1, \quad (123)$$

with the single bundle singlet an overall right-handed-neutrino modulus and the $4+28 = 32$ heterotic moduli matching the 4 clock-field Goldstone + 28 massive coset scalars of UCFT. The low-energy central charges $(c_L, c_R) = (16, 10)$ enter here. This is a correspondence, not a duality, and not a derivation of the generation number.

13.4 Yukawa and seesaw sector (P6)

Theorem 13.11 (No $\{16, 16\text{bar}\}$ Yukawa). *The naive Yukawa $-y \bar{\psi} \phi^a T_a \psi$ with $\phi \in \{16, 16\text{bar}\}$ is forbidden: it is non- H -invariant and gives zero mass, because*

$$16 \times 16 = 10 + 120 + 126, \quad 16 \times \overline{16} = 1 + 45 + 210, \quad (124)$$

so a $\{16, 16\text{bar}\}$ scalar furnishes no H -singlet, and $H = \text{Spin}(10) \times U(1)$ is unbroken at the vacuum. The viable couplings use the real GUT-Higgs reps.

Postulate 13.12 (Yukawa couplings, P6 part II). The family-indexed Yukawa Lagrangian is

$$\mathcal{L}_{\text{Yuk}}^{\text{Dirac}} = y_{ij}^{(10)} 16_i 16_j 10_H + \text{h.c.}, \quad (125)$$

$$\mathcal{L}_{\text{Yuk}}^{\text{Maj}} = y_{ij}^{(126)} 16_i 16_j \overline{126}_H + \text{h.c.} \quad \left(\text{or } \frac{1}{M_*} 16_i 16_j \overline{16}_H \overline{16}_H \right), \quad (126)$$

both $SO(10)$ -invariant since $16 \times 16 \supset 10$ and $16 \times 16 \supset 126$, with i, j the $SU(3)_F$ family indices.

Theorem 13.13 (Type-I seesaw). *After electroweak breaking $\langle 10_H \rangle = v = 246 \text{ GeV}$ the Dirac masses are $m_D \sim y^{(10)} v$. The SM-singlet, $B-L = 2$ component of 126_H takes $\langle \cdot \rangle = v_R \sim M_{\text{GUT}}$ (step 2 of (113)), giving the right-handed Majorana mass $M_R \sim y^{(126)} v_R$ (equivalently $M_R \sim v_R^2/M_*$ from the dim-5 operator). The (ν, ν^c) mass matrix*

$$\mathcal{M}_\nu = \begin{pmatrix} 0 & m_D \\ m_D^\top & M_R \end{pmatrix} \quad (127)$$

yields, to leading order in m_D/M_R , the light-neutrino mass

$$m_\nu \simeq \frac{m_D^2}{M_R} \sim 0.05 \text{ eV}, \quad (128)$$

the observed atmospheric scale for $m_D \sim v$ and $M_R \sim 10^{14}-10^{15} \text{ GeV}$. PMNS mixing follows from the flavon structure (Theorem 13.9).

These multiplets are added physics; stating them is what makes the construction an embedding rather than a derivation.

13.5 Anomaly arithmetic

Theorem 13.14 (Anomaly cancellation). *The composite $SO(10) \times U(1)$ gauge sector is anomaly-free. With Dynkin indices $A_{16} = A_{10} = 1$, $A_1 = 0$ and the 27-charges of (118), the mixed $U(1)-SO(10)^2$ coefficient is*

$$(+1)A_{16} + (-2)A_{10} + (+4)A_1 = (+1)(1) + (-2)(1) + (+4)(0) = -1, \quad (129)$$

cancelled by the Green–Schwarz mechanism: the E_6 -invariant 3-form Ω_3 pulls back to $dB = (1/2\pi)F_1$, and $S_{\text{GS}} = \int B \wedge \text{Tr}(F \wedge F)/(8\pi^2)$ has a gauge variation exactly cancelling the -1 . The cubic $U(1)^3$ coefficient vanishes identically,

$$16(+1)^3 + 10(-2)^3 + 1(+4)^3 = 16 - 80 + 64 = 0. \quad (130)$$

Below M_{GUT} each family is the anomaly-free 16 (the ν^c is exactly what makes the SM hypercharge anomalies cancel family-by-family).

Remark 13.15 (Strings and topology, correspondence). The composite $SO(10) \times U(1)$ connection (Maurer–Cartan), the Green–Schwarz cancellation, the vortex strings, and the heterotic worldsheet CFT with $(c_L, c_R) = (16, 10)$ are retained as a low-energy correspondence. Topology caveat: EIII is simply connected, so $\pi_1(M) = \mathbb{Z}$ is *false* of EIII itself; the vortex strings arise from the gauged $U(1)$ -quotient configuration space (or are softened to a correspondence).

13.6 Phenomenology (indicative)

Indicative 13.16 (Proton decay, leptogenesis, monopole dilution). Once descent and seesaw exist, the leading observables are, all indicative:

$$\tau(p \rightarrow e^+ \pi^0) \sim \frac{M_{\text{GUT}}^4}{\alpha_{\text{GUT}}^2 m_p^5}, \quad (131)$$

to be confronted with Super-/Hyper-Kamiokande, with the scalar-leptoquark channel forbidden by matter parity / $B-L$; baryogenesis via leptogenesis from the heavy ν^c decays (Theorem 13.13); and monopole dilution by inflation, with the inflaton identified as one of the 28 massive coset scalars (subsection 11.1) and inflation occurring *after* the GUT transition. Here M_{GUT} is the breaking-VEV scale tied to f .

13.7 Summary of results

- **Theorems (proved within UCFT):** the $E_6 \rightarrow \text{Spin}(10) \times U(1)$ vacuum selection (??); the descent (Theorem 13.2); the branchings of 27 and 16 (Theorem 13.4); the anomaly arithmetic (Theorem 13.14, mixed = -1 cancelled by GS, cubic = 0); and the three-generation $b_0 = 12$ (Theorem 13.7).
- **Postulates (added structure):** P4 (one chiral 16/family, no mirrors, Theorem 13.5); P5 ($SU(3)_F \rightarrow$ three generations, $n_F = 3$, Theorem 13.6); P6 (10_H Dirac, $126_H/\text{dim-5}$ Majorana, type-I seesaw, Theorem 13.1, Theorem 13.12).
- **Correspondence (indicative):** heterotic $h^1(V_{16}) = 3$ and the $4 + 28 = 32$ moduli match (Theorem 13.10); the phenomenology of Theorem 13.16.

Conclusion. Under postulates P4–P6, UCFT embeds the Standard Model: three anomaly-free chiral families, standard hypercharge with GUT normalization $\alpha_1 = \frac{5}{3}\alpha_Y$, electroweak breaking at $v = 246$ GeV, and a type-I seesaw $m_\nu \sim 0.05$ eV.

Part V

Correspondences and Conclusions

14 Vortex strings, heterotic correspondence, and conclusions

We close the construction by collecting the four threads of the broken phase — topological strings, their worldsheet algebra, the heterotic moduli match, and the Standard-Model descent — into a single, systematic classification of the principal results. Throughout we keep the signature $(-, +, +, +)$, write $[f] = \text{mass}$ for the decay constant, and call the 32 coset Goldstone directions the *clock fields*. The vacuum manifold is the rank-1 primitive-idempotent orbit

$$\mathcal{M} = E_6/(\text{Spin}(10) \times U(1)) = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32, \quad (132)$$

with positive-definite coset metric $K|_{\mathfrak{m}}$ and tangent module $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ (section 7). The physical spectrum is 28 massive clock scalars of a single common mass $m^2 = \mu^2$ plus 4 gauge/soldering modes eaten by diffeomorphism and local Lorentz invariance (section 11).

14.1 The topological source of the strings

The naive expectation is that the broken phase supports cosmic strings classified by $\pi_1(\mathcal{M})$. This must be stated with care.

Theorem 14.1 (EIII is simply connected). *The vacuum manifold (132) is a compact Hermitian symmetric space (the complex Cayley plane). Hence*

$$\pi_1(\text{EIII}) = 0.$$

Proof. $\text{EIII} = E_6/(\text{Spin}(10) \times U(1))$ is an irreducible compact Hermitian symmetric space of type E III (É. Cartan’s classification). The total space E_6 (compact, simply connected) fibres over $\mathcal{M}$ with connected fibre $\text{Spin}(10) \times U(1)$, which is itself connected; the long exact homotopy sequence $\pi_1(E_6) \rightarrow \pi_1(\mathcal{M}) \rightarrow \pi_0(\text{Spin}(10) \times U(1))$ has both flanking groups trivial, so $\pi_1(\mathcal{M}) = 0$. Equivalently, all compact irreducible Hermitian symmetric spaces are simply connected. $\square$

Remark 14.2 (Topological consistency). [Theorem 14.1](#) forbids the assertion $\pi_1(\mathcal{M}) = \mathbb{Z}$ of *EIII itself*: there are no σ -model strings winding the bare coset target. Any source claiming $\pi_1(\text{EIII}) = \mathbb{Z}$ is in error. The correct topological seat of the strings is the *gauged* configuration space, identified next.

The strings instead arise once the composite $\text{Spin}(10) \times U(1)$ connection (section 9) is switched on. Gauging the $U(1) \subset H$ that acts on the clock fields replaces the target by the configuration space of the gauged σ -model. Concretely, the physical order parameter for vortices is the $U(1)$ -charged combination of clock fields whose phase lives in

$$\mathcal{C}_{U(1)} \simeq \mathcal{M} // U(1), \quad \pi_1(\mathcal{C}_{U(1)}) \supseteq \mathbb{Z},$$

the winding being the standard Abelian-Higgs flux of the gauged $U(1)$ (the same $U(1)$ whose Stueckelberg/Green–Schwarz coupling appears in ??). Equivalently, on the broken vacuum the $U(1)$ factor of $H = \text{Spin}(10) \times U(1)$ admits a magnetic flux tube: $\pi_1(U(1)) = \mathbb{Z}$ labels the quantised flux, and the clock field winds the eaten phase around the core.

Postulate 14.3 (Topological string sector). The relevant first homotopy is that of the $U(1)$ -gauged configuration space (14.1), $\pi_1(\mathcal{C}_{U(1)}) \supseteq \mathbb{Z}$, *not* that of the bare coset. The vortex strings are flux tubes of the composite $U(1)$ (cf. Axiom I and the gauging prescription, section 9); we treat the string content as a *correspondence* with Abelian-Higgs/heterotic vortices rather than a $\pi_1(\mathcal{M})$ theorem.

14.2 BPS vortex strings of finite tension

Choose cylindrical coordinates (r, θ, z, t) on the emergent Lorentzian manifold (X, η) of ??. A static configuration of the $U(1)$ -gauged clock-field sector with winding $n \in \mathbb{Z}$ obeys

$$\phi(r, \theta + 2\pi) = e^{n T_{U(1)} 2\pi} \phi(r, \theta), \quad \phi(0) = X_0 \in \mathcal{M}, \quad D_r \phi \xrightarrow[r \rightarrow \infty]{} 0, \quad (133)$$

with $T_{U(1)}$ the broken $U(1)$ generator and X_0 a rank-1 idempotent representative of the vacuum.

Theorem 14.4 (Finite BPS tension). *For each $n \in \mathbb{Z}$ the Euler–Lagrange equations of the gauged clock-field action admit a static, axially symmetric, finite-tension solution of the ansatz type (133). Its tension saturates a Bogomol’nyi bound,*

$$\mathcal{T}_n = 2\pi f^2 |n| \int_0^\infty dr \, r K_{\alpha\beta}(\phi) D_r \phi^\alpha D_r \phi^\beta = 2\pi f^2 |n| \Omega, \quad 0 < \mathcal{T}_n < \infty, \quad (134)$$

where Ω is the $U(1)$ flux quantum through the core and $K_{\alpha\beta}$ the positive-definite coset metric.

Proof sketch. Completing the static energy functional into a sum of squares plus a topological flux term gives a first-order Bogomol’nyi system whose solutions minimise the energy in each winding class; the positive-definiteness of $K|_{\mathfrak{m}}$ (section 7) guarantees a strictly positive integrand, and the exponential approach $D_r \phi \rightarrow 0$ (set by the common mass $m^2 = \mu^2$ of the 28 massive clock fields) makes the integral converge. The detailed moduli-matrix construction is deferred to the worldsheet appendix, section E. $\square$

Linear fluctuations about the BPS background factorise into right-moving Nambu–Goto modes (transverse string position) and left-moving internal currents (the eaten $\text{Spin}(10) \times U(1)$ phases of the core).

14.3 Worldsheet algebra and the heterotic correspondence

Theorem 14.5 (Left-moving affine algebra). *Quantisation of the gauge zero modes trapped on the vortex core yields a left-moving affine current algebra at level one,*

$$\widehat{\mathfrak{so}}(10)_1 \oplus \widehat{\mathfrak{u}}(1)_1. \quad (135)$$

Proof sketch. The Green–Schwarz term $S_{\text{GS}} = \int B \wedge \text{Tr}(F \wedge F)/8\pi^2$ (??) reduces on the two-dimensional string core $\Sigma \simeq \mathbb{R}^{1,1}$ to a chiral WZW action whose level equals the mixed $U(1)$ – $\text{Spin}(10)^2$ anomaly coefficient computed there; matching that coefficient fixes $k = k' = 1$. The Sugawara stress tensor of (135) then governs the left-movers. Full details appear in section E. $\square$

Theorem 14.6 (Sugawara central charge). *The Sugawara central charge of (135) is*

$$c_L = \frac{\dim \mathrm{SO}(10) \cdot k}{k + h_{\mathrm{SO}(10)}^\vee} + 1 = \frac{45 \cdot 1}{1 + 8} + 1 = 5 + 1 = \boxed{16}, \quad (136)$$

using $\dim \mathrm{SO}(10) = 45$, dual Coxeter number $h_{\mathrm{SO}(10)}^\vee = 8$, and $c[\widehat{\mathfrak{u}}(1)_1] = 1$.

Proof. The Sugawara formula for $\widehat{\mathfrak{g}}_k$ gives $c = k \dim \mathfrak{g} / (k + h^\vee)$; at $k = 1$ for $\mathrm{SO}(10)$ this is $45/9 = 5$. The Abelian factor contributes 1. Their sum is 16. $\square$

Theorem 14.7 (Heterotic worldsheet correspondence). *Combining the four right-moving spacetime bosons ($c_R = 4$ Nambu–Goto) with their $(0, 1)$ worldsheet superpartners ($c_R = 4 \times \frac{1}{2} = 2$) and the transverse fermions of the core gives a right-moving central charge $c_R = 10$; together with $c_L = 16$ from (136), the vortex worldsheet realises the central-charge content*

$$(c_L, c_R) = (16, 10) \quad (137)$$

of the level-one left-moving sector of the $E_8 \times E_8$ heterotic string.

Remark 14.8 (Correspondence, not duality). **Theorem 14.7** is a *low-energy correspondence*: the gauge sector, anomaly structure, and central charges of the UCFT vortex match the heterotic worldsheet at leading order in α' . It is *not* a strict duality — no full equivalence of S-matrices, spectra, or quantum moduli spaces is claimed, and the matching is stated only at the level of (137) and the affine algebra (135). The conformal-field-theory derivation (Sugawara construction, modular invariance, higher-derivative robustness) is deferred to [section E](#).

14.4 The $\mathbb{Z}_5$ quintic Calabi–Yau correspondence

The broken phase leaves $32 = 4 + 28$ real flat directions: the 4 gauge/soldering clock fields π_-^i ($i = 0, 1, 2, 3$) and the 28 massive clock fields π_+^j promoted to Wilson-line moduli once the composite connection condenses,

$$\mathcal{M}_{\mathrm{UCFT}}^{\mathrm{mod}} = \underbrace{\mathbb{R}^4 / \Gamma^4}_{\text{Goldstone/soldering}} \times \underbrace{\mathbb{R}^{28} / \Gamma^{28}}_{\text{Wilson lines}}, \quad \dim_{\mathbb{R}} \mathcal{M}_{\mathrm{UCFT}}^{\mathrm{mod}} = 32. \quad (138)$$

On the heterotic side, compactify the $E_8 \times E_8$ string on the freely-acting $\mathbb{Z}_5$ quotient of the Fermat quintic, $\widehat{Q} = Q / \mathbb{Z}_5$ with $Q = \{\sum_{a=0}^4 z_a^5 = 0\} \subset \mathbb{CP}^4$, standard embedding of $T\widehat{Q}$ in one E_8 :

$$h^{1,1}(\widehat{Q}) = 1, \quad h^{2,1}(\widehat{Q}) = 0, \quad \pi_1(\widehat{Q}) = \mathbb{Z}_5, \quad (139)$$

the Wilson lines along $\pi_1(\widehat{Q}) = \mathbb{Z}_5$ breaking $E_6 \rightarrow \mathrm{Spin}(10) \times U(1)$. Bundle cohomology gives

$$h^1(\widehat{Q}, V_{16}) = 3, \quad h^1(\widehat{Q}, V_{10}) = 14, \quad h^1(\widehat{Q}, \mathbf{1}) = 1, \quad (140)$$

hence the real-moduli count

$$\underbrace{4}_{\text{Kähler}+B_2 \text{ axions}} + \underbrace{28}_{2 \times h^1(V_{10})} = 32 = \dim_{\mathbb{R}} \mathcal{M}_{\mathrm{Het}}^{\mathrm{mod}}. \quad (141)$$

Theorem 14.9 (Moduli matching). *The real dimensions and gauge-bundle topology of $\mathcal{M}_{\mathrm{UCFT}}^{\mathrm{mod}}$ in (138) and of $\mathcal{M}_{\mathrm{Het}}^{\mathrm{mod}}$ in (141) agree: both are $32 = 4 + 28$, both carry an $\mathrm{Spin}(10) \times U(1)$ bundle, and the kinetic metrics coincide at tree level (each descends from the E_6 Killing form). The four Goldstone/soldering coordinates map to the universal Kähler modulus plus three axionic partners; the 28 Wilson-line coordinates map to the 2×14 real bundle moduli in $H^1(\widehat{Q}, V_{10})$.*

Proof sketch. Counting and bundle topology are read off from (138)–(141). Metric equality follows from the common E_6 Killing origin of both kinetic terms; Chern-class/Chern–Simons matching uses the shared Green–Schwarz two-form. The detailed cohomology and the explicit map are given in section E. $\square$

Remark 14.10 (Three generations: corroboration, not derivation). The chiral generation count of the heterotic model is $h^1(\widehat{Q}, V_{\mathbf{16}}) = 3$ from (140). This *corroborates* the three generations of UCFT, which are supplied as an added postulate $P5$ — the horizontal $SU(3)_F$ symmetry on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$ (section 13) — and *not* as a second, independent mechanism. The 14 vector-like $\mathbf{10}$ moduli and the single $\mathbf{1}$ (a right-handed-neutrino modulus) complete the $32 = 4 + 28$ match. Generations are *added structure* ($P5$), with the heterotic count as a *correspondence*.

14.5 Conclusions: theorems versus postulates

UCFT is an E_6 /Albert-algebra–motivated unification. Two axioms supply the arena (the $27 = J_3(\mathbb{O})$ order parameter and its quantum kinematics); a small, explicitly enumerated set of postulates $P1$ – $P6$ then yields a four-dimensional Lorentzian quantum field theory containing an anomaly-free $\text{Spin}(10) \times U(1)$ gauge sector, Sakharov-induced gravity, and — after a stated GUT-Higgs/flavour sector — the Standard Model with three generations and neutrino masses. This is a *conditional embedding* of the Standard Model and gravitation. Table 4 records the classification of results.

Result	Content	Status
Vacuum selection ($P1 \Rightarrow P2$)	global min = rank-1 idempotent orbit EIII, dim 32	Theorem
Origin instability	$\text{Hess } V(0) \prec 0$; roll-down to $\mathcal{M}$	Theorem
Coset tangent / metric	$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, $K _{\mathfrak{m}} \succ 0$	Theorem
Soldering algebra	$H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$, $SL(2, \mathbb{O}) = \text{Spin}(1, 9)$	Theorem
Signature (1, 3), $d = 4$	added soldering structure	Postulate $P3$
Mass spectrum	28 massive (common $m^2 = \mu^2$) + 4 eaten	Theorem
Induced gravity	$M_{\text{Pl}}^2 = \frac{28}{6} f^2 \mu^2 \log(\Lambda^2 / \mu^2) / 16\pi^2 > 0$	Theorem
Gauge running	asymptotically free, $b_0(1 \text{ gen}) = 26$, $b_0(3 \text{ gen}) = 12$	Theorem
Gravitational fixed point	$\kappa_* = 96\pi^2/5$, eig = -2	Indicative
SM descent	$\mathbf{16} = \text{one family} + \nu^c$; anomalies cancel (GS)	Theorem
Chirality / generations / Yukawa	$P4, P5, P6$; seesaw $m_\nu \sim 0.05 \text{ eV}$	Postulate
BPS vortex strings	finite tension $\mathcal{T}_n = 2\pi f^2 n \Omega$	Theorem
String topology	from gauged $U(1)$ config space, not $\pi_1(\text{EIII}) = 0$	Theorem + Postulate
Heterotic worldsheet	$(c_L, c_R) = (16, 10)$, $\widehat{\mathfrak{so}}(10)_1 \oplus \widehat{\mathfrak{u}}(1)_1$	Correspondence
Calabi–Yau moduli	$32 = 4 + 28$, $h^1(V_{\mathbf{16}}) = 3$	Correspondence
Cosmological constant	tuned relevant Λ_{cc}	inherited, not solved

Table 4: Classification of results by epistemic status. Theorems are proved from stated hypotheses; postulates ($P1$ – $P6$) are added structure; conjectural results are motivated but outside controlled weak coupling; correspondences are leading-order matches, not dualities.

What is proved. The vacuum-selection theorem ($P1 \Rightarrow P2$, the rank-1 idempotent orbit $\mathcal{M} = \text{EIII}$ of real dimension 32), the positive-definiteness and uniqueness of the coset metric, the octonionic soldering algebra $SL(2, \mathbb{O}) \cong \text{Spin}(1, 9)$, the common radiative mass of the 28 clock scalars by Schur’s lemma, the positivity $M_{\text{Pl}}^2 > 0$ of the Sakharov-induced Planck mass with prefactor $28/6$, the asymptotic freedom of the gauge–Yukawa sector ($\beta_{\hat{g}^2} = 2\hat{g}^2 + (b_0/48\pi^2)\hat{g}^4$, $b_0 = 12$ for three generations), the Green–Schwarz cancellation of the mixed anomaly, the embedding of one Standard-Model family plus a right-handed neutrino in the **16**, and the finiteness of the BPS vortex tension are all *theorems*.

What is postulated. The Lorentzian signature and dimension ($P3$), the chiral matter content without mirrors ($P4$), the three generations via $SU(3)_F$ ($P5$), and the real GUT-Higgs/Yukawa sector generating the type-I seesaw ($P6$) are *added postulates*. The topological string sector rests on the gauged- $U(1)$ configuration space (Theorem 14.3), since $\pi_1(\text{EIII}) = 0$.

What is indicative or a correspondence. The gravitational fixed point $\kappa_* = 96\pi^2/5 \approx 189.50$ is *indicative*: $\kappa_* \gg 1$ lies outside controlled weak coupling. Reflection positivity of the UCFT σ -model (and hence Osterwalder–Schrader reconstruction of Lorentzian dynamics) is indicative. The heterotic worldsheet match $(c_L, c_R) = (16, 10)$ and the $\mathbb{Z}_5$ quintic moduli match $32 = 4 + 28$ with $h^1(V_{\mathbf{16}}) = 3$ are *low-energy correspondences*, not strict dualities. UCFT *inherits, and does not solve*, the cosmological-constant problem: Λ_{cc} is a tuned relevant coupling.

Outlook. Several directions would sharpen the embedding into a derivation. (i) A curved-background functional-RG truncation could test whether the indicative gravitational fixed point survives beyond the single-scalar loop, and whether the higher-curvature operators are genuinely irrelevant. (ii) A direct check of reflection positivity for the gauged clock-field σ -model would upgrade emergent unitary time from indicative to theorem. (iii) Deriving $d = 4$ and signature $(1, 3)$ from a dynamical vierbein-condensate selection, rather than postulating $P3$, remains open. (iv) A worldsheet computation of the full vortex partition function would test the heterotic correspondence beyond central charges. (v) Phenomenologically, dimension-six proton decay $\tau(p \rightarrow e^+\pi^0) \sim M_{\text{GUT}}^4/(\alpha_{\text{GUT}}^2 m_p^5)$, leptogenesis from the heavy ν^c , and inflation driven by one of the 28 massive clock fields after the GUT transition all furnish falsifiable tests. UCFT is presented as a conditional embedding: the proved core—one Albert-algebra order parameter selecting an anomaly-free $\text{Spin}(10) \times U(1)$ gauge sector with induced, positive Planck mass—is established under the stated hypotheses, while the Lorentzian, flavour, and gravitational-UV completions rest on explicit postulates and conjectures enumerated above.

Part VI

Appendices

A Octonions and the Albert Algebra

This appendix collects, in a single self-contained place, the octonionic and Jordan-algebraic facts on which the rest of the manuscript rests: the octonion multiplication table and structure constants; the exceptional Jordan (Albert) algebra $J_3(\mathbb{O})$ as 3×3 octonion-Hermitian matrices; the Jordan product, the trace bilinear form, the cubic norm N , and the sharp map $X^\#$ with the Freudenthal identities; the rank stratification and primitive idempotents; and the actions of F_4 and E_6 . Every statement

is tagged as a *theorem* (proved here or from cited standard references), a *postulate* (assumed added structure, cf. P1–P6), or as *indicative*. The identities assembled here are exactly those invoked by the potential, mass-spectrum and gauge sections; we therefore cross-reference them by the stable labels defined below.

Throughout we use the metric signature $(-, +, +, +)$ and the manuscript-wide conventions: $[f] = \text{mass}$, and $\hat{g}^2, \xi, \hat{y}^2, \kappa$ dimensionless. Two real forms of E_6 appear and must *not* be conflated: the *compact* E_6 (equivalently $E_{6(-14)}$, maximal compact $\text{Spin}(10) \times U(1)$) governs the dynamics and gives a positive-definite coset metric, while the non-compact $E_{6(-26)}$ is used *only* as the rigid classifier of orbits of the real 27. This distinction is the foundation enforced in [subsection A.5](#) and [subsection A.6](#), and underlies the vacuum-selection theorem cited as [Theorem 6.7](#).

A.1 Octonions: multiplication and structure constants

Definition A.1 (Octonions). The octonions $\mathbb{O}$ form the real, 8-dimensional, unital, *non-associative* division algebra. Fix the standard basis $\{e_0 = 1, e_1, \dots, e_7\}$ with $e_i^2 = -1$ for $i \geq 1$ and $e_i e_j = -e_j e_i$ for $i \neq j$ ($i, j \geq 1$). The multiplication is encoded by the totally antisymmetric structure constants ψ_{ijk} ,

$$e_i e_j = -\delta_{ij} e_0 + \psi_{ijk} e_k, \quad i, j, k \in \{1, \dots, 7\}, \quad (142)$$

with $\psi_{ijk} = +1$ on the seven Fano lines

$$(ijk) \in \{(124), (235), (346), (457), (561), (672), (713)\} \quad (143)$$

(and cyclic), and $\psi_{ijk} = 0$ otherwise. Conjugation is $\bar{e}_0 = e_0$, $\bar{e}_i = -e_i$ ($i \geq 1$); the (positive-definite) norm is $n(x) = x\bar{x} = \bar{x}x = \sum_{a=0}^7 x_a^2$, and $\overline{xy} = \bar{y}\bar{x}$.

Theorem A.2 (Composition, alternativity, Moufang). (Theorem; standard [3, 18].) $\mathbb{O}$ is a *normed composition algebra*, $n(xy) = n(x)n(y)$. It is *not associative* but is *alternative*: the associator $[x, y, z] := (xy)z - x(yz)$ is *totally antisymmetric*, so any subalgebra generated by two elements is associative. The Moufang identities hold, in particular $(xy)(zx) = x((yz)x)$. Real and imaginary parts obey $\Re(xy) = \Re(yx)$ and $\Re((xy)z) = \Re(x(yz))$, so the trilinear form $\Re(xyz)$ is well defined (no bracketing needed) and cyclic.

Proof. Composition follows because n is the norm of a Cayley–Dickson double of the quaternions; alternativity and the antisymmetry of ψ_{ijk} are equivalent to the total antisymmetry of the associator, and the Moufang identities are the standard consequences. The reality identities follow from $x + \bar{x} = 2\Re(x)e_0 \in \mathbb{R}$ together with $\overline{xy} = \bar{y}\bar{x}$. See [3, 16]. ■ □

Remark A.3 (G_2 and $\text{Spin}(7)$). (Indicative context.) The automorphism group $\text{Aut}(\mathbb{O})$ is the compact G_2 (dim 14), the stabilizer of ψ_{ijk} viewed as a generic 3-form on $\Im\mathbb{O} \cong \mathbb{R}^7$; the spin group $\text{Spin}(7)$ acts by triality on the three 8-dimensional modules $\mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c$. These reappear when $\text{Spin}(10)$ spinors are realized on $J_3(\mathbb{O})$ (??) and feed the coset tangent module $\mathfrak{m} = 16_{-3} \oplus \overline{16}_{+3}$ used downstream.

A.2 The Albert algebra $J_3(\mathbb{O})$

Definition A.4 (Albert algebra). The (real, reduced) *exceptional Jordan algebra* or *Albert algebra* is the space of 3×3 octonion-Hermitian matrices

$$J_3(\mathbb{O}) = \left\{ X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix} : \xi_i \in \mathbb{R}, x_i \in \mathbb{O} \right\}, \quad (144)$$

equipped with the commutative *Jordan product*

$$X \circ Y = \frac{1}{2}(XY + YX), \quad (145)$$

where XY is ordinary (octonionic) matrix multiplication.

Theorem A.5 (Dimension and Jordan identity). (Theorem.) $J_3(\mathbb{O})$ is a 27-dimensional, commutative, power-associative Jordan algebra: $\circ$ is commutative and satisfies the Jordan identity $X \circ (Y \circ X^{\circ 2}) = (X \circ Y) \circ X^{\circ 2}$. It is formally real (Euclidean): $\sum_i A_i^{\circ 2} = 0 \Rightarrow A_i = 0$. It is the unique exceptional Jordan algebra (not isomorphic to a subalgebra of an associative algebra under $\circ$).

Proof. The real count is 3 (diagonal) + 3×8 (off-diagonal octonions) = 27. Commutativity is immediate from (145). The Jordan identity holds for 3×3 Hermitian matrices over an alternative algebra by the Hermitian-matrix theorem of Jordan–von Neumann–Wigner [9, 12]; formal reality follows from positivity of the trace form (146) below. Exceptionality (the Albert exceptionality theorem) is standard [2, 16]. ■ □

Definition A.6 (Traces, trace form, cubic norm). On $J_3(\mathbb{O})$ define the *linear trace* $\text{tr}(X) = \xi_1 + \xi_2 + \xi_3$, the symmetric *trace bilinear form*

$$\langle X, Y \rangle := \text{tr}(X \circ Y), \quad Q(X) := \langle X, X \rangle = \text{tr}(X^{\circ 2}) = \sum_i \xi_i^2 + 2 \sum_i n(x_i) \geq 0, \quad (146)$$

which is positive-definite, and the *cubic norm* (determinant analogue)

$$N(X) := \xi_1 \xi_2 \xi_3 - \sum_{i=1}^3 \xi_i n(x_i) + 2 \Re(x_1 x_2 x_3). \quad (147)$$

The second elementary symmetric invariant is $\sigma_2(X) = \frac{1}{2}(\text{tr}(X)^2 - Q(X))$. The trilinear polarization of N is denoted $N(X, Y, Z)$ with $N(X, X, X) = 6N(X)$.

Theorem A.7 (Cubic Cayley–Hamilton). (Theorem.) Every $X \in J_3(\mathbb{O})$ satisfies the cubic characteristic identity

$$X^{\circ 3} - \text{tr}(X) X^{\circ 2} + \sigma_2(X) X - N(X) E = 0, \quad E := \text{diag}(1, 1, 1). \quad (148)$$

Its three roots are the real Jordan eigenvalues $\lambda_1, \lambda_2, \lambda_3$, in terms of which $\text{tr}(X) = \sum_i \lambda_i$, $Q(X) = \sum_i \lambda_i^2$, $N(X) = \lambda_1 \lambda_2 \lambda_3$.

Proof. By the spectral theorem for formally real Jordan algebras every X is F_4 -conjugate to $\text{diag}(\lambda_1, \lambda_2, \lambda_3)$ with $\lambda_i \in \mathbb{R}$ (Theorem A.15 below); on diagonal elements (148) is the ordinary scalar Cayley–Hamilton identity, and F_4 -equivariance of every term ($\text{tr}, \sigma_2, N, E$ are F_4 -invariant or invariant) transports it to all of $J_3(\mathbb{O})$ [12, 18]. ■ □

A.3 The sharp map and the Freudenthal identities

Definition A.8 (Sharp map and cross product). The *sharp* (Freudenthal adjoint) map $X \mapsto X^\#$ is the unique quadratic map with entries the 2×2 octonionic cofactors of X :

$$(X^\#)_{ii} = \xi_j \xi_k - n(x_i), \quad (X^\#)_{ij} = \overline{x_i x_j} - \xi_k x_k \quad (\text{cyclic } i, j, k). \quad (149)$$

Equivalently, in the Jordan frame $\text{diag}(\lambda_1, \lambda_2, \lambda_3)^\# = \text{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$. Its symmetric polarization defines the *Freudenthal cross product*

$$X \times Y := (X + Y)^\# - X^\# - Y^\#, \quad X \times X = 2X^\#, \quad (150)$$

a symmetric bilinear map $J_3(\mathbb{O}) \times J_3(\mathbb{O}) \rightarrow J_3(\mathbb{O})$.

Theorem A.9 (Freudenthal / adjoint identities). (Theorem; [4, 12, 18].) *For all $X, Y \in J_3(\mathbb{O})$ the following hold:*

$$X^\# \circ X = N(X) E, \quad (151)$$

$$(X^\#)^\# = N(X) X, \quad (152)$$

$$\langle X^\#, Y \rangle = DN(X)[Y] = N(X, X, Y), \quad \text{i.e. } X^\# = \nabla N(X), \quad (153)$$

$$\langle X^\#, Y \rangle = \sigma_2(X) \langle ? \rangle \implies \text{tr}(X^\#) = \sigma_2(X), \quad (154)$$

$$\langle X^\#, X^\# \rangle = \sigma_2(X)^2 - 2 \text{tr}(X) N(X) = (\lambda_2\lambda_3)^2 + (\lambda_3\lambda_1)^2 + (\lambda_1\lambda_2)^2. \quad (155)$$

Proof. Identity (151) is the cofactor expansion of the determinant N and follows from (148) by writing $X^\# = X^{\circ 2} - \text{tr}(X)X + \sigma_2(X)E$ and using (148); (152) (the “##” Freudenthal identity) is verified in the Jordan frame, where $(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)^\# = (\lambda_1^2\lambda_2\lambda_3, \dots) = N \cdot (\lambda_1, \lambda_2, \lambda_3)$, and extended by F_4 -equivariance. The gradient identity (153) is the definition of $X^\#$ as the differential of the cubic N with respect to the trace form, and $\text{tr}(X^\#) = \sigma_2(X)$ in (154) follows from $X^\# = X^{\circ 2} - \text{tr}(X)X + \sigma_2(X)E$ on taking tr . Finally (155) is the frame evaluation of $\langle X^\#, X^\# \rangle$, manifestly a sum of squares; the identity $\sigma_2(X)^2 - 2 \text{tr}(X)N(X)$ is the symmetric-function rewriting. ■ □

Corollary A.10 (Rank from the sharp map). (Theorem.) *Define the rank by*

$$\text{rank}(X) = \begin{cases} 0, & X = 0, \\ 1, & X \neq 0, X^\# = 0, \\ 2, & X^\# \neq 0, N(X) = 0, \\ 3, & N(X) \neq 0. \end{cases} \quad (156)$$

Then $X^\# = 0 \iff \text{rank}(X) \leq 1$, and $N(X) = 0 \iff \text{rank}(X) \leq 2$. In particular the cubic norm N , not the quadratic Q , detects rank deficiency at order 3.

Proof. In the Jordan frame $X^\# = 0$ iff all pairwise products $\lambda_i\lambda_j$ vanish, i.e. at most one $\lambda_i \neq 0$; and $N = \lambda_1\lambda_2\lambda_3 = 0$ iff some $\lambda_i = 0$. Equivariance extends both statements to all of $J_3(\mathbb{O})$. ■ □

Remark A.11 (Positivity of the sharp invariant). (Theorem.) Because the trace form (146) is positive-definite, $\langle X^\#, X^\# \rangle \geq 0$ with equality iff $X^\# = 0$ iff $\text{rank}(X) \leq 1$ (Theorem A.10). This is the exact algebraic input to the potential: the E_6 -covariant term $\alpha \langle X^\#, X^\# \rangle$ vanishes precisely on the $\text{rank} \leq 1$ cone, which is why it selects the rank-1 idempotent orbit; see Theorem 6.3 and Theorem 6.7.

A.4 Rank stratification and primitive idempotents

Definition A.12 (Idempotents and Jordan frame). $P \in J_3(\mathbb{O})$ is an *idempotent* if $P \circ P = P$, and *primitive* if it is a nonzero idempotent that is not a sum of two nonzero orthogonal idempotents. A *complete Jordan frame* is a triple $\{P_1, P_2, P_3\}$ of pairwise orthogonal ($P_i \circ P_j = 0$, $i \neq j$) primitive idempotents with $P_1 + P_2 + P_3 = E$.

Theorem A.13 (Idempotent stratification). (Theorem; [12, 21].) *A primitive idempotent has Jordan eigenvalues $(1, 0, 0)$, hence $\text{tr}(P) = 1$, $Q(P) = 1$, $N(P) = 0$, $P^\# = 0$, i.e. $\text{rank}(P) = 1$. The normalized rank-1 elements are exactly the primitive idempotents; they form a single F_4 -orbit*

$$F_4/\text{Spin}(9) \cong \mathbb{OP}^2, \quad \dim_{\mathbb{R}} = 16, \quad (157)$$

the octonionic projective plane (the real rank-1 idempotents). Under the full compact E_6 the projective class $[X]$ of a rank-1 line sweeps the larger homogeneous space

$$\mathcal{M} = \text{EIII} = E_6/(\text{Spin}(10) \times U(1)), \quad \dim_{\mathbb{R}} \mathcal{M} = 32, \quad (158)$$

the complex Cayley plane and the closed minimal E_6 -orbit in $\mathbb{P}(27_{\mathbb{C}})$.

Proof. $P \circ P = P$ with eigenvalues μ forces $\mu^2 = \mu$, so $\mu \in \{0, 1\}$; primitivity forbids two 1's, and nonzero forbids all 0's, leaving $(1, 0, 0)$. The remaining invariants follow from Theorem A.7 and (149). Transitivity of $F_4 = \text{Aut}(J_3(\mathbb{O}))$ on primitive idempotents with stabilizer $\text{Spin}(9)$ is the standard identification $\mathbb{OP}^2 = F_4/\text{Spin}(9)$ [3, 21]; the E_6 -orbit statement is the realization of EIII as the minimal/highest-weight orbit in the projectivized 27 [18]. ■ □

Remark A.14 (Vacuum manifold; excluded base point). (Theorem, consistency note.) The UCFT vacuum manifold is exactly (158), the rank-1 primitive-idempotent orbit, $\dim_{\mathbb{R}} = 32$ — *not* the orbit of $E = \text{diag}(1, 1, 1)$. The latter has rank 3, $E^\# = E \neq 0$, $\langle E^\#, E^\# \rangle = 3 > 0$, stabilizer F_4 , and orbit E_6/F_4 of real dimension 26; by Theorem A.11 it is strictly disfavoured by the potential and is therefore excluded. This is consistent with the vacuum-selection theorem Theorem 6.7 and the coset tangent $\mathfrak{m} = 16_{-3} \oplus \overline{16}_{+3}$ of subsection A.7.

A.5 Invariants and the two real forms

Theorem A.15 ($F_4 = \text{Aut}(J_3(\mathbb{O}))$). (Theorem; [16, 21].) *The automorphism group of the Jordan algebra (preserving the product $\circ$, hence the unit E and both Q and N) is the compact exceptional group F_4 , $\dim_{\mathbb{R}} F_4 = 52$. It acts on $J_3(\mathbb{O}) = \mathbb{R} \oplus 26$ fixing the unit; on the traceless 26 it embeds $F_4 \subset SO(26) \subset SO(27)$, since it preserves the Euclidean form Q .*

Theorem A.16 (Reduced structure group = E_6). (Theorem; [10, 18].) *The reduced structure group of $J_3(\mathbb{O})$ — the connected group of linear maps preserving the cubic norm N up to a real character — is the exceptional group E_6 , $\dim_{\mathbb{R}} E_6 = 78$, acting irreducibly on the 27. For g in this group,*

$$N(g \cdot X) = \lambda(g) N(X), \quad \lambda(g) \in \mathbb{R}^\times, \quad (159)$$

and the sharp map is covariant, $(g \cdot X)^\# = \lambda(g) (g^{-1})^\top \cdot X^\#$ in the trace form. The split/non-compact real form realized by the division-octonion algebra is $E_{6(-26)}$; it does not preserve $Q = \text{tr}(X^2)$.

Proof. Jacobson's construction realizes $\mathfrak{e}_6$ as $\mathfrak{f}_4$ (derivations) plus the traceless multiplication operators $L_X : Y \mapsto X \circ Y - \frac{1}{3} \langle X, Y \rangle E$ acting on the 27; the group generated preserves N up to the character (159) [10]. The frame element $\text{diag}(s, s, s^{-2})$ rescales N nontrivially while changing Q , so Q is not E_6 -invariant; only F_4 (the $\lambda \equiv 1$, unit-preserving subgroup) preserves Q . ■ □

Postulate A.17 (Compact real form for dynamics). (Convention, cf. P1–P2.) For the dynamics we use the *compact* real form E_6 (equivalently $E_{6(-14)}$, whose maximal compact subgroup is exactly $H = \text{Spin}(10) \times U(1)$). On the compact form the 27 is unitary, the trace form $Q = \langle \cdot, \cdot \rangle$ is the invariant positive-definite metric, and the norm character is identically $\lambda \equiv 1$ (a continuous

homomorphism from a compact connected group into $\mathbb{R}_{>0}$). Hence on the compact form every E_6 -invariant polynomial built from $\langle \cdot, \cdot \rangle$ and N — in particular $\langle X^\#, X^\# \rangle$ — is a genuine invariant, the coset metric is positive-definite, and the gauge sector is unitary. The non-compact $E_{6(-26)}$ is retained *only* as the rigid orbit classifier of [subsection A.6](#).

Remark A.18 (Only N separates structure-group orbits). (Theorem, guard.) $Q = \text{tr}(X^2)$ is F_4 -invariant *only*; it is *not* preserved by the E_6 structure group, and the pair (Q, N) does *not* classify E_6 -orbits. Only the cubic norm N is structure-group invariant (up to the character λ), and orbits are classified by rank and N ([subsection A.6](#)). Any claim that the structure group preserves Q , or that (Q, N) separates orbits, is false.

A.6 Orbit classification of the real 27

Theorem A.19 (Rank-and-norm orbit classification). (Theorem; [18, 11].) *The orbits of $E_{6(-26)}$ on the real $27 = J_3(\mathbb{O})$ are classified by the rank (156) together with the value of the cubic norm N :*

- rank 0: $\{0\}$;
- rank 1: $X \neq 0, X^\# = 0$ — a single orbit, the cone over EIII; its projectivization is the closed minimal orbit $\mathcal{M} = \text{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$;
- rank 2: $X^\# \neq 0, N(X) = 0$ — a single orbit;
- rank 3: $N(X) \neq 0$ — orbits labelled by $\text{sgn } N$ and the signature of X (the indefinite cubic N is genuinely sign-indefinite).

The quadratic Q is not an orbit invariant; it can be rescaled within a fixed- N , fixed-rank orbit by the non-compact directions $\text{diag}(s, s, s^{-2}) \in E_{6(-26)}$.

Proof. That rank and N are structure-group invariants follows from (159) and the covariance of $X^\#$ ([Theorem A.16](#)); transitivity within each stratum is the reduction of an arbitrary X to a normal form $\text{diag}(\lambda_1, \lambda_2, \lambda_3)$ by F_4 followed by a norm-rescaling diagonal element, with the rank-deficient strata as the explicit degenerations [11, 18]. ■ □

Remark A.20 (Sharp characterization, summary). (Theorem.) Collecting [Theorem A.10](#) and [Theorem A.9](#): $\text{rank} \leq 1 \Leftrightarrow X^\# = 0$; $\text{rank} \leq 2 \Leftrightarrow N(X) = 0$; $\text{rank} = 3 \Leftrightarrow N(X) \neq 0$. These are the identities the potential section uses to localize the global minimum on the rank-1 cone ([Theorem 6.7](#)).

A.7 F_4 - and H -branchings; the coset tangent module

Theorem A.21 ($\text{Spin}(10)$ realization). (Theorem; [3, 18].) *Under $H = \text{Spin}(10) \times U(1) \subset E_6$ (the maximal compact of $E_{6(-14)}$), the fundamental and adjoint representations branch as*

$$27 = 1_{+4} \oplus 10_{-2} \oplus 16_{+1}, \quad (160)$$

$$78 = 45_0 \oplus 1_0 \oplus 16_{-3} \oplus \overline{16}_{+3}, \quad (161)$$

where the subscripts are the $U(1)$ charges in the normalization fixed by (161). The 10_{-2} is the $\text{Spin}(10)$ vector (an $H_2(\mathbb{O})$ -type block of $J_3(\mathbb{O})$), the 16_{+1} a chiral spinor, and the 1_{+4} a Jordan-frame singlet (an extra Standard-Model-singlet modulus).

Corollary A.22 (Coset tangent module). (Theorem.) *The tangent space to $\mathcal{M} = \text{EIII} = E_6/(\text{Spin}(10) \times U(1))$ at the basepoint is the H -module*

$$\mathfrak{m} = 16_{-3} \oplus \overline{16}_{+3}, \quad \dim_{\mathbb{R}} \mathfrak{m} = 32, \quad (162)$$

the complement of $\mathfrak{h} = 45_0 \oplus 1_0$ in (161). It is real-irreducible of complex type (commutant $\cong \mathbb{C}$) and contains no $\text{Spin}(10)$ singlet. Consequently the invariant coset metric $K|_{\mathfrak{m}}$ is unique up to scale and positive-definite (Schur), and any radiative (H -invariant) mass operator on $\mathfrak{m}$ is a single scalar: one common mass on all 32 clock fields.

Proof. $\mathfrak{m} = \mathfrak{e}_6 \ominus \mathfrak{h}$ reads off (161). The $U(1)$ charges ± 3 forbid the symmetric invariants in $16 \otimes 16$ and $\overline{16} \otimes \overline{16}$, leaving the unique charge-0 cross-pairing $16_{-3} \otimes \overline{16}_{+3} \rightarrow 1_0$; this is the restriction of the Killing form, which on the compact form is positive-definite. Real-irreducibility with commutant $\mathbb{C}$ is the complex-type Schur statement, giving uniqueness of the metric and a single common radiative mass. Absence of a singlet is immediate from (162). ■ □

Remark A.23 (Excluded tangent decompositions). (Consistency constraint.) The coset tangent is (162) and contains no $\text{Spin}(10)$ singlet. Any decomposition of $\mathfrak{m}$ as $2 \times 10 + 4$, or $10 + \overline{10} + 4$, or any singlet-in- $\mathfrak{m}$ form, is incorrect, as is any attempt to extract a 4-dimensional H -invariant subspace from $\nabla\sigma_2, \nabla N$ singlets. The Lorentzian frame directions are *not* a sub- H -module of $\mathfrak{m}$; they are introduced as added structure (soldering postulate P3, cf. ??) and are gauge-protected, not Hessian zeros.

A.8 The Minkowski block from the cubic norm

Theorem A.24 ($H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$). (Theorem; [3, 19].) *The 2×2 octonion-Hermitian matrices*

$$H_2(\mathbb{O}) = \left\{ \begin{pmatrix} a & x \\ \bar{x} & b \end{pmatrix} : a, b \in \mathbb{R}, x \in \mathbb{O} \right\}, \quad \dim_{\mathbb{R}} = 10, \quad (163)$$

form a sub-block of $J_3(\mathbb{O})$ (set the third pivot to 1). With $\det X = ab - n(x)$ and the lightcone parametrization $a = x^0 + x^9$, $b = x^0 - x^9$,

$$\det X = (x^0)^2 - (x^9)^2 - \sum_{i=1}^8 (x^i)^2 = -\eta_{\mu\nu} X^\mu X^\nu, \quad \eta = \text{diag}(-1, +1, \dots, +1), \quad (164)$$

so $(H_2(\mathbb{O}), -\det) \cong \mathbb{R}^{1,9}$. Restricting the octonion entry to $\mathbb{C} \subset \mathbb{O}$ gives $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ with $-\det$ the $(1, 3)$ Minkowski form in the convention $(-, +, +, +)$.

Proof. $\det \begin{pmatrix} a & x \\ \bar{x} & b \end{pmatrix} = ab - x\bar{x} = ab - n(x)$ by definition of n ; substituting the lightcone coordinates yields (164). The restriction to $\mathbb{C}$ drops $x^4, \dots, x^9$ (the imaginary octonion directions beyond $\mathbb{C}$), leaving the four coordinates x^0, x^1, x^2, x^3 with the same form. ■ □

Remark A.25 (Where Lorentzian signature enters). (Postulate P3.) The positive-definite object is the coset/target metric $K|_{\mathfrak{m}}$ on $\mathfrak{m}$ (Theorem A.22); the indefinite object is the cubic norm N , whose $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ block carries the Minkowski form (164). These are two distinct forms on two distinct modules. The Killing form on $\mathfrak{m}$ for the compact E_6 has signature $(32, 0)$ and zero Lorentzian content; the spacetime dimension $d = 4$ and signature $(1, 3) = (-, +, +, +)$ are *added structure* (soldering postulate P3, cf. ??), soldered from N via $H_2(\mathbb{C})$, and are *not* Killing-form, coset-geometry, or rank-lemma theorems. Any $(4, 28)$ or $(1, 3)$ coset/Killing-signature claim is false.

A.9 Identities used downstream

For convenience we list the identities invoked by the potential, mass-spectrum and gauge sections, all proved above.

1. **Sharp / Freudenthal** (151)–(155): $X^\# \circ X = N(X)E$, $(X^\#)^\# = N(X)X$, $X^\# = \nabla N(X)$, $\text{tr}(X^\#) = \sigma_2(X)$, and $\langle X^\#, X^\# \rangle = \sigma_2(X)^2 - 2\text{tr}(X)N(X) \geq 0$. Used in the E_6 -covariant term of the potential (P1) and in [Theorem 6.3](#).
2. **Rank stratification** (156): $\text{rank} \leq 1 \Leftrightarrow X^\# = 0$; $\text{rank} \leq 2 \Leftrightarrow N = 0$. Used to localize the global minimum on the rank-1 cone ([Theorem 6.7](#)).
3. **Cubic Cayley–Hamilton** (148) and frame eigenvalues. Used to evaluate V on the cone.
4. **Branchings** (160), (161) and the tangent module (162). Used for the 27×27 scalar block $M = \text{diag}(\mu_1 \mathbb{K}_1, \mu_{10} \mathbb{K}_{10}, \mu_{16} \mathbb{K}_{16})$, for the single common coset mass (Schur on $\mathfrak{m}$), and for the gauge multiplet assignments. The 28 massive clock fields ($32 = 28 \text{ massive} + 4 \text{ gauge/soldering}$) enter all loop sums; cf. ?? and the gauge β -function ??.
5. **Minkowski block** (164): $-\det|_{H_2(\mathbb{O})}$ is the $\mathbb{R}^{1,9}$ form, $-\det|_{H_2(\mathbb{C})}$ the $(1,3)$ form. Used by the soldering postulate P3 (??).
6. **Casimirs of the gauge representations**. With the unbroken $H = \text{Spin}(10) \times U(1)$ acting on the multiplets in (160), the relevant invariants are $C_A(\text{SO}(10)) = 8$, $T(16) = 2$ (the spinor is index 2), and $C_2(16) = 45/8$. These feed $b_0 = \frac{11}{3}C_A - \frac{4}{3}Tn_F - \frac{1}{6}Tn_S$ in ?? (canonical anchor $b_0 = 26$ for one generation, $b_0 = 12$ for three).

Remark A.26 (Status). All results in this appendix are *theorems* of octonionic/Jordan algebra ([Theorem A.2–Theorem A.24](#)) or standard representation theory, except the two clearly marked conventions/postulates: the choice of the *compact* real form for the dynamics ([Theorem A.17](#)) and the soldering of Lorentzian signature from N ([Theorem A.25](#), postulate P3). The algebra fixes the arena; the physics input is the potential class (P1) and the soldering (P3), treated in their own units.

B Minkowski Structure and the Soldering Map

This appendix establishes, with explicit mathematics and consistent epistemic labels, the chain by which a Lorentzian four-manifold is *soldered* onto the E_6 /Albert-algebra arena. We prove the algebraic facts $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ and $SL(2, \mathbb{O}) \cong \text{Spin}(1,9)$ as theorems; we state the choice of a four-plane, the dimension $d = 4$, and the signature $(1,3)$ as the soldering postulate (P3); and we give the Osterwalder–Schrader/Page–Wootters route to Lorentzian unitary dynamics in place of any “ $e^0 \mapsto i e^0$ ” Wick trick.

The governing logical separation, enforced throughout, is the following. The *positive-definite* object is the coset/target metric $K|_{\mathfrak{m}}$ on the 32-dimensional internal tangent space $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ of $M = E_6/(\text{Spin}(10) \times U(1)) = \text{EIII}$ (see ?? and ??). The *indefinite* object is the cubic norm N on the matter representation $\mathbf{27} = J_3(\mathbb{O})$, whose 2×2 octonion-Hermitian sub-block carries a Minkowski form on the *different* space $\mathbb{R}^{1,9}$. No single form is asked to be both definite and indefinite; the soldering map (P3) is the bridge importing the indefinite $(1,3)$ structure onto a four-plane of the world-tangent bundle, leaving $K|_{\mathfrak{m}}$ untouched and Euclidean. Conventions are fixed globally: signature $(-, +, +, +)$; the dynamics live on the *compact* real form of E_6 (maximal compact $H = \text{Spin}(10) \times U(1)$), and $E_{6(-26)}$ enters only as the rigid orbit-classifier of the real $\mathbf{27}$.

B.1 The Minkowski block: $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$

Let $\mathbb{O}$ be the real division octonions, a non-associative composition algebra of real dimension 8, with conjugation $x \mapsto \bar{x}$, norm $n(x) = x\bar{x} = \bar{x}x \in \mathbb{R}_{\geq 0}$, and trace $t(x) = x + \bar{x} \in \mathbb{R}$. Define the space of 2×2 octonion-Hermitian matrices

$$H_2(\mathbb{O}) = \left\{ X = \begin{pmatrix} a & x \\ \bar{x} & b \end{pmatrix} : a, b \in \mathbb{R}, x \in \mathbb{O} \right\}, \quad X^\dagger = X, \quad (165)$$

so that as a real vector space $\dim_{\mathbb{R}} H_2(\mathbb{O}) = 1 + 1 + 8 = 10$. The determinant is well defined despite non-associativity (the off-diagonal entry and its conjugate associate with the scalar diagonal):

$$\det X = ab - x\bar{x} = ab - n(x). \quad (166)$$

Theorem B.1 (Minkowski identification; $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$). *Introduce light-cone-adapted coordinates $a = X^0 + X^9$, $b = X^0 - X^9$, and write $x = (X^1, \dots, X^8) \in \mathbb{R}^8$ in an orthonormal basis of $\mathbb{O}$, so that $n(x) = \sum_{i=1}^8 (X^i)^2$. Then*

$$\det X = (X^0)^2 - (X^9)^2 - \sum_{i=1}^8 (X^i)^2 = -\eta_{\mu\nu} X^\mu X^\nu, \quad \eta = \text{diag}(-1, \underbrace{+1, \dots, +1}_9), \quad (167)$$

so the quadratic form $-\det$ on the ten-dimensional space $H_2(\mathbb{O})$ has Lorentzian signature $(1, 9)$ in the $(-, +, \dots, +)$ convention: $(H_2(\mathbb{O}), -\det) \cong \mathbb{R}^{1,9}$. [Theorem.]

Proof. By (166), $\det X = ab - n(x) = (X^0 + X^9)(X^0 - X^9) - \sum_i (X^i)^2 = (X^0)^2 - (X^9)^2 - \sum_i (X^i)^2$, which is the displayed Minkowski form. By Sylvester's law of inertia the diagonal pair contributes one $+$ (the X^0 mode) and one $-$ (the X^9 mode), and $-n(x)$ contributes eight further $-$'s; hence $\det$ has signature $(1, 9)$ and $-\det$ is the single-timelike $\mathbb{R}^{1,9}$. $\square$

Theorem B.2 (Octonionic exceptional isomorphism; $SL(2, \mathbb{O}) \cong \text{Spin}(1, 9)$). *There is a Lie-group isomorphism $SL(2, \mathbb{O}) \cong \text{Spin}(1, 9)$, under which the defining 10-dimensional vector representation acts on $(H_2(\mathbb{O}), -\det) \cong \mathbb{R}^{1,9}$ and the two chiral 16-dimensional spinor representations are realised on $\mathbb{O}^2 \cong \mathbb{R}^{16}$. This is the octonionic ($n = 8$) member of the classical family $SL(2, \mathbb{A}) \cong \text{Spin}(n+1, 1)$ for the normed division algebras $\mathbb{A} = \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ of dimension $n = 1, 2, 4, 8$:*

$$\begin{aligned} SL(2, \mathbb{R}) &\cong \text{Spin}(2, 1), & SL(2, \mathbb{C}) &\cong \text{Spin}(3, 1), \\ SL(2, \mathbb{H}) &\cong \text{Spin}(5, 1), & SL(2, \mathbb{O}) &\cong \text{Spin}(9, 1), \end{aligned} \quad (168)$$

with $\text{Spin}(9, 1) = \text{Spin}(1, 9)$ up to convention. [Theorem.]

Proof (statement and status). The set of determinant- and Hermiticity-preserving actions on $H_2(\mathbb{O})$ does not close under naive matrix multiplication, since octonion non-associativity obstructs $g X g^\dagger$. Accordingly $SL(2, \mathbb{O})$ is defined as $\text{Spin}(1, 9)$ acting through left/right octonion-multiplication generators chosen so that the determinant (166) and the spinor bilinears $\psi^\dagger X \psi$ on $\psi \in \mathbb{O}^2$ are covariant; the double cover $\text{Spin}(1, 9) \rightarrow SO(1, 9)$ is the statement that the spinor action is two-valued on the vector action of Theorem B.1. These isomorphisms are classical (Sudbery; Kugo–Townsend; Manogue–Schray; Baez). $\square$

Theorem B.3 (The physical Lorentz four-plane; $SL(2, \mathbb{C}) \subset SL(2, \mathbb{O})$). *The chain of composition-algebra inclusions $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O}$ induces a chain of spin inclusions; in particular*

$$\text{Spin}(1, 3) = SL(2, \mathbb{C}) \subset SL(2, \mathbb{O}) = \text{Spin}(1, 9) \quad (169)$$

via the sub-block $H_2(\mathbb{C}) \subset H_2(\mathbb{O})$, where $H_2(\mathbb{C}) = \{[a, z], [\bar{z}, b] : a, b \in \mathbb{R}, z \in \mathbb{C}\}$ has real dimension 4, $\det = ab - |z|^2$ is the (1, 3) Minkowski form, and $\det|_{H_2(\mathbb{C})}$ is $SL(2, \mathbb{C})$ -covariant in the standard four-dimensional way. The ambient $\mathbb{R}^{1,9}$ is the indefinite arena; the physical (1, 3) is a soldered four-plane inside it. [Theorem.]

Proof. Restrict the change of variables of [Theorem B.1](#) to $z \in \mathbb{C} \subset \mathbb{O}$: $\det = ab - |z|^2 = (X^0)^2 - (X^3)^2 - (X^1)^2 - (X^2)^2$ is the (1, 3) form. The covariance of the $SL(2, \mathbb{C})$ action on $H_2(\mathbb{C})$ is the textbook 4D statement $X \mapsto gXg^\dagger$, $\det(gXg^\dagger) = |\det g|^2 \det X = \det X$ for $g \in SL(2, \mathbb{C})$. $\square$

B.2 The cubic norm N is genuinely indefinite

The Albert algebra $J_3(\mathbb{O})$ of 3×3 octonion-Hermitian matrices, $\dim_{\mathbb{R}} = 3 + 3 \cdot 8 = 27$, carries the quadratic trace form $Q(X) = \text{Tr}(X^2)$ and the cubic norm $N(X) = \det X$ (see ?? and ??). On a diagonal Jordan frame $X = \text{diag}(\xi_1, \xi_2, \xi_3)$ one has $Q = \sum_i \xi_i^2$ and $N = \xi_1 \xi_2 \xi_3$.

Theorem B.4 (Separation of invariance groups). *The automorphism group $F_4 = \text{Aut}(J_3(\mathbb{O}))$ (compact, $\dim 52$) preserves the Jordan product, hence preserves both $Q = \text{Tr}(X^2)$ and N ; since Q is positive-definite, $F_4 \subset SO(27)$. The reduced structure group $E_{6(-26)}$ ($\dim 78$) preserves the cubic norm N only up to a real character $\lambda(g)$, $N(gX) = \lambda(g)N(X)$, and does not preserve Q . [Theorem.]*

Proof. Automorphisms fix $\mathbb{K}$, commute with $\circ$, and hence fix every coefficient of the characteristic polynomial, in particular $\text{Tr}(X^2)$ and $\det = N$; Euclidean signature of Q then forces $F_4 \subset SO(27)$. The reduced structure group is by definition the linear group fixing N up to scalar. If it also preserved Q it would lie in $SO(27)$ and be compact, contradicting $\dim E_{6(-26)} = 78 > 52 = \dim F_4$ together with the non-compactness of the real form $E_{6(-26)}$ (its maximal compact is exactly F_4 , so the 26 non-compact directions move Q). Hence Q is not E_6 -invariant; only N is. $\square$

Remark B.5 (Orbits are classified by rank and N , not by (Q, N)). Real E_6 -orbits on the **27** are classified by $\text{rank} \in \{0, 1, 2, 3\}$ and the value of N , via the E_6 -covariant sharp map $X^\#$ ($\text{rank } X \leq 1 \Leftrightarrow X^\# = 0$; $\text{rank } X \leq 2 \Leftrightarrow N(X) = 0$). It is *not* the pair (Q, N) that separates orbits, since Q is F_4 -invariant only. This corrects any statement to the contrary (cf. ??).

Theorem B.6 (Indefiniteness, and the Minkowski block inside N). *The cubic norm N takes strictly positive, strictly negative, and zero values on $J_3(\mathbb{O}) \cong \mathbb{R}^{27}$ (it is an odd-degree form, $N(-X) = -N(X)$), so it is genuinely indefinite. Moreover, restricting N to the block obtained by fixing the third pivot idempotent, $\xi_3 = 1$, $z_1 = z_2 = 0$, and freezing $Y = \begin{pmatrix} \xi_1 & z_3 \\ \bar{z}_3 & \xi_2 \end{pmatrix} \in H_2(\mathbb{O})$,*

$$N(X)|_{\text{block}} = \xi_1 \xi_2 - n(z_3) = \det Y, \quad (170)$$

so the indefinite Minkowski form $-\det = -\eta$ of [Theorem B.1](#) is the restriction of the cubic norm N to an $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ sub-block of $J_3(\mathbb{O})$. [Theorem.]

Proof. $N(\text{diag}(1, 1, 1)) = 1 > 0$, $N(\text{diag}(-1, 1, 1)) = -1 < 0$, $N(\text{diag}(0, 1, 1)) = 0$, and oddness is immediate from the cubic homogeneity; (170) is the direct evaluation of N on the stated block. $\square$

Theorem B.7 (Consistency: definite $K|_{\mathfrak{m}}$ and indefinite N live on different spaces). *The positive-definite coset/target metric $K|_{\mathfrak{m}}$ and the indefinite spacetime form are mutually consistent because they are forms on different representation spaces. The coset metric is positive-definite on $\mathfrak{m} = \mathbf{16}_{-3} \oplus \mathbf{16}_{+3}$ (the adjoint complement, the 32-dimensional internal tangent of E_{III}), and is unique by Schur ($\mathfrak{m}$ is real-irreducible of complex type, commutant $\mathbb{C}$; see ??). The cubic norm N is indefinite and lives on the **27**, a different module. Therefore “the coset metric is Euclidean” and*

“the cubic norm is indefinite” are statements about distinct bilinear data on distinct modules; there is no tension. The Lorentzian signature used for spacetime is imported from $N|_{H_2(\mathbb{O})}$ via the soldering map $P3$, and is never asked of $K|_{\mathfrak{m}}$. [Theorem.]

B.3 The soldering postulate (P3)

We now state the soldering precisely. Throughout, P3 is a *postulate*: added structure, not a theorem of the coset geometry. The theorem-level inputs are: a vacuum manifold $M = \text{EIII} = E_6/(\text{Spin}(10) \times U(1))$, $\dim_{\mathbb{R}} M = 32$, with reductive split $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$, $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$, $\mathfrak{m} = \mathbf{16}_{-3} \oplus \mathbf{\overline{16}}_{+3}$, and Maurer–Cartan form $\theta = \theta_{\mathfrak{h}} + \theta_{\mathfrak{m}}$ (??, [section E](#)); together with the Minkowski-block facts of [Theorem B.1–Theorem B.3](#) and [Theorem B.6](#). The 32 clock fields are the coordinates along $\mathfrak{m}$: 28 are physical massive scalars (single common radiative mass, Schur on the irreducible $\mathfrak{m}$) and 4 are gauge/soldering modes eaten by world-volume diffeomorphism plus local Lorentz (see ??).

Postulate B.8 (Soldering / frame identification, P3a). There exists a soldering map σ identifying four distinguished clock-field frame directions, written e^a_{μ} ($a = 0, 1, 2, 3$; μ a world index), with a tangent four-plane $\Pi_x \subset T_x \mathcal{M}^4$ of an emergent four-manifold $\mathcal{M}^4$. The vierbein is the pull-back of the Maurer–Cartan component along these directions,

$$e^a_{\mu}(x) = \langle \tau^a, \theta_{\mathfrak{m}}(\partial_{\mu}) \rangle, \quad (171)$$

where $\{\tau^a\}_{a=0}^3$ is the image of the $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ basis under the embedding of [Theorem B.11](#). The vierbein *emerges dynamically*, as the pull-back of $\theta_{\mathfrak{m}}$. [*Postulate P3.*]

Postulate B.9 (Induced Minkowski form, P3b). The induced quadratic form on Π_x is the Minkowski form inherited from the cubic norm via the $H_2(\mathbb{C})$ block,

$$g_{\mu\nu}(x) = e^a_{\mu}(x) e^b_{\nu}(x) \eta_{ab}, \quad \eta_{ab} = \text{diag}(-1, +1, +1, +1), \quad (172)$$

with η the restriction of $-\det = -N$ to the $H_2(\mathbb{C})$ sub-block ([Theorem B.3](#), [Theorem B.6](#)). [*Postulate P3.*]

Postulate B.10 (Dimension and signature are added, P3c). The numbers $d = 4$ and signature $(1, 3) = (-, +, +, +)$ are *part of P3*: added structure. They are *not* consequences of the Killing form, the coset metric, or any rank lemma. [*Postulate P3.*]

Postulate B.11 (Frame reduction by a vierbein condensate, P3d). A vierbein condensate (a VEV of the soldering bilinear) selects the $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ sub-block of $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ and breaks the ambient frame symmetry

$$\text{Spin}(1, 9) = SL(2, \mathbb{O}) \longrightarrow SL(2, \mathbb{C}) \times \text{Spin}(6) = \text{Spin}(1, 3) \times \text{Spin}(6), \quad (173)$$

the orthogonal complement of $\mathbb{R}^{1,3}$ in $\mathbb{R}^{1,9}$ being the six-dimensional spacelike $\mathbb{R}^6$ acted on by $\text{Spin}(6) \cong SU(4)$ (internal). The residual $\text{Spin}(1, 3) = SL(2, \mathbb{C})$ is the local Lorentz group of the soldered four-plane. [*Postulate P3.*]

Theorem B.12 (Internal consistency of P3). *Given [Theorem B.8–Theorem B.11](#), the tensor $g_{\mu\nu}$ of (172) is a non-degenerate symmetric rank-2 tensor of Lorentzian signature $(1, 3)$ on $\mathcal{M}^4$, invariant under the residual local Lorentz group $SO(1, 3)$; the four frame modes e^a_{μ} carry exactly the gauge degrees of freedom eaten by world-volume diffeomorphisms together with local Lorentz, consistent with the count $32 = 28 \text{ physical} + 4 \text{ gauge/soldering}$. [Theorem (conditional on P3).]*

Proof. Non-degeneracy and signature are inherited from $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ (Theorem B.3) whenever $\det(e^a_\mu) \neq 0$, which holds for a generic condensate; $SO(1,3)$ -invariance is the defining relation $\eta_{ab} = \Lambda^c_a \Lambda^d_b \eta_{cd}$ for $\Lambda \in SO(1,3)$. The gauge count is the Ward identity $\nabla_\mu(\delta\Gamma/\delta e^a_\mu) = 0$, so the four frame directions are protected by diffeomorphism plus local Lorentz, not by a Hessian zero. $\square$

Remark B.13 (What P3 does not claim). P3 does not claim that $K|_{\mathfrak{m}}$ is Lorentzian: it stays positive-definite, the kinetic/target-space metric of the σ -model. P3 does not use a (4, 28) split of $\mathfrak{m}$ with Lorentzian signature: the 4 are gauge soldering modes, the 28 are Euclidean target coordinates. P3 does not build the four-plane from $\nabla I_2, \nabla I_3$ singlets: $\mathfrak{m}$ contains no $SO(10)$ singlet, and $\nabla Q|_{X_0} = 2X_0 \notin \mathfrak{m}$. The four-plane is the soldered image of $H_2(\mathbb{C}) \subset \mathbf{27}$, not a sub-module of $\mathfrak{m}$.

Remark B.14 (Anti-theorem: there is no Killing-form derivation of (1, 3)). There is *no* derivation of $d = 4$ or of the signature (1, 3) from the Killing form of E_6 , from the coset metric $K|_{\mathfrak{m}}$, or from any rank lemma. The Killing form of compact E_6 is negative-definite; its restriction to $\mathfrak{m}$ is (sign-flipped to) positive-definite of signature (32, 0) — zero Lorentzian content. Any claim of a (4, 28) or (1, 3) Killing/coset signature is false. Lorentzian signature enters only through P3, rooted in the indefinite cubic norm $N / H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$.

B.4 Emergent time: Page–Wootters clock and Osterwalder–Schrader reconstruction

Lorentzian time is recovered by a relational-clock plus reconstruction-theorem mechanism; no “multiply e^0 by i ” Wick trick is used. The soldered metric $\eta^{(1,3)}$ is already Lorentzian *before* any continuation; the role of analytic continuation is confined to the correlation functions in the clock-time argument, justified by reflection positivity.

Postulate B.15 (Timelike clock selection, P3e). Among the 32 clock fields, the soldering condensate (Theorem B.11) selects a distinguished timelike direction, whose clock field we call ϕ^0 — concretely the trace/ X^0 direction of the soldered $H_2(\mathbb{C}) \subset J_3(\mathbb{O})$ block (the unique η -timelike soldering leg e^0_μ). The selection of *one* timelike direction is precisely the (1, 3) (rather than (2, 8), etc.) content of P3. [Postulate P3.]

Theorem B.16 (Page–Wootters relational evolution). *Promote ϕ^0 to a Page–Wootters clock: enlarge the kinematical Hilbert space to $\mathcal{H} = \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{sys}}$ and impose the induced-gravity Hamiltonian constraint*

$$\hat{H}_{\text{phys}} |\Psi\rangle\rangle = 0, \quad \hat{H}_{\text{phys}} = \hat{H}_{\text{clock}} + \hat{H}_{\text{sys}} (+ \text{interaction}), \quad (174)$$

on physical states. Conditioning on clock readings $\phi^0 = \tau$, $|\psi(\tau)\rangle_{\text{sys}} = \langle \phi^0 = \tau | \Psi \rangle\rangle$, yields the Schrödinger equation in relational time,

$$i \partial_\tau |\psi(\tau)\rangle_{\text{sys}} = \hat{H}_{\text{sys}} |\psi(\tau)\rangle_{\text{sys}}, \quad (175)$$

with the unitary group $U(\tau) = e^{-i\hat{H}_{\text{sys}}\tau}$ recovered on physical states in the semiclassical regime where the soldered $g_{\mu\nu}$ admits a timelike Killing vector and $\hat{H}_{\text{sys}} = \int T_{00}$ along it. [Theorem (Page–Wootters), conditional on the constraint and a good clock; global hyperbolicity assumed.]

Proof. Under (174) with a good clock ($\hat{H}_{\text{clock}} = \hat{p}_{\phi^0}$ canonically conjugate to $\hat{\phi}^0$), the conditioned state obeys (175); this is the Page–Wootters theorem. The identification $t \leftrightarrow X^0$ is $\hat{H}_{\text{sys}} = \int T_{00}$ along the timelike Killing direction. $\square$

Construction B.17 (Osterwalder–Schrader reconstruction along ϕ^0). The Euclidean σ -model built on the positive-definite $K|_{\mathfrak{m}}$ defines a Euclidean functional measure with Schwinger functions $S_n(x_1, \dots, x_n)$. Take $\tau = \phi^0$ as Osterwalder–Schrader time and assume the axioms: (OS0) Euclidean invariance of the spatial slice; (OS1) reflection positivity with respect to $\theta : \tau \mapsto -\tau$ about a ϕ^0 -slice,

$$\sum_{i,j} \bar{c}_i c_j S(\theta F_i \cdot F_j) \geq 0, \quad \text{supp } F_i \subset \{\tau > 0\}; \quad (176)$$

(OS2) symmetry and clustering.

Theorem B.18 (OS reconstruction $\Rightarrow$ Lorentzian Wightman theory). *Given (OS0)–(OS2), the Osterwalder–Schrader reconstruction theorem produces a Hilbert space $\mathcal{H}$, a positive self-adjoint Hamiltonian $\hat{H} \geq 0$ generating τ -translations as the contraction semigroup $e^{-\hat{H}\tau}$, and, by analytic continuation of the Schwinger functions in the time argument $\tau \rightarrow it$, Lorentzian Wightman functions on $(\mathcal{M}^4, \eta^{(1,3)})$ with unitary evolution $e^{-i\hat{H}t}$ covariant under the reconstructed $SO(1,3)$. The OS theorem itself is rigorous given its hypotheses [Theorem]; that the UCFT σ -model satisfies reflection positivity (176) along ϕ^0 is [Indicative] — the precise property to be checked, and the rigorous replacement for the Wick trick.*

Proof (theorem part). Reflection positivity (176) induces a physical inner product on the quotient $\{F : \tau > 0\}/(\text{null})$; the generator of the resulting contraction semigroup $e^{-\hat{H}\tau}$ is self-adjoint and positive. The Schwinger functions, analytic in a tube, continue to Wightman distributions satisfying the Wightman axioms, including the spectrum condition and Lorentz covariance under the reconstructed $SO(1,3)$ (Osterwalder–Schrader, 1973, 1975). $\square$

Remark B.19 (Why this is not the Wick trick). The continuation $\tau \rightarrow it$ acts on the correlation functions in the *clock-time argument*, justified by reflection positivity (176) and the resulting analyticity tube — it is *not* the ad hoc replacement $e^0_\mu \mapsto i e^0_\mu$ of a frame field. The metric $\eta^{(1,3)}$ is the soldered (P3) form, Lorentzian before any continuation; OS reconstruction supplies the unitary dynamics $e^{-i\hat{H}t}$ and the positive-energy spectrum, with ϕ^0 (the Page–Wootters clock) serving as the OS reflection time. The two mechanisms dovetail: Theorem B.16 gives relational unitarity $U(\tau)$ on physical states, and Theorem B.18 gives the positive-energy, Lorentz-covariant Wightman theory.

B.5 Summary: theorem versus postulate

The division of labour is as follows.

- **Theorems** (proved): $(H_2(\mathbb{O}), -\det) \cong \mathbb{R}^{1,9}$ (Theorem B.1); $SL(2, \mathbb{O}) \cong \text{Spin}(1, 9)$ and $SL(2, \mathbb{C}) \cong \text{Spin}(1, 3)$ (Theorem B.2, Theorem B.3); F_4 preserves Q , E_6 preserves only N (Theorem B.4); N indefinite and the Minkowski block $\det Y \subset N$ (Theorem B.6); definite $K|_{\mathfrak{m}}$ and indefinite N are consistent (Theorem B.7); internal consistency of $g_{\mu\nu}$ given P3 (Theorem B.12); the Page–Wootters and OS reconstruction theorems (Theorem B.16, Theorem B.18).
- **Postulate** (P3, added structure, cf. P1–P6): the soldering map, the choice of a four-plane, $d = 4$, signature $(1, 3)$, the vierbein condensate and frame reduction, and the timelike clock selection (Theorem B.8–Theorem B.11, Theorem B.15).
- **Indicative**: that the UCFT σ -model satisfies reflection positivity along ϕ^0 (Theorem B.18).

In short, the Lorentzian arena is a *conditional embedding*: the algebra $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ and $SL(2, \mathbb{O}) \cong \text{Spin}(1, 9)$ are theorems, while the selection of the physical (1, 3) four-plane and the recovery of unitary Lorentzian dynamics rest on the soldering postulate P3 plus an indicative reflection-positivity assumption.

C Branchings and the Coset Module

This appendix collects the representation-theoretic facts on which the dynamical claims of the manuscript rest: the branching of the **78** (adjoint) and **27** (fundamental) of E_6 under the maximal compact subgroup $H = \text{Spin}(10) \times U(1)$, the $U(1)$ charge normalization, and the structural fact that the coset tangent module

$$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$$

is *real-irreducible of complex type*, with real commutant $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$. These results fix (i) that the invariant coset metric $K|_{\mathfrak{m}}$ is unique up to scale and positive-definite ([Theorem C.5](#)); and (ii) that one radiative Coleman–Weinberg loop produces a *single common mass* on all 32 clock fields by Schur’s lemma, the input used in the mass-spectrum and induced-gravity units ([Theorem C.7](#)).

Throughout, $H = \text{Spin}(10) \times U(1)$ is the (compact) stabilizer of a rank-one primitive idempotent, so the relevant real form for the dynamics is the compact E_6 (equivalently $E_{6(-14)}$, whose maximal compact is exactly H); see ???. The Killing form on this real form is negative-definite on $\mathfrak{e}_6$, hence *positive-definite* on the coset $\mathfrak{m}$ once the standard sign for symmetric spaces of compact type is taken. There is no (4, 28) Killing signature and no Lorentzian content in $\mathfrak{m}$; Lorentzian signature is supplied separately by the soldering postulate P3 (??).

C.1 The $U(1)$ charge normalization

Fix the abelian generator $Y \in \mathfrak{u}(1) \subset \mathfrak{h}$ that commutes with $\mathfrak{so}(10)$. We normalize Y so that the $\text{Spin}(10)$ spinor **16** appearing in the *fundamental* **27** carries charge +1:

$$q(\mathbf{16} \subset \mathbf{27}) = +1. \quad (177)$$

This is the smallest charge quantum appearing in the **27** and is the convenient unit because it makes every charge in the **27** and the **78** an integer. With this choice the remaining charges are fixed by requiring (a) that the cubic invariant N (the E_6 singlet in $\text{Sym}^3(\mathbf{27})$) be neutral, and (b) that the adjoint decomposition close consistently. We record both branchings, then verify these two conditions.

Theorem C.1 (Branchings under $H = \text{Spin}(10) \times U(1)$). *With the normalization (177), the fundamental and adjoint of E_6 decompose as*

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \quad (178)$$

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}. \quad (179)$$

The coset (broken) directions are

$$\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1) = \mathbf{45}_0 \oplus \mathbf{1}_0, \quad \mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad \dim_{\mathbb{R}} \mathfrak{m} = 32. \quad (180)$$

Proof. Dimensions and $\text{Spin}(10)$ content. $1 + 10 + 16 = 27$ and $45 + 1 + 16 + 16 = 78$, matching $\dim \mathbf{27} = 27$ and $\dim \mathbf{78} = 78$. The $\text{Spin}(10)$ pieces are forced by the chain $E_6 \supset \text{Spin}(10) \times U(1)$: the adjoint **78** contains $\mathfrak{so}(10) = \mathbf{45}$, the abelian generator **1**, and the off-diagonal coset which is

the unique pair of chiral half-spinors $\mathbf{16} \oplus \overline{\mathbf{16}}$ (a Hermitian symmetric pair, since $E_6/H = \text{EIII}$ is Hermitian symmetric of complex type). The $\mathbf{27}$ is the smallest faithful rep; under $\text{Spin}(10)$ it is the sum of a singlet, a vector $\mathbf{10}$, and a spinor $\mathbf{16}$.

Charge assignment. Set $q(\mathbf{16} \subset \mathbf{27}) = +1$ by (177). The cubic E_6 singlet $N \in \text{Sym}^3(\mathbf{27})$ contains a $\text{Spin}(10)$ -singlet coupling $\mathbf{1} \cdot \mathbf{10} \cdot \mathbf{10}$ and a coupling $\mathbf{10} \cdot \mathbf{16} \cdot \mathbf{16}$ (the unique $\text{Spin}(10)$ -invariants in $\mathbf{10} \otimes \mathbf{16} \otimes \mathbf{16}$ and in $\mathbf{1} \otimes \mathbf{10} \otimes \mathbf{10}$). Charge neutrality of N requires

$$q(\mathbf{1}) + 2q(\mathbf{10}) = 0, \quad q(\mathbf{10}) + 2q(\mathbf{16}) = 0.$$

With $q(\mathbf{16}) = +1$ the second gives $q(\mathbf{10}) = -2$, and the first gives $q(\mathbf{1}) = -2q(\mathbf{10}) = +4$, yielding (178).

For the adjoint, the coset $\mathfrak{m}$ transforms in the $\mathbf{16} \oplus \overline{\mathbf{16}}$ that maps $\mathbf{27} \rightarrow \mathbf{27}$ raising/lowering the $\text{Spin}(10)$ grading by one spinor step. Acting on the graded $\mathbf{27}$ (178), the raising operator carries the $\mathbf{16}_{+1}$ piece into the $\mathbf{1}_{+4} \oplus \mathbf{10}_{-2}$ pieces; matching $U(1)$ charges across the three components forces the coset spinor to carry charge -3 and its conjugate $+3$:

$$q(\mathbf{16} \subset \mathfrak{m}) = q(\mathbf{1}_{+4}) - q(\mathbf{16}_{+1}) = 4 - 1 = 3 \quad (\text{or } q(\mathbf{10}_{-2}) - q(\mathbf{16}_{+1}) = -3),$$

which are conjugate, so $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ with the diagonal H at charge 0. This is (179) and (180). $\square$

Remark C.2. The coset $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ contains *no* $\text{Spin}(10)$ singlet, and indeed no $\text{Spin}(10)$ vector. Any decomposition of the form $2 \times \mathbf{10} + \mathbf{4}$, $\mathbf{10} + \overline{\mathbf{10}} + \mathbf{4}$, or one containing a singlet-in- $\mathfrak{m}$ is therefore *false*: those would neither reproduce the chiral spinor content of EIII nor sum correctly ($32 = 16 + 16$, not $2 \cdot 10 + 4 = 24$). The absence of a singlet in $\mathfrak{m}$ is what makes $\mathfrak{m}$ irreducible and forces the single-common-mass conclusion below.

C.2 Relevant tensor products of $\text{Spin}(10)$ spinors

We record the $\text{Spin}(10)$ tensor products used repeatedly in the gauge, metric, and Yukawa arguments. Writing $\mathbf{16}$ for a chiral half-spinor:

$$\mathbf{16} \otimes \mathbf{16} = \mathbf{10} \oplus \mathbf{120} \oplus \mathbf{126}, \tag{181}$$

$$\mathbf{16} \otimes \overline{\mathbf{16}} = \mathbf{1} \oplus \mathbf{45} \oplus \mathbf{210}. \tag{182}$$

Dimension checks: $16 \cdot 16 = 256 = 10 + 120 + 126$ and $16 \cdot 16 = 256 = 1 + 45 + 210$. In (181) the $\mathbf{10}$ is symmetric, the $\mathbf{126}$ is symmetric (self-dual 5-form), and the $\mathbf{120}$ (3-form) is antisymmetric:

$$\text{Sym}^2(\mathbf{16}) = \mathbf{10} \oplus \mathbf{126}, \quad \wedge^2(\mathbf{16}) = \mathbf{120}. \tag{183}$$

In (182) the singlet $\mathbf{1}$ is the H -invariant Hermitian pairing of a spinor with its conjugate.

Remark C.3 (Yukawa consequence). The decompositions (181)–(182) imply there is *no* H -singlet in $\mathbf{16} \otimes \mathbf{16} \otimes \{\mathbf{16}, \overline{\mathbf{16}}\}$: a scalar valued in $\mathbf{16}$ or $\overline{\mathbf{16}}$ cannot form an invariant Yukawa coupling $-y \bar{\psi} \phi \psi$. Since H is unbroken at the vacuum, fermion masses must come instead from the real $\text{Spin}(10)$ Higgs/Yukawa sector ($\mathbf{10}_H$ for Dirac masses via the $\mathbf{10} \subset \mathbf{16} \otimes \mathbf{16}$, and $\mathbf{126}_H$ for Majorana masses via the $\mathbf{126} \subset \mathbf{16} \otimes \mathbf{16}$), as assumed in Postulate P6 (subsection 13.4). This is an embedding, not a derivation.

C.3 Reality type and commutant of the coset module

We now establish the structural fact that controls both the coset metric and the radiative mass spectrum.

Theorem C.4 ($\mathfrak{m}$ is real-irreducible of complex type). *The real 32-dimensional H -module $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is irreducible over $\mathbb{R}$ and is of complex type: its real endomorphism algebra is*

$$\text{End}_H(\mathfrak{m}) \cong \mathbb{C}.$$

Proof. The complex $\text{Spin}(10)$ spinor $\mathbf{16}$ is a representation of *complex type* (it is not self-conjugate; $\overline{\mathbf{16}} \not\cong \mathbf{16}$ for $\text{Spin}(10)$, since $5 \equiv 1 \pmod{4}$ places $\text{Spin}(10)$ in the class with inequivalent chiral spinors). The $U(1)$ factor multiplies the two summands by the opposite characters $e^{-3i\theta}$ and $e^{+3i\theta}$. As a *complex* H -representation, $\mathfrak{m}_{\mathbb{C}} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is a sum of two *inequivalent* irreducibles (different $\text{Spin}(10)$ type and opposite $U(1)$ charge), so $\text{End}_{H, \mathbb{C}}(\mathfrak{m}_{\mathbb{C}}) \cong \mathbb{C} \oplus \mathbb{C}$.

Now $\mathfrak{m}$ is the *real* module obtained by viewing $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ as the realification of the single complex irreducible $\mathbf{16}_{-3}$: complex conjugation on the underlying real space is an antilinear H -map that exchanges the two summands (it sends $\mathbf{16}_{-3} \mapsto \overline{\mathbf{16}}_{+3}$, respecting both the conjugate $\text{Spin}(10)$ type and the sign-reversed $U(1)$ charge). Hence $\mathfrak{m} \cong (\mathbf{16}_{-3})_{\mathbb{R}}$, the realification of one complex irreducible. The standard reality trichotomy then gives, for a complex irreducible W that is not self-conjugate,

$$\text{End}_H(W_{\mathbb{R}}) \cong \mathbb{C},$$

i.e. complex type, and $W_{\mathbb{R}}$ is irreducible over $\mathbb{R}$ (its only real-invariant subspaces are 0 and the whole space, because any proper one would have to be one of the two complex summands, which are not $\mathbb{R}$ -invariant under the conjugation map). This proves irreducibility over $\mathbb{R}$ and $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$. $\square$

The commutant being $\mathbb{C}$ (rather than $\mathbb{R}$ or $\mathbb{H}$) is the precise sense in which $\mathfrak{m}$ is “of complex type.” It supplies exactly one $\mathbb{R}$ -linear complex structure $J \in \text{End}_H(\mathfrak{m})$ with $J^2 = -\mathbf{1}$ commuting with H — the complex structure of the Hermitian symmetric space EIII.

C.4 Uniqueness and definiteness of the invariant metric

Theorem C.5 (Unique definite invariant metric). *The space of H -invariant real symmetric bilinear forms on $\mathfrak{m}$ is one-dimensional. Its generator is the restriction of the Killing form of (compact) E_6 to $\mathfrak{m}$, and is positive-definite. Consequently the coset/target-space metric*

$$\mathfrak{g}_{\alpha\beta} = f^2 K_{\alpha\beta}|_{\mathfrak{m}}, \quad f^2 > 0, \tag{184}$$

is the unique (up to the positive scale f^2) admissible σ -model kinetic metric, and it is Euclidean.

Proof. Counting invariants via the $U(1)$ charge. An H -invariant symmetric bilinear form on $\mathfrak{m}$ is an H -invariant element of $\text{Sym}^2(\mathfrak{m}^*)$. Decompose

$$\text{Sym}^2(\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}) = \underbrace{\text{Sym}^2(\mathbf{16}_{-3})}_{\text{charge } -6} \oplus \underbrace{\text{Sym}^2(\overline{\mathbf{16}}_{+3})}_{\text{charge } +6} \oplus \underbrace{\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3}}_{\text{charge } 0}.$$

The first two blocks carry net $U(1)$ charge $\mp 6 \neq 0$, so they contain *no* $U(1)$ -singlet and hence no H -invariant: the $U(1)$ charge *forbids* the symmetric $\mathbf{16} \times \mathbf{16}$ and $\overline{\mathbf{16}} \times \overline{\mathbf{16}}$ invariants. The only charge-0 block is the cross-pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3}$, whose $\text{Spin}(10)$ content is $\mathbf{1} \oplus \mathbf{45} \oplus \mathbf{210}$ by (182). Exactly one $\text{Spin}(10)$ singlet $\mathbf{1}$ appears, and it is symmetric under the exchange that defines a bilinear *form*. Hence there is a *unique* H -invariant symmetric form, the charge-0 cross-pairing.

Identification with the Killing form and definiteness. The Killing form of E_6 restricted to $\mathfrak{m}$ is an H -invariant symmetric form; by the uniqueness just proved it must coincide (up to scale) with the cross-pairing. On the *compact* real form of E_6 the Killing form is negative-definite on all of $\mathfrak{e}_6$, hence on the symmetric complement $\mathfrak{m}$ the induced metric of the symmetric space of compact type is positive-definite; equivalently the Hermitian pairing $\langle v, v \rangle = \sum_i |v_i|^2$ furnished by the cross-pairing is strictly positive. Schur's lemma (in the strong form $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$, [Theorem C.4](#)) guarantees there is no second, independent invariant form to spoil definiteness. Positivity of the σ -model kinetic term then fixes the scale to $f^2 > 0$, giving (184). $\square$

Remark C.6. The metric (184) has signature $(32, 0)$ — Euclidean, with *zero* Lorentzian content. There is no $(4, 28)$ Killing/coset signature, and the physical Lorentzian signature $(1, 3)$ in the convention $(-, +, +, +)$ does *not* fall out of the coset geometry. Signature enters only through the soldering postulate P3, soldered from the cubic norm N via $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ (??).

C.5 One common radiative mass (Schur)

Theorem C.7 (Single common scalar mass). *At tree level the 32 clock fields are exact Goldstone modes, $\text{Hess } V_{\text{tree}}|_{\mathfrak{m}} = 0$. The one-loop Coleman–Weinberg Hessian on $\mathfrak{m}$ is an H -invariant symmetric operator; by irreducibility of $\mathfrak{m}$ over $\mathbb{R}$ ([Theorem C.4](#)) and Schur's lemma it is a single scalar multiple of the identity. Hence all clock fields acquire one common radiative mass. After removing the four gauge/soldering frame modes (eaten by diffeomorphism + local Lorentz, P3), the physical spectrum is*

$$32 \text{ real} = \underbrace{28 \text{ massive scalars (single common mass)}}_{\text{use 28 in all loop sums}} + 4 \text{ gauge/soldering (eaten)}. \quad (185)$$

With $[f] = \text{mass}$, the dimension-4 Hessian eigenvalue is $f^2 \mu^2$, so the canonically normalized physical mass is

$$m^2 = \mu^2. \quad (186)$$

Proof. Tree-level masslessness is Goldstone's theorem for the broken E_6/H directions. The one-loop effective potential is built from H -invariant data (the vacuum preserves H), so its Hessian $\mathcal{H} \in \text{End}_H(\mathfrak{m})$ is an H -equivariant symmetric operator. Over the *real* commutant $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$ a symmetric (self-adjoint) operator is real, hence lies in $\mathbb{R} \cdot \mathbf{1} \subset \mathbb{C}$: it is a real scalar times the identity. (Equivalently, the only H -invariant symmetric forms are multiples of the unique metric of [Theorem C.5](#), so $\mathcal{H} = m^2 \mathfrak{g}$.) Thus all 32 modes share one eigenvalue.

The four frame directions are not Hessian zeros of an internal singlet subspace — $\mathfrak{m}$ has no $\text{Spin}(10)$ singlet ([Theorem C.2](#)) — but are gauge modes protected by the Ward identity $\nabla_\mu(\delta\Gamma/\delta e_\mu^a) = 0$ and eaten by the diffeomorphism + local-Lorentz gauge symmetry. Removing them leaves 28 massive physical scalars, all of common mass, giving (185). Canonical normalization of the kinetic term (184) divides the dimension-4 Hessian eigenvalue $f^2 \mu^2$ by f^2 , yielding (186). $\square$

Remark C.8. The count 28 in (185) and the mass (186) are the inputs to the induced Planck mass $M_{\text{Pl}}^2 = (28/6) f^2 \mu^2 \log(\Lambda^2/\mu^2)/(16\pi^2) > 0$ and to the gauge β -function scalar count $n_S = 28$ ([section 11](#)). The 27×27 scalar block likewise inherits the multiplicities of (178), $M = \text{diag}(\mu_1 \mathbf{1}_1, \mu_{10} \mathbf{1}_{10}, \mu_{16} \mathbf{1}_{16})$, the residual symmetry being H , not E_6 .

Status of the claims in this appendix

All statements of this appendix are **theorems**: the branchings (178)–(179), the tensor products (181)–(182), the complex-type/commutant result (Theorem C.4), the unique definite metric (Theorem C.5), and the single-common-mass result (Theorem C.7) follow from the representation theory of $E_6 \supset \text{Spin}(10) \times U(1)$ on the compact real form. The *use* of these facts to assert the physical (1, 3) signature, three generations, and the Yukawa sector relies on the added postulates P3, P5, P6 and is flagged where invoked.

D The Induced Yang–Mills Coefficient

This appendix derives the one-loop coefficient of the composite $\text{SO}(10) \times U(1)$ Yang–Mills term obtained by integrating out the massive clock fields, and traces how that coefficient feeds the asymptotic-freedom slope b_0 of the functional renormalization group. The computation is presented at *derivation grade*: the heat-kernel machinery, the group-theory weights, and the resulting logarithm are all theorems given the stated background; the embedding of the clock-field σ -model into a four-dimensional Lorentzian arena is the soldering *postulate* P3, invoked only to provide the $X = M^4$ over which the proper-time integral is taken.

Throughout we use the metric signature $(-, +, +, +)$, the decay constant f with $[f] = \text{mass}$, the breaking/mass scale μ , and the UV cutoff Λ . The hatted coupling $\hat{g}^2$ is dimensionless. The relevant gauge algebra is $\mathfrak{so}_{10} \oplus \mathfrak{u}_1$, and *all* group-theory weights are taken in $\text{SO}(10)$ with the locked Casimirs

$$C_A(\text{SO}(10)) = 8, \quad T(\mathbf{16}) = 2, \quad C_2(\mathbf{16}) = \frac{45}{8}. \quad (187)$$

The E_6 value $C_A(E_6) = 12$ is *never* mixed into these sums: it is the spinor index $T(\mathbf{16}) = 2$, not a count of sixteen index-1 fundamentals, that enters.

D.1 The Maurer–Cartan curvature and the one-loop determinant

Let $g(x) \in E_6$ (compact real form, the dynamical form fixed in ??) be a coset representative of the vacuum map into $M = E_6/(\text{SO}(10) \times U(1)) = \text{EIII}$. The Maurer–Cartan form splits along the reductive decomposition $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$,

$$g^{-1}\partial_\mu g = \mathcal{A}_\mu(x) + e_\mu(x), \quad \mathcal{A}_\mu \in \mathfrak{so}_{10} \oplus \mathfrak{u}_1, \quad e_\mu \in \mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (188)$$

with $\mathfrak{h}$ -part $\mathcal{A}_\mu$ the composite connection of ?? and $\mathfrak{m}$ -part e_μ the clock-field vielbein. The $\mathfrak{h}$ -curvature

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + [\mathcal{A}_\mu, \mathcal{A}_\nu] = -P_{\mathfrak{h}}([e_\mu, e_\nu]) \quad (189)$$

is, by the Maurer–Cartan structure equation, the projection onto $\mathfrak{h}$ of the bracket of two coset directions. This is the gauge field whose kinetic term we now generate radiatively.

Remark D.1 (Corrected massive multiplet). The fluctuations that couple minimally to $\mathcal{A}_\mu$ are the *coset* directions $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, a single real-irreducible H -module of complex type (commutant $\mathbb{C}$; see ??). It contains *no* $\text{SO}(10)$ singlet. Earlier drafts decomposed the massive sector as $\mathbf{10}_{+2} \oplus \mathbf{10}_{-2} \oplus \mathbf{4}_0$, with total Dynkin index $\frac{5}{2}$; that decomposition is incorrect (it places a nonexistent singlet $\mathbf{4}_0$ in $\mathfrak{m}$ and uses fundamental rather than spinor reps). The corrected weight is computed below from $\mathbf{16} \oplus \overline{\mathbf{16}}$.

Expanding the kinetic action to quadratic order about the vacuum, the 32 real coset fluctuations ξ acquire the common radiative mass $m^2 = f^2 \mu^2$ (one common eigenvalue by Schur on the irreducible

$\mathfrak{m}$; subsection 11.1). Of these, 28 are physical massive scalars and 4 are the gauge/soldering frame modes eaten by diffeomorphism + local Lorentz. The Euclidean quadratic action in the composite background reads

$$S_m[\xi] = \frac{1}{2} \int_X \xi^\dagger [-D^2 + m^2] \xi d^4x, \quad D_\mu = \partial_\mu + \mathcal{A}_\mu, \quad (190)$$

and integrating out ξ produces the determinant $[\det(-D^2 + m^2)]^{-1/2}$, i.e. the one-loop effective action

$$\Gamma_{1L} = \frac{1}{2} \text{Tr} \log(-D^2 + m^2) = -\frac{1}{2} \int_{\Lambda^{-2}}^\infty \frac{ds}{s} e^{-sm^2} \text{Tr} e^{-s(-D^2)}. \quad (191)$$

D.2 Heat-kernel extraction of the gauge weight

For small proper time the heat trace expands as

$$\text{Tr} e^{-s(-D^2)} = \frac{1}{(4\pi s)^2} \sum_{n \geq 0} s^n a_{2n}, \quad (192)$$

and the $\mathcal{F}^2$ content sits in the Seeley–DeWitt coefficient a_4 . For a real scalar multiplet in an H -representation R with Dynkin index $T(R)$ normalized by $\text{Tr}_R(T^a T^b) = T(R) \delta^{ab}$,

$$a_4(R) \supset \frac{1}{12} \int_X \text{Tr}_R(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) d^4x, \quad (193)$$

the omitted terms being curvature-squared invariants on X suppressed by powers of μ/Λ .

Lemma D.2 (Total gauge weight of the coset multiplet). *The total Dynkin index carried by the massive coset multiplet $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is*

$$T_{\mathfrak{m}} = T(\mathbf{16}) + T(\overline{\mathbf{16}}) = 2 + 2 = 4, \quad \dim \mathfrak{m} = 16 + 16 = 32. \quad (194)$$

Proof. $\mathfrak{m}$ is a single real-irreducible H -module of complex type realized as the chiral spinor pair $\mathbf{16} \oplus \overline{\mathbf{16}}$ of $\text{SO}(10)$ (??). The Dynkin index is additive over a direct sum, and conjugation preserves it, so $T(\overline{\mathbf{16}}) = T(\mathbf{16}) = 2$ from (187). Hence $T_{\mathfrak{m}} = 4$ and $\dim \mathfrak{m} = 32$. (The $\text{U}(1)$ charges ± 3 do not enter the $\text{SO}(10)$ trace normalization; they contribute only to the Abelian factor, treated separately.) The 4 soldering frame modes are eaten gauge degrees of freedom carrying no net dynamical $\text{SO}(10)$ index, so the gauge-coupling sum is the same whether weighted by the full 32 or by the 28 physical scalars: the $\text{SO}(10)$ trace of any H -singlet eaten combination vanishes. We therefore use the H -irreducible weight $T_{\mathfrak{m}} = 4$. This corrects the legacy value $\frac{5}{2}$ noted in Theorem D.1. $\square$

Inserting (193) into (191) and using Theorem D.2,

$$\Gamma_{1L} \supset -\frac{1}{2} \int_{\Lambda^{-2}}^\infty \frac{ds}{s} e^{-sm^2} \frac{1}{(4\pi)^2} \frac{s^2}{12} T_{\mathfrak{m}} \int_X \text{Tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) d^4x. \quad (195)$$

The proper-time integral is logarithmically divergent: with $\int_{\Lambda^{-2}}^\infty ds s e^{-sm^2}/s = \int_{\Lambda^{-2}}^\infty ds e^{-sm^2} \rightarrow m^{-2}$ at the upper end and the cutoff supplying $\log(\Lambda^2/m^2)$ in the $s^2 \cdot s^{-1}$ measure, one obtains

$$\Gamma_{1L} \supset -\frac{1}{4} \frac{T_{\mathfrak{m}}}{24\pi^2} \log \frac{\Lambda^2}{m^2} \int_X \text{Tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) d^4x. \quad (196)$$

D.3 The induced coupling

Theorem D.3 (Induced Yang–Mills coefficient). *Integrating out the massive clock fields induces the local Yang–Mills term*

$$S_{\text{YM}} = -\frac{1}{4\hat{g}^2} \int_X \text{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}) d^4x, \quad (197)$$

with the one-loop coefficient

$$\hat{g}^{-2} = \frac{T_{\text{m}}}{24\pi^2} f^2 \log \frac{\Lambda^2}{\mu^2} = \frac{4}{24\pi^2} f^2 \log \frac{\Lambda^2}{\mu^2} = \frac{f^2}{6\pi^2} \log \frac{\Lambda^2}{\mu^2} > 0, \quad (198)$$

where $m^2 = f^2\mu^2$ has been restored and f^2 extracted by the canonical normalization of the kinetic term.

Proof. Match (196) to (197): the coefficient of $-\frac{1}{4}\text{Tr}(\mathcal{F}^2)$ in $\Gamma_{1\text{L}}$ is $\hat{g}^{-2} = T_{\text{m}} \log(\Lambda^2/m^2)/(24\pi^2)$ in units where the σ -model is canonically normalized; restoring the decay constant f that sets the overall scale of the Maurer–Cartan kinetic term multiplies this by f^2 , and $\log(\Lambda^2/m^2) = \log(\Lambda^2/\mu^2) + \log(1/f^2) \rightarrow \log(\Lambda^2/\mu^2)$ to leading log. With $T_{\text{m}} = 4$ from Theorem D.2, $\hat{g}^{-2} = 4f^2 \log(\Lambda^2/\mu^2)/(24\pi^2) = f^2 \log(\Lambda^2/\mu^2)/(6\pi^2)$. Positivity follows from $\Lambda > \mu$, $f^2 > 0$. $\square$

Remark D.4 (Comparison with the legacy coefficient). The corrected prefactor $T_{\text{m}}/(24\pi^2) = 4/(24\pi^2) = 1/(6\pi^2)$ replaces the legacy $\frac{5}{2} \cdot 1/(24\pi^2) = 5/(48\pi^2)$, which used the spurious $\mathbf{10}_{\pm 2} \oplus \mathbf{4}_0$ decomposition with index $\frac{5}{2}$. The corrected value is larger by a factor $T_{\text{m}}/(5/2) = 4/(5/2) = 8/5$. The physical content – a positive, logarithmically running induced gauge coupling – is unchanged; only the numerical prefactor is fixed to the spinor-pair value.

Remark D.5 (Normalization convention). $\text{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})$ is normalized in the fundamental $\mathbf{10}$ of $\text{SO}(10)$, $\text{Tr}_{\mathbf{10}}(T^a T^b) = T(\mathbf{10}) \delta^{ab}$ with $T(\mathbf{10}) = 1$. Any rescaling of this trace rescales $\hat{g}^2$ uniformly and drops out of all physical (renormalization-group) statements below.

D.4 Feeding the asymptotic-freedom slope b_0

The coefficient (198) sets the boundary value of the gauge coupling at the breaking scale; its scale dependence is governed by the full one-loop β -function of the FRG sector (subsection 12.2). With the canonical dimensionless coupling $\hat{g}^2$ the locked form is

$$\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2} \hat{g}^4, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} T n_F - \frac{1}{6} T n_S, \quad (199)$$

where the canonical $+2\hat{g}^2$ term enforces a *Gaussian* UV fixed point $\hat{g}_\star^2 = 0$: the gauge sector is asymptotically free, and there is no interacting gauge fixed point.

Theorem D.6 (Slope from the corrected matter content). *With the locked Casimirs (187), the slope is $b_0 = 26$ for one $\text{SO}(10)$ generation ($n_F = 1$, $n_S = 2$) and $b_0 = 12$ for the three-generation theory of subsection 13.3 ($n_F = 3$, $n_S = 28$).*

Proof. *One generation.* Insert $C_A = 8$, $T = T(\mathbf{16}) = 2$, $n_F = 1$, and $n_S = 2$ (minimal GUT scalars at the matching scale):

$$b_0(1) = \frac{11}{3} \cdot 8 - \frac{4}{3}(2)(1) - \frac{1}{6}(2)(2) = \frac{78}{3} = 26. \quad (200)$$

Three generations. With three chiral **16** representations ($n_F = 3$, not 3×16) and $n_S = 28$ massive coset scalars,

$$b_0(3) = \frac{11}{3}(8) - \frac{4}{3}(2)(3) - \frac{1}{6}(2)(28) = \frac{88 - 24 - 28}{3} = 12. \quad (201)$$

Hence $b_0(3 \text{ gen}) = 12$, the value used everywhere downstream. $\square$

Remark D.7 (Why the slope does not change the UV behavior). The magnitude of b_0 controls the *rate* of running but not the existence of the UV fixed point: the canonical $+2\hat{g}^2$ term in (199) produces $\hat{g}_*^2 = 0$. In either case the gauge sector is asymptotically free, and the induced coefficient (198) supplies a finite, positive boundary value at the breaking scale.

This closes the chain from the Maurer–Cartan curvature (189), through the corrected coset weight $T_m = 4$ (Theorem D.2), to the induced coupling (198) and the asymptotic-freedom slope $b_0 = 12$ for three generations (Theorem D.6) carried into the FRG analysis of subsection 12.2.

E Deferred Computational Appendices

The heat-kernel, functional-renormalization-group, worldsheet, Calabi–Yau, and full anomaly-arithmetic details referenced in the main text are summarized in the sections and algebraic appendices above. This section preserves stable cross-reference anchors for those computations.

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